



North Coast Regional Water Quality Control Board

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD NORTH COAST REGION

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ORDER No. R1-2023-0022
NPDES No. CA0022748
WDID NO. 1B83134OHUM

Waste Discharge Requirements and Water Recycling Requirements

The following Permittee is subject to waste discharge requirements (WDRs) set forth in this Order:

Permittee	City of Rio Dell
Name of Facility	City of Rio Dell Wastewater Treatment Facility
Facility Address	475 Hilltop Drive Rio Dell, CA 95562 Humboldt County

Table 1. Discharge Location

Discharge Point	Effluent Description	Discharge Point Latitude (North-South)	Discharge Point Longitude (East-West)	Receiving Water
001	Disinfected Secondary Treated Wastewater	40° 29' 48.7"	-124° 5' 42.2"	Lower Eel River

Discharge Point	Effluent Description	Discharge Point Latitude (North-South)	Discharge Point Longitude (East-West)	Receiving Water
003	Disinfected Secondary Treated Recycled Water	40° 30' 47.79"	-124° 7' 54.54"	Groundwater

This Order was adopted on:

October 5, 2023

This Order shall become effective on:

December 1, 2023

This Order shall expire on:

November 31, 2028

The Permittee shall file a Report of Waste Discharge as an application for reissuance of WDRs in accordance with title 23, California Code of Regulations, and an application for reissuance of a National Pollutant Discharge Elimination System (NPDES) permit no later than: **October 31, 2027**. The U.S. Environmental Protection Agency (U.S. EPA) and the California Regional Water Quality Control Board, North Coast Region have classified this discharge as follows: **Minor discharge**.

I, Valerie Quinto, Executive Officer, do hereby certify that this Order with all attachments is a full, true, and correct copy of the Order adopted by the California Regional Water Quality Control Board, North Coast Region on October 5, 2023.

Valerie Quinto, Executive Officer

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TABLE OF CONTENTS

1. Facility Information	5
2. Findings.....	5
2.1. Legal Authorities.....	5
2.2. Background and Rationale for Requirements.....	5
2.3. Provisions and Requirements Implementing State Law	5
2.4. Notification of Interested Parties.....	5
2.5. Compliance Schedule Recension.....	6
2.6. Anticipated Water Quality Impacts in Disadvantaged or Tribal Communities.....	6
3. Discharge Prohibitions	7
4. Effluent Limitations and Discharge Specifications.....	9
4.1. Effluent Limitations – Discharge Point 001	9
4.2. Land Discharge Specifications – Not Applicable	11
4.3. Recycling Specifications – Discharge Point 003.....	12
4.4. Other Requirements	14
5. Receiving Water Limitations.....	14
5.1. Surface Water Limitations	14
5.2. Groundwater Limitations.....	17
6. Provisions.....	18
6.1. Standard Provisions	18
6.2. Monitoring and Reporting Program (MRP) Requirements	19
6.3. Special Provisions	19
7. Compliance Determination	29

TABLE OF TABLES

Table 1. Discharge Location.....	1
Table 2. Effluent Limitations ¹ – Discharge Point 001.....	9
Table 3. Interim Effluent Limitations – Discharge Point 001	11
Table 4. Recycled Water Discharge Limitations ¹ – Discharge Point 003	12
Table 5. Schedule for Compliance with Final Effluent Limitations for Disinfection Byproducts	28

TABLE OF ATTACHMENTS

Attachment A – Definitions	A-1
Attachment B – Map.....	B-1
Attachment C – Flow Schematic	C-1

Attachment D – Standard Provisions.....	D-1
Attachment E – Monitoring and Reporting Program.....	E-1
Attachment F – Fact Sheet	F-1
Attachment G – AMEL and MDEL Ammonia Standards Based on 2013	
Freshwater Acute Criteria.....	G-1
Attachment H – Example Ammonia Impact Ratio (AIR) Calculator	H-1

1. FACILITY INFORMATION

Information describing the City of Rio Dell (Permittee) Wastewater Treatment Facility (Facility) is summarized on the cover page and in sections 1 and 2 of the Fact Sheet (Attachment F). Section 1 of the Fact Sheet also includes information regarding the Facility's permit application.

2. FINDINGS

The California Regional Water Quality Control Board, North Coast Region (Regional Water Board), finds:

2.1. Legal Authorities

This Order serves as waste discharge requirements (WDRs) pursuant to article 4, chapter 4, division 7 of the California Water Code (commencing with section 13260). This Order is also issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the U.S. EPA and chapter 5.5, division 7 of the Water Code (commencing with section 13370). It shall serve as a National Pollutant Discharge Elimination System (NPDES) permit authorizing the Permittee to discharge into waters of the United States at the discharge location described in Table 1 subject to the WDRs in this Order. This Order also serves as water recycling requirements pursuant to article 4, chapter 7, division 7 of the Water Code (commencing with section 13500).

2.2. Background and Rationale for Requirements

The Regional Water Board developed the requirements in this Order based on information submitted as part of the application, through monitoring and reporting programs, and other available information. The Fact Sheet (Attachment F), which contains background information and rationale for the requirements in this Order, is hereby incorporated into and constitutes Findings for this Order. Attachments A through E and Attachment H are also incorporated into this Order.

2.3. Provisions and Requirements Implementing State Law

The provisions/requirements in subsections 3.1, 3.2, 3.5, 3.6, 3.7, 4.2, 4.3, 5.2, and 6.3.5.1 and sections 6, 7, 10.5 of the MRP are included to implement state law only. These provisions/requirements are not required or authorized under the federal CWA; consequently, violations of these provisions/requirements are not subject to the enforcement remedies that are available for NPDES violations.

2.4. Notification of Interested Parties

The Regional Water Board has notified the Permittee and interested agencies and persons of its intent to prescribe WDRs for discharge and has provided them with an opportunity to submit their written comments and recommendations. Details of the notification are provided in the Fact Sheet.

2.5. Compliance Schedule Recension

The requirements and interim effluent limitations and compliance schedule of TSO Order No. R1-2023-0030 are implemented in this order in section 4.1.2 and 6.3.5.2, respectively. Therefore, R1-2023-0030 is no longer necessary to implement interim effluent limitations for disinfection byproducts.

2.6. Anticipated Water Quality Impacts in Disadvantaged or Tribal Communities

The Permittee, City of Rio Dell, operates a wastewater treatment facility within a disadvantaged community located along the Lower Eel River in Humboldt County. The discharge is classified as "minor" under federal regulations, and during the term of the prior permit, Order No. R1-2017-0007, from October 1 through May 14, the Facility discharged treated wastewater to the Lower Eel River. In addition, among other updates, this renewed permit contains new requirements to implement bacteria and dissolved oxygen limitations and implement provisions for chronic toxicity. This Order includes a time schedule in accordance with section 13263, subdivision (c) of the Water Code for achieving the applicable water quality objectives for dichlorobromomethane, chlorodibromomethane, total trihalomethanes, and Haloacetic Acids. Expanded monitoring and reporting requirements are included in the renewed Order to ensure waste discharges do not exceed water quality objectives. The Permittee is also currently seeking funding for significant upgrades and improvements to its wastewater treatment facility and collection system. The Regional Water Board publicly noticed the permit and provided opportunities for public comment. Public notice was provided to interested persons and public agencies in the region with jurisdiction over natural resources in the affected area, including the Humboldt County Health Department. The discharge regulated by this Order is not expected to result in a disproportionate impact to tribal or disadvantaged communities. The Regional Water Board has satisfied the outreach and findings requirements set forth in Water Code section 189.7 and Water Code section 13149.2, subd. (d).

The Regional Water Board, in a public meeting, heard and considered all comments pertaining to the discharge. Details of the Public Hearing are provided in the Fact Sheet.

THEREFORE, IT IS HEREBY ORDERED, that Order No. R1-2017-0007 and Time Schedule Order No. R1-2023-0030 are rescinded upon the effective date of this Order except for enforcement purposes, and, in order to meet the provisions contained in division 7 of the Water Code (commencing with section 13000) and regulations adopted thereunder, and the provisions of the CWA and regulations and guidelines adopted thereunder, the Permittee shall comply with the requirements in this Order. This action in no way prevents the Regional Water Board from taking enforcement action for violations of the previous Order.

3. DISCHARGE PROHIBITIONS

- 3.1 The discharge of any waste not disclosed by the Permittee or not within the reasonable contemplation of the Regional Water Board is prohibited.
- 3.2 Creation of pollution, contamination, or nuisance, as defined by section 13050 of the Water Code is prohibited.
- 3.3 The discharge of sludge or digester supernatant is prohibited, except as authorized under section 6.3.4.3 of this Order (Sludge Disposal and Handling Requirements).
- 3.4 The discharge or recycling use of untreated or partially treated waste (receiving a lower level of treatment than described in section 2.1 of the Fact Sheet) from anywhere within the collection, treatment, or disposal systems is prohibited, except as provided for in Attachment D, Standard Provisions 1.7 (Bypass) and 1.8 (Upset).
- 3.5 The discharge of waste to land that is not owned by the Permittee, governed by City ordinance, or under agreement to use by the Permittee, or for which the Permittee has explicitly permitted such use, is prohibited, except for use for fire suppression as provided in title 22, sections 60307(a) and 60307(b) of the California Code of Regulations (CCR).
- 3.6 The discharge of recycled wastewater to any use area not addressed in a DDW-accepted title 22 Recycled Water Engineering Report is prohibited.
- 3.7 The discharge of waste at any point not described in Finding 2.2 of the Fact Sheet or authorized by a permit issued by the State Water Resources Control Board (State Water Board) or another Regional Water Board is prohibited.
- 3.8 The average dry weather flow of waste through the Facility shall not exceed 0.40 mgd, measured daily and averaged over a calendar month. The average monthly wet weather flow of waste through the Facility shall not exceed 1.25 mgd. The peak daily wet weather flow of waste through the Facility shall not exceed 2.51 mgd, measured daily. Compliance with this prohibition shall be determined as defined in sections 7.10 and 7.11 of this Order.
- 3.9 The discharge of waste to the Lower Eel River and its tributaries is prohibited during the period from May 15 through September 30 of each year.
- 3.10 During the period from October 1 through May 14, discharges of treated wastewater to the Lower Eel River shall not exceed one percent of the flow of the Lower Eel River, as measured at United States Geological Survey (USGS) Gage No. 11477000 in the Lower Eel River near Scotia. For the purposes of this Order, compliance with this discharge prohibition shall be determined as follows:
 - 3.10.1 The discharge of secondary treated wastewater shall be adjusted at least once daily to avoid exceeding, to the extent practicable, one percent of the most recent daily flow measurement of the Lower Eel River. Daily

flow shall be based on flow meter comparisons reasonably read between the hours of 12:01 am and 12:00 midnight.

- 3.10.2 In no case shall the total volume of secondary treated wastewater discharged in a calendar month exceed one percent of the total volume of the Lower Eel River's flow that occurs in the same calendar month, as measured per Section 3.10. At the beginning of the discharge season, the monthly flow volume comparison shall be based on the date when the discharge commenced to the end of the calendar month. At the end of the discharge season, the monthly flow volume comparison shall be based on the first day of the calendar month to the date when the discharge ceases for the season.
- 3.11 The discharge of any radiological, chemical, or biological warfare agent into waters of the state under Water Code section 13375.
- 3.12 The acceptance of septage to a location other than an approved septage receiving station and in accordance with a septage management program approved by the Regional Water Board Executive Officer is prohibited.

4. EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

4.1. Effluent Limitations – Discharge Point 001

4.1.1. The Permittee shall maintain compliance with the below effluent limitations at Discharge Point 001, with compliance measured at Monitoring Location EFF-001 as described in the Monitoring and Reporting Program, Attachment E:

Table 2. Effluent Limitations ¹ – Discharge Point 001

Parameter	Units	Average Monthly	Average Weekly	Maximum Daily	Instantaneous Minimum	Instantaneous Maximum
Biochemical Oxygen Demand 5-day @ 20°C (BOD ₅)	mg/L	30	45	--	--	--
Total Suspended Solids	mg/L	30	45	--	--	--
pH	standard units	--	--	--	6.5	8.5
Settleable Solids	ml/L	0.1	--	0.2	--	--
Chlorine Total Residual ²	mg/L	0.01	--	0.02	--	--
Chlorodibromomethane	µg/L	0.41	--	0.89	--	--
Dichlorobromomethane	µg/L	0.56	--	1.1	--	--
Total Trihalomethanes ³	µg/L	80	--	--	--	--
Haloacetic Acids ⁴	µg/L	60	--	--	--	--
Nitrogen, Total (as N)	mg/L	10	--	--	--	--
Ammonia Impact Ratio ⁵	Ratio	1.0	--	1.0	--	--
4,4-DDT	ug/L	0.00059	--	0.00118	--	--

Parameter	Units	Average Monthly	Average Weekly	Maximum Daily	Instantaneous Minimum	Instantaneous Maximum
Antimony, Total Recoverable	ug/L	6	--	12	--	--

Table Notes

1. See Definitions in Attachment A and Compliance Determination discussion in section 7 of this Order.
2. See section 7.14 of this Order regarding compliance with chlorine residual effluent limitations.
3. The sum of the concentrations of bromoform, chlorodibromomethane, chloroform, and dichlorobromomethane.
4. The sum of the concentrations of Monochloroacetic Acid, Dichloroacetic Acid, Trichloroacetic Acid, Monobromoacetic Acid, and Dibromoacetic Acid.
5. The Ammonia Impact Ratio (AIR) is calculated as the ratio of the ammonia concentration in the effluent and the applicable ammonia standard (AMEL and MDEL). Attachment I is a PDF example of the calculator that will be sent to the Permittee to determine compliance with the AMEL/MDEL AIR. For each of the applicable ammonia standards, Attachment G includes two tables that provide the variable AMEL and MDEL ammonia standards used in calculating the AIR. The AIR is the ammonia effluent limit and must be reported in the self-monitoring reports in addition to ammonia concentrations, and pH and temperature values in the effluent and receiving water. Monitoring for ammonia, pH and temperature must be conducted concurrently in order for the AIR to be calculated properly.

- 4.1.1.1. **Percent Removal.** The average monthly percent removal of BOD₅ and total suspended solids shall not be less than 85 percent. Percent removal shall be determined from the monthly average value of influent wastewater concentration in comparison to the monthly average value of effluent concentration for the same constituent over the same time period as measured at Monitoring Locations INF-001 and EFF-001, respectively.
- 4.1.1.2. **Disinfection.** Disinfected effluent discharged from the Facility through Discharge Point 001 to Lower Eel River shall not contain total coliform bacteria exceeding the following concentrations, as measured at Monitoring Location EFF-001:

- 4.1.1.2.1. The median concentration shall not exceed a Most Probable Number (MPN) of 23 per 100 milliliters (mL) in a calendar month; and
- 4.1.1.2.2. No sample shall exceed an MPN of 240 per 100 mL.
- 4.1.1.3. **Chronic Toxicity.** To determine compliance with the water quality objective for toxicity in the Basin Plan, the discharge, as measured at Monitoring Location EFF-001, shall meet the following effluent limitations:
- 4.1.1.3.1. **Maximum Daily Effluent Limitation.** No chronic toxicity test shall result in a “fail” at the IWC for the sub-lethal endpoint measured in the test and a percent effect for the survival endpoint greater than or equal to 50 percent.
- 4.1.1.3.2. **Median Monthly Effluent Target.** No more than one chronic toxicity test initiated in a calendar month shall result in a “fail” at the IWC for any endpoint, effect only through December 31, 2023.
- 4.1.1.3.3. **Median Monthly Effluent Limitation.** No more than one chronic aquatic toxicity test initiated in a calendar month shall result in a “fail” at the IWC for any endpoint, starting January 1, 2024.

4.1.2. **Interim Effluent Limitations for Disinfection Byproducts**

For the duration of Compliance Schedule outlined in section 6.3.5.2 , the Permittee shall maintain compliance with the following interim effluent limitations at Discharge Point 001, with compliance measured at Monitoring Location EFF-001, as described in the attached MRP (Attachment E). Upon completion of the compliance schedule, the Permittee shall maintain compliance with final effluent limitations in section 4.1.1 of this order.

Table 3. Interim Effluent Limitations – Discharge Point 001

Parameter	Unit	Average Monthly
Chlorodibromomethane	µg/L	2.34
Dichlorobromomethane	µg/L	20.0
Total Trihalomethanes	µg/L	113
Haloacetic Acids	µg/L	196

4.2. **Land Discharge Specifications – Not Applicable**

This Order does not authorize discharges to land.

4.3. Recycling Specifications – Discharge Point 003

4.3.1. Water Recycling Specifications

The Permittee shall maintain compliance with the following limitations at Discharge Point 003, with compliance measured at Monitoring Location EFF-003, as described in the attached MRP.

Table 4. Recycled Water Discharge Limitations ¹ – Discharge Point 003

Parameter	Units	Average Monthly	Average Weekly	Maximum Daily	Instantaneous Minimum	Instantaneous Maximum
Biochemical Oxygen Demand 5-day @ 20°C (BOD ₅)	mg/L	30	45	---	---	---
Total Suspended Solids (TSS)	mg/L	30	45	---	---	---
pH	standard units	---	---	---	6.5	8.5
Nitrogen, Total (as N) ²	mg/L	10	---	---	---	---

Table Notes

1. See Definitions in Attachment A and Compliance Determination discussion in section 7 of this Order.
2. The sum of the concentrations of nitrate nitrogen, nitrite nitrogen, total organic nitrogen and ammonia.

4.3.1.1. **Disinfection.** Disinfected effluent discharged from the Facility through Discharge Point 003 to the irrigation site shall not contain total coliform bacteria exceeding the following concentrations, as measured at Monitoring Location EFF-003:

4.3.1.1.1. The median concentration shall not exceed a Most Probable Number (MPN) of 23 per 100 milliliters (mL) in a calendar month; and

4.3.1.1.2. No sample shall exceed an MPN of 240 per 100 mL.

4.3.2. Water Recycling Requirements

- 4.3.2.1. The Permittee shall install, operate, and maintain the irrigation system in a manner that ensures compliance with all requirements of this Order.
- 4.3.2.2. The Permittee shall conduct periodic inspections of the irrigation system, facilities, and operations to monitor and ensure compliance with the conditions of this Order.
- 4.3.2.3. The use of disinfected recycled water for irrigation shall not cause or contribute to an exceedance of any applicable water quality standard. The Permittee shall be responsible for ensuring that all discharges to the irrigation system meet all terms and conditions of this Order, including the Recycled Water Discharge Limitations in section 4.3.1 of this Order.
- 4.3.2.4. The Permittee shall discontinue delivery of recycled water for irrigation during any period that there is reason to believe that the requirements for use, as specified in this Order are not being met. The delivery of recycled water for irrigation shall not resume until all conditions have been corrected.
- 4.3.2.5. Disinfected recycled water shall not be irrigated within 100 feet of any domestic water supply well.
- 4.3.2.6. The use of disinfected recycled water for irrigation shall not cause degradation of any water supply.
- 4.3.2.7. Irrigation areas shall be managed to prevent ponding and conditions conducive to the proliferation of mosquitoes and other disease vectors, and to avoid creation of a public nuisance or health hazard. The following practices shall be implemented, at a minimum:
 - 4.3.2.7.1. Irrigation water shall infiltrate completely within a 48-hour period; and
 - 4.3.2.7.2. Low-pressure and unpressurized pipelines and ditches that may be accessible to mosquitoes shall not be used to store effluent.
- 4.3.2.8. Recycled water used for irrigation shall not be allowed to escape the irrigation area in the form of surface runoff. Where appropriate, practices and strategies to prevent the occurrence of runoff shall include, but not be limited to the parameters outlined below:
 - 4.3.2.8.1. A minimum 50-foot setback to all surface waters or implementation of Best Management Practices (BMPs) designed to prevent the potential for runoff discharging to surface water;

- 4.3.2.8.2. Proper design and operation of the irrigation system, including the use of the drainage collection system to pump runoff back onto the irrigation fields;
- 4.3.2.8.3. Refraining from application during precipitation events; and
- 4.3.2.8.4. Maintenance of irrigation infrastructure (e.g., pipelines, pumps, etc.) to prevent and minimize breakage and leaks.
- 4.3.2.9. All irrigation equipment, pumps, piping, valves, quick couplers and outlets shall be a type or secured in a manner that only permits operation by authorized personnel and shall be appropriately marked to differentiate them from potable facilities.
- 4.3.2.10. The Permittee shall comply with applicable state and local requirements regarding the production of recycled water, including requirements of Water Code sections 13500-13577 (Water Reclamation) and State Water Board, Division of Drinking Water (DDW) regulations at title 22, sections 60301 – 60357 of the CCR (Water Recycling Criteria).
- 4.3.2.11. The Permittee shall implement its DDW title 22 Recycled Water Engineering Report, accepted April 2022, (and any subsequent amendments thereto). The Permittee shall submit revisions and updates to the title 22 Recycled Water Engineering Report to reflect any changes in operations and recycled water management or new use types.

4.4. Other Requirements

- 4.4.1. **Total Residual Chlorine, Monitoring Location INT-001.** As measured at the end of the chlorine contact chamber at Monitoring Location INT-001, the total residual chlorine concentration shall be maintained at a level that ensures the discharge meets the total coliform effluent limitation at the end of the disinfection process for discharges to Discharge Points 001 and 003.

5. RECEIVING WATER LIMITATIONS

5.1. Surface Water Limitations

Receiving water limitations are based on water quality objectives contained in the Basin Plan and are a required part of this Order. Receiving water conditions not in conformance with the limitations are not necessarily a violation of this Order. Compliance with receiving water limitations shall be measured at monitoring locations described in the MRP (Attachment E). The Regional Water Board may require an investigation and/or consider other available information to determine cause and culpability prior to asserting that a violation has occurred.

The discharge shall not cause the following in the receiving water:

- 5.1.1. The discharge shall not cause the dissolved oxygen (DO) concentration of the receiving water to be depressed below 9.0 mg/L daily and 11.0 mg/L as a 7-day rolling average. In those waterbodies for which the aquatic life-based DO requirements are unachievable due to natural conditions, site-specific background DO requirements can be applied as water quality objectives by calculating the daily minimum DO necessary to maintain 85% DO saturation during the dry season and 90% DO saturation during the wet season under site salinity, site atmospheric pressure, and natural receiving water temperature. In no event may controllable factors reduce the daily minimum DO below 6.0 mg/L.

Natural conditions are conditions or circumstances affecting the physical, chemical, or biological integrity of water that are not influenced by past or present anthropogenic activities. Site specific DO requirements can be applied upon approval from the Executive Officer. The method(s) used to estimate natural temperatures for a given waterbody or stream length must be approved by the Executive Officer and may include, as appropriate, comparison with reference streams, simple calculation, or computer models.

- 5.1.2. The discharge shall not cause the specific conductance (micromhos) concentration of the receiving waters to increase above 225 micromhos more than 50 percent of the time, or above 375 micromhos more than 10 percent of the time.
- 5.1.3. The discharge shall not cause the total dissolved solids concentration of the receiving waters to increase above 140 mg/L more than 50 percent of the time, or above 275 mg/L more than 10 percent of the time.
- 5.1.4. The discharge shall not cause the pH of receiving waters to be depressed below 6.5 nor raised above 8.5. Within this range, the discharge shall not cause the pH of the receiving waters to be changed at any time more than 0.5 units from that which occurs naturally.
- 5.1.5. The discharge shall not cause the turbidity of receiving waters to be increased more than 20 percent above naturally occurring background levels.
- 5.1.6. The discharge shall not cause receiving waters to contain suspended material in concentrations that cause nuisance or adversely affect beneficial uses.
- 5.1.7. The discharge shall not cause receiving waters to contain floating materials, including solids, liquids, foams, and scum, in concentrations that cause nuisance or adversely affect beneficial uses.
- 5.1.8. The discharge shall not cause receiving waters to contain taste- or odor-producing substances in concentrations that impart undesirable tastes or odors to fish flesh or other edible products of aquatic origin, that cause nuisance, or that adversely affect beneficial uses.

- 5.1.9. The discharge shall not cause coloration of receiving waters that causes nuisance or adversely affects beneficial uses.
- 5.1.10. The discharge shall not cause bottom deposits in receiving waters to the extent that such deposits cause nuisance or adversely affect beneficial uses.
- 5.1.11. The discharge shall not cause receiving waters to contain concentrations of biostimulatory substances that promote objectionable aquatic growth to the extent that such growth causes nuisance or adversely affects beneficial uses.
- 5.1.12. The discharge shall not cause receiving waters to contain toxic substances in concentrations that are toxic to, or that produce detrimental physiological responses in humans, plants, animals, or aquatic life. Compliance with this objective will be determined by use of indicator organisms, analyses of species diversity, population density, growth anomalies, bioassays of appropriate duration, or other appropriate methods, as specified by the Regional Water Board.
- 5.1.13. The discharge shall not cause a measurable temperature change in the receiving water at any time.
- 5.1.14. The discharge shall not cause an individual pesticide or combination of pesticides to be present in concentrations that adversely affect beneficial uses. The discharge shall not cause bioaccumulation of pesticide concentrations in bottom sediments or aquatic life.
- 5.1.15. The discharge shall not cause receiving waters to contain concentrations of pesticides in excess of Maximum Contaminant Levels (MCLs) established for these pollutants in title 22, division 4, chapter 15, articles 4 and 5.5 of the CCR.
- 5.1.16. The discharge shall not cause receiving waters to contain oils, greases, waxes, or other materials in concentrations that result in a visible film or coating on the surface of the water or on objects in the water, that cause nuisance, or that otherwise affect beneficial uses.
- 5.1.17. The discharge shall not cause a violation of any applicable water quality standard for receiving waters adopted by the Regional Water Board or the State Water Board, as required by the federal Clean Water Act and regulations adopted thereunder. If more stringent applicable water quality standards are promulgated or approved pursuant to section 303 of the Clean Water Act, or amendments thereto, the Regional Water Board will revise and modify this Order in accordance with such more stringent standards.
- 5.1.18. The discharge shall not cause concentrations of chemical constituents to occur in excess of MCLs established for these pollutants in title 22, division 4, chapter 15, articles 4 and 5.5 of the CCR.

- 5.1.19. The discharge shall not cause receiving waters to contain radionuclides in concentrations which are deleterious to human, plant, animal or aquatic life, nor which result in the accumulation of radionuclides in the food web to an extent which presents a hazard to human, plant, animal or indigenous aquatic life.
- 5.1.20. The bacteria water quality objective for all waters where the salinity is equal to or less than 1 part per thousand (ppt) 95 percent or more of the time during the calendar year is: a six week rolling geometric mean of *Escherichia coli* (*E. coli*) not to exceed 100 colony forming units (cfu) per 100 milliliter (mL), calculated weekly, and a statistical threshold value (STV) of 320 cfu/100 mL not to be exceeded by more than 10 percent of the samples collected in a calendar month, calculated in a static manner.

5.2. Groundwater Limitations

- 5.2.1. The collection, treatment, storage, and disposal of wastewater or use of recycled water shall not cause degradation of groundwater quality unless a technical evaluation is performed that demonstrates that any degradation that could reasonably be expected to occur, after implementation of all reasonable BMPs, will not violate groundwater quality objectives or cause impacts to beneficial uses of groundwater.
- 5.2.2. The collection, treatment, storage, and disposal of wastewater or use of recycled water shall not cause or contribute to levels of chemical constituents in groundwater that exceed the maximum and secondary maximum contaminant levels (MCLs and SMCLs) established for these pollutants in the title 22, division 4, chapter 15, article 4, section 64431; article 5.5, section 64444; and article 16, section 64449 of the CCR.
- 5.2.3. The collection, treatment, storage, and disposal of wastewater or use of recycled water shall not cause or contribute to levels of radionuclides in groundwater in concentrations that cause nuisance or adversely affect beneficial uses, nor in excess of the MCLs and SMCLs established for these pollutants in title 22, division 4, chapter 15, article 5, sections 64442 and 64443 of the CCR.
- 5.2.4. The collection, treatment, storage, and disposal of wastewater or use of recycled water shall not cause groundwater to contain taste- or odor-producing substances in concentrations that cause nuisance or adversely affect beneficial uses.
- 5.2.5. The collection, treatment, storage, and disposal of wastewater or use of recycled water shall not cause the median of the most probable number of coliform organisms over any 7-day period to exceed 1.1 MPN/100 mL or 1 colony/100 mL in groundwaters used for domestic or municipal supply (MUN).
- 5.2.6. The collection, treatment, and disposal of wastewater or use of recycled water shall not cause groundwater to contain toxic substances in concentrations that

are toxic to, or that produce detrimental physiological responses in humans, or that adversely affects beneficial uses. This limitation applies regardless of whether the toxicity is caused by a single substance or the synergistic effect of multiple substances.

6. PROVISIONS

6.1. Standard Provisions

- 6.1.1. **Federal Standard Provisions.** The Permittee shall comply with all Standard Provisions included in Attachment D of this Order.
- 6.1.2. **Regional Water Board Standard Provisions.** The Permittee shall comply with the following Regional Water Board standard provisions. In the event that there is any conflict, duplication, or overlap between provisions specified by this Order, the more stringent provision shall apply:
- 6.1.2.1. Failure to comply with provisions or requirements of this Order, or violation of other applicable laws or regulations governing discharges from this facility, may subject the Permittee to administrative or civil liabilities, criminal penalties, and/or other enforcement remedies to ensure compliance. Additionally, certain violations may subject the Permittee to civil or criminal enforcement from appropriate local, state, or federal law enforcement entities.
 - 6.1.2.2. In the event the Permittee does not comply or will be unable to comply for any reason, with any prohibition, final effluent limitation, recycled water specification, other specification, receiving water limitation, or provision of this Order, that may result in significant threat to human health or the environment, such as inundation of treatment infrastructure, breach of pond containment, sanitary sewer overflow, recycled water main break or equivalent release, irrigation runoff, etc., that results in a discharge to a drainage channel or a surface water, the Permittee shall:
 - 6.1.2.2.1. Notify the Regional Water Board within 24 hours of having knowledge of such noncompliance. Spill notification and reporting shall be conducted in accordance with section 5.5 of Attachment D and section 10.5 of the MRP (Attachment E).
 - 6.1.2.2.2. Investigate the cause(s) of final effluent limitation and discharge specification exceedances and failures to comply with any prohibition, specification, or provision of this Order that may result in significant threat to human health or the environment.
 - 6.1.2.2.3. Identify and implement corrective actions to prevent future exceedances or failures to comply with Order requirements.

- 6.1.2.2.4. Report the results of such investigations and corrective actions implemented in the quarterly SMR as required by MRP section 10.2.6.2.5 and 10.2.6.2.6.

6.2. Monitoring and Reporting Program (MRP) Requirements

The Permittee shall comply with the MRP, and future revisions thereto, in Attachment E.

6.3. Special Provisions

6.3.1. Reopener Provisions

- 6.3.1.1. **Standard Revisions.** If applicable water quality standards are promulgated or approved pursuant to section 303 of the CWA, or amendments thereto, the Regional Water Board may reopen this Order and make modifications in accordance with such revised standards.
- 6.3.1.2. **Reasonable Potential.** This Order may be reopened for modification to include an effluent limitation if monitoring establishes that the discharge causes or has the reasonable potential to cause or contribute to, an excursion above a water quality criterion or objective applicable to the receiving water.
- 6.3.1.3. **Species Sensitivity Screening.** Upon completion of the species sensitivity screening, this Order may be reopened to specify the most sensitive species. Furthermore, the MDEL and MMEL, as identified in sections 4.1.1.3.1 and 4.1.1.3.2, respectively, may be modified to reflect the identified most sensitive species. Reopening of the permit is not required if the species sensitivity screening indicates that the most sensitive species is *Ceriodaphnia dubia*.
- 6.3.1.4. **Whole Effluent Toxicity.** As a result of a Toxicity Reduction Evaluation (TRE), this Order may be reopened to include a narrative or numeric chronic toxicity limitation, a new acute toxicity limitation, and/or a limitation for a specific toxicant identified in the TRE.
- 6.3.1.5. **Acute Aquatic Toxicity.** This Order may be reopened to allow the reevaluation of reasonable potential for the Permittee to cause or contribute to an exceedance of the acute aquatic toxicity water quality objective, and add the resulting MDEL and MMEL, if warranted, after the evaluation of new data and information.
- 6.3.1.6. **303(d)-Listed Pollutants.** If an applicable total maximum daily load (TMDL) (see Fact Sheet, section 3.4) program is adopted, this Order may be reopened and effluent limitations for the pollutant(s) that are the subject of the TMDL may be modified or imposed to conform this Order to the TMDL requirements.

- 6.3.1.7. **Water Effects Ratios (WERs) and Metal Translators.** A default WER of 1.0 has been used in this Order for calculating CTR criteria for applicable priority pollutant inorganic constituents. If the Permittee performs studies to determine site-specific WERs and/or site-specific dissolved-to-total metal translators and submits a report that demonstrates that WER or translator studies were performed in accordance with U.S. EPA or other approved guidance, this Order may be reopened to modify the effluent limitations for the applicable constituents.
- 6.3.1.8. **Nutrients.** This Order contains effluent limitations for ammonia and effluent and receiving water monitoring requirements for nutrients (ammonia, nitrate, and phosphorus). If new water quality objectives for nutrients are established, if monitoring data indicate the need for new or revised effluent limitations for any of these parameters, or if new or revised methods for compliance with effluent limitations for any of these parameters are developed, this Order may be reopened and modified to include new or modified effluent limitations or other requirements, as necessary.
- 6.3.1.9. **Salt and Nutrient Management Plans (SNMPs).** The Recycled Water Policy adopted by the State Water Board on February 3, 2009, and effective May 14, 2009, recognizes the fact that some groundwater basins in the state contain salts and nutrients that exceed or threaten to exceed water quality objectives in the applicable Basin Plans, and that not all Basin Plans include adequate implementation procedures for achieving or ensuring compliance with the water quality objectives for salt or nutrients. The Recycled Water Policy finds that the appropriate way to address salt and nutrient issues is through the development of regional or subregional SNMPs rather than through imposing requirements solely on individual recycled water projects. This Order may be reopened to incorporate provisions consistent with any SNMP(s) adopted by the Regional Water Board.
- 6.3.1.10. **Title 22 Recycled Water Engineering Report.** This Order implements title 22 requirements to protect public health. If the Permittee's title 22 Recycled Water Engineering Report requires modifications to this Order to adequately implement title 22, this Order may be reopened and modified as necessary.
- 6.3.1.11. **Mixing Zone Study.** This Order may be reopened to consider modifying effluent limitations and receiving water monitoring locations if the Permittee demonstrates to the satisfaction of the Regional Water Board Executive Officer that it has evaluated all reasonable alternatives for compliance with human health-based effluent limitations and conducts a mixing zone study that provides a basis for determining that permit conditions should be modified.

6.3.2. **Best Management Practices and Pollution Prevention**

6.3.2.1. **Pollutant Minimization Program (PMP)**

The Permittee shall, as required by the Regional Water Board Executive Officer (Hereafter Executive Officer), develop and conduct a PMP, as further described below, when there is evidence (e.g., sample results reported as DNQ when the effluent limitation is less than the MDL, sample results from analytical methods more sensitive than those methods required by this Order, presence of whole effluent toxicity, health advisories for fish consumption, results of benthic or aquatic organism tissue sampling) that a priority pollutant is present in the effluent above an effluent limitation and either:

- 6.3.2.1.1. A sample result is reported as DNQ and the effluent limitation is less than the RL; or
- 6.3.2.1.2. A sample result is reported as ND and the effluent limitation is less than the MDL, using definitions described in Attachment A and reporting protocols described in MRP section 10.2.4.
- 6.3.2.2. The PMP shall include, but not be limited to, the following actions and submittals acceptable to the Regional Water Board:
 - 6.3.2.2.1. An annual review and semi-annual monitoring of potential sources of the reportable priority pollutant(s), which may include fish tissue monitoring and other bio-uptake sampling;
 - 6.3.2.2.2. Quarterly monitoring for the reportable priority pollutant(s) in the influent to the wastewater treatment system;
 - 6.3.2.2.3. Submittal of a control strategy designed to proceed toward the goal of maintaining concentrations of the reportable priority pollutant(s) in the effluent at or below the effluent limitation;
 - 6.3.2.2.4. Implementation of appropriate cost-effective control measures for the reportable priority pollutant(s), consistent with the control strategy; and
 - 6.3.2.2.5. An annual status report that shall be submitted as part of the Annual Facility Report due March 1st to the Regional Water Board and shall include:
 - 6.3.2.2.5.1.1. All PMP monitoring results for the previous year;
 - 6.3.2.2.5.1.2. A list of potential sources of the reportable priority pollutant(s);
 - 6.3.2.2.5.1.3. A summary of all actions undertaken pursuant to the control strategy; and

6.3.2.2.5.1.4. A description of actions to be taken in the following year.

6.3.3. Construction, Operation and Maintenance Specifications

- 6.3.3.1. **Proper Operation and Maintenance.** This Order (Attachment D, Standard Provision 1.4) requires that the Permittee at all times properly operate and maintain all facilities and systems of treatment and control (and related appurtenances) that are installed or used by the Permittee to achieve compliance with this Order. Proper operation and maintenance includes adequate laboratory quality control and appropriate quality assurance procedures.
- 6.3.3.2. **Operation and Maintenance Manual.** The Permittee shall maintain an updated Operation and Maintenance (O&M) Manual for the operational components of the Facility. The Permittee shall update the O&M Manual, as necessary, to conform to changes in operation and maintenance of the Facility. The Permittee shall operate and maintain the Facility in accordance with the most recently updated O&M Manual. The O&M Manual shall be readily available to operating personnel onsite and for review by state or federal inspectors. The O&M Manual shall include the following:
- 6.3.3.2.1. Description of the Facility's organizational structure showing the number of employees, duties and qualifications, and facility attendance schedules (daily, weekends and holidays, part-time, etc.). The description should include documentation that the personnel are knowledgeable and qualified to operate the Facility so as to achieve the required level of treatment at all times.
- 6.3.3.2.2. Detailed description of safe and effective operation and maintenance of treatment processes, process control instrumentation and equipment.
- 6.3.3.2.3. Description of laboratory and quality assurance procedures.
- 6.3.3.2.4. Process and equipment inspection and maintenance schedules.
- 6.3.3.2.5. Description of safeguards to assure that, should there be reduction, loss, or failure of electric power, the Permittee will be able to comply with requirements of this Order.
- 6.3.3.2.6. Description of preventive (fail-safe) and contingency (response and cleanup) plans for controlling accidental discharges, and for minimizing the effect of such events. These plans shall identify the possible sources (such as loading and storage areas, power outage, waste treatment unit failure, process equipment failure, tank and piping failure) of accidental discharges, untreated or partially treated waste bypass, and polluted drainage.
- 6.3.3.3. **Operating Records.** The Permittee shall maintain operating records at the Facility or at the Permittee's central records depository. The records shall

include: all analyses specified in the reclamation criteria; any documentation of operational problems, facility and equipment breakdowns, and diversions to emergency storage or disposal; and documentation of all corrective or preventive actions taken.

6.3.4. Special Provisions for Publicly-Owned Treatment Works (POTWs)

6.3.4.1. Wastewater Collection Systems

The Permittee has coverage under, and is separately subject to the requirements of, State Water Board Order No. 2022-0103-DWQ, Statewide General WDRs for Sanitary Sewer Systems, and any subsequent revisions. As such, the Permittee provides notification and reporting of SSOs in accordance with the requirements of Order No. 2022-0103-DWQ and any revisions thereto for operation of its wastewater collection system.

6.3.4.2. Source Control and Pretreatment Provisions

6.3.4.2.1. The Permittee shall perform source control functions and provide a summary of source control activities conducted in the Annual Report (due **March 1st** to the Regional Water Board). Source control functions and requirements shall include the following:

6.3.4.2.1.1. Implement the necessary legal authorities to monitor and enforce source control standards, restrict discharges of toxic materials to the collection system and inspect facilities connected to the system.

6.3.4.2.1.2. If waste haulers are allowed to discharge to the Facility, establish a waste hauler permit system, to be reviewed and approved by the Executive Officer, to regulate waste haulers discharging to the collection system or Facility.

6.3.4.2.1.3. Perform public outreach to educate industrial, commercial, and residential users about the importance of preventing discharges of industrial and toxic wastes to the wastewater treatment facility, at least once per year.

6.3.4.2.1.4. Perform ongoing inspections and monitoring, as necessary, to ensure adequate source control.

6.3.4.2.1.5. **General Prohibitions.** Pollutants introduced into wastewater treatment facilities (WWTFs) by a non-domestic source shall not pass through [40 CFR 403.3(n)] the WWTF or interfere [40 CFR 403.3(i)] with the operation or performance of the works. These general prohibitions and the specific prohibitions in section 6.3.5.2.1.6 of this provision apply to all non-domestic sources introducing pollutants into a WWTF whether or not the source is subject to other National Pretreatment Standards or any national, state, or local pretreatment requirements.

6.3.4.2.1.6. **Specific Prohibitions.** In addition, the following pollutants shall not be introduced into a WWTF:

- 6.3.4.2.1.6.1. Pollutants that create a fire or explosion hazard in the WWTF;
- 6.3.4.2.1.6.2. Pollutants that will cause corrosive structural damage to the WWTF, but in no case discharges with pH lower than 5.0, unless the WWTF is specifically designed to accommodate such discharges;
- 6.3.4.2.1.6.3. Solid or viscous pollutants in amounts that will cause obstruction to the flow in the WWTF resulting in interference;
- 6.3.4.2.1.6.4. Any pollutant, including oxygen demanding pollutants (BOD, etc.) released in a discharge at a flow rate and/or pollutant concentration that will cause interference with the WWTF;
- 6.3.4.2.1.6.5. Heat in amounts that will inhibit biological activity in the WWTF resulting in interference, but in no case heat in such quantities that the temperature at the WWTF exceeds 40°C (104°F) unless the Regional Water Board, upon request of the Permittee, approves alternate temperature limits;
- 6.3.4.2.1.6.6. Petroleum oil, non-biodegradable cutting oil, or products of mineral oil origin in amounts that will cause interferences or pass through; and
- 6.3.4.2.1.6.7. Pollutants that result in the presence of toxic gases, vapors, or fumes within the WWTF in a quantity that may cause acute worker health and safety problems

6.3.4.2.2. **Industrial Waste Additions**

In the event that the Permittee identifies industrial wastes subject to regulation under the NPDES Pretreatment Program being discharged to the wastewater treatment facility, or the Regional Water Board or its Executive Officer determines that circumstances warrant pretreatment requirements in order to prevent interference [40 C.F.R. §403.3(j)] with the wastewater treatment Facility or Pass Through [40 C.F.R. §403.3(n)], then:

- 6.3.4.2.2.1. The Permittee shall notify the Regional Water Board within 30 days after there are discharges that trigger the pretreatment requirements;
- 6.3.4.2.2.2. The Permittee shall submit a revised ROWD and the pretreatment program for the Regional Water Board's review and approval as soon as possible, but not more than one year after the Permittee's notification to the Regional Water Board of the need for pretreatment requirements being triggered;

- 6.3.4.2.2.3. The Permittee shall enforce the federal categorical pretreatment standards on all categorical industrial users (CIUs);
- 6.3.4.2.2.4. The Permittee shall notify each CIU of its discharge effluent limits. The limits must be as stringent as the pretreatment standards contained in the applicable federal category (40 C.F.R. Part 400-699). The Permittee may develop more stringent, technology-based local limits if it can show cause; and
- 6.3.4.2.2.5. The Permittee shall notify the Regional Water Board if any CIU violates its discharge effluent limits.
- 6.3.4.2.3. The Regional Water Board retains the right to take legal action against an industrial user and/or the Permittee where a user fails to meet the approved applicable federal, state, or local pretreatment standards.
- 6.3.4.2.4. The Regional Water Board may amend this Order, at any time, to require the Permittee to develop and implement an industrial pretreatment program pursuant to the requirements of 40 C.F.R. Part 403 if the Regional Water Board finds that the Facility receives pollutants from an IU that is subject to pretreatment standards, or if other circumstances so warrant.

6.3.4.3. Sludge Disposal and Handling Requirements

- 6.3.4.3.1. Sludge, as used in this Order, means the solid, semisolid, and liquid residues removed during primary, secondary, or secondary wastewater treatment processes. Solid waste refers to grit and screenings generated during preliminary treatment. Biosolids refers to sludge that has been treated, tested, and demonstrated to be capable of being beneficially and legally used pursuant to federal and state regulations as a soil amendment for agriculture, silviculture, horticulture, and land reclamation activities.
- 6.3.4.3.2. All collected sludges and other solid waste removed from liquid wastes shall be removed from screens, sumps, ponds, and tanks as needed to ensure optimal facility operation and disposed of in accordance with applicable federal and state regulations.
- 6.3.4.3.3. The use and disposal of biosolids shall comply with all of the land application and disposal requirements in 40 C.F.R. part 503, which are enforceable by the U.S. EPA, not the Regional Water Board. If during the life of this Order, the state accepts primacy for implementation of 40 C.F.R. part 503, the Regional Water Board may also initiate enforcement where appropriate.
- 6.3.4.3.4. The Permittee shall ensure that any biosolids it has land applied is incorporated within six hours in order to meet Vector Attraction Reduction requirements in 40 C.F.R. 503.33.

- 6.3.4.3.5. Sludge or biosolids that are disposed of in a municipal solid waste landfill or used as daily landfill cover shall meet the applicable requirements of 40 C.F.R. part 258. In the annual self-monitoring report, the Permittee shall report the amount of sludge placed in a landfill and the landfill(s) which received the sludge or biosolids.
- 6.3.4.3.6. The Permittee shall take all reasonable steps to prevent and minimize any sludge use or disposal in violation of this Order that may adversely affect human health or the environment.
- 6.3.4.3.7. Solids and sludge treatment, storage, and disposal or reuse shall not create a nuisance, such as objectionable odors or flies, and shall not result in groundwater contamination.
- 6.3.4.3.8. Solids and sludge treatment and storage sites shall have facilities adequate to divert surface water runoff from adjacent areas, protect the boundaries of the site from erosion, and prevent drainage from the treatment and storage site. Adequate protection is defined as protection from a design storm with a 100-year recurrence interval and 24-hour duration.
- 6.3.4.3.9. The discharge of sewage sludge and solids shall not cause waste material to be in a position where it is, or can be, conveyed from the treatment and storage sites and deposited in the waters of the state.
- 6.3.4.3.10. For the land application of biosolids as soil amendment, the Permittee shall submit a report of waste discharge or the Permittee may dispose of biosolids at another appropriately permitted facility.
- 6.3.4.3.11. New sludge treatment and storage facilities must comply with the requirements of the Water Code and title 27 of the CCR for the protection of water quality.
- 6.3.4.3.12. The Permittee currently gives away all dewatered Class A biosolids to residents as a soil amendment. The Permittee shall notify the Regional Water Board and U.S. EPA at NorthCoast@waterboards.ca.gov and R9NPDES@epa.gov, respectively, prior to changing biosolids use or disposal practices.

6.3.4.4. Biosolids Management

For any discharge of biosolids from the Facility, the Permittee shall comply with the following requirements:

- 6.3.4.4.1. For the land application of biosolids as soil amendment within the North Coast Region, the Permittee shall obtain or maintain coverage under the State Water Board Water Quality Order No. 2004-0012-DWQ General Waste Discharge Requirements for the Discharge of Biosolids to Land or

Use as a Soil Amendment in Agricultural, Silvicultural, Horticultural, and Land Reclamation Activities, or

- 6.3.4.4.2. Alternatively, the Permittee may dispose of biosolids at another appropriately permitted facility.
- 6.3.4.4.3. New sludge treatment and storage facilities must comply with the requirements of the Water Code and title 27 of the CCR for the protection of water quality.

6.3.4.5. **Operator Certification**

Supervisors and operators of municipal wastewater treatment facilities shall possess a certificate of appropriate grade in accordance with title 23, CCR, section 3680. The State Water Board may accept experience in lieu of qualification training. In lieu of a properly certified wastewater treatment facility operator, the State Water Board may approve use of a water treatment facility operator of appropriate grade certified by DDW where water recycling is involved.

6.3.4.6. **Adequate Capacity**

If the Facility is projected to reach capacity within 4 years, the Permittee shall notify the Regional Water Board. A copy of such notification shall be sent to appropriate local elected officials, local permitting agencies, and the press. Factors to be evaluated in assessing reserve capacity shall include, at a minimum, (1) comparison of the wet weather design flow with the highest daily flow, and (2) comparison of the average dry weather design flow with the lowest 30-day flow. The Permittee shall demonstrate that adequate steps are being taken to address the capacity problem. The Permittee shall submit a technical report to the Regional Water Board showing how flow volumes will be prevented from exceeding capacity, or how capacity will be increased, within 120 days after providing notification to the Regional Water Board, or **within 120 days after receipt of Regional Water Board notification** that the Facility will reach capacity within 4 years. The time for filing the required technical report may be extended by the Regional Water Board. An extension of 30 days may be granted by the Executive Officer, and longer extensions may be granted by the Regional Water Board itself. [Cal. Code Regs., tit. 23, § 2232].

6.3.5. **Other Special Provisions**

6.3.5.1. **Storm Water**

For the control of storm water discharges from the Facility, if required, the Permittee shall seek separate authorization to discharge under the requirements of the State Water Board's Water Quality Order No. 2014-0057-DWQ, NPDES General Permit No. CAS000001, General Permit for

Storm Water Discharges Associated with Industrial Activities (or subsequent renewed versions of the NPDES General Permit CAS000001), which is not incorporated by reference in this Order.

BMPs to control the run-on of storm water to the Facility site shall be maintained and upgraded as necessary. The Permittee shall describe the effectiveness of these storm water BMPs, as well as activities to maintain and upgrade these BMPs during the previous year, in its annual self-monitoring report to the Regional Water Board.

6.3.5.2. Compliance Schedules

6.3.5.2.1. Compliance Schedule for Final Effluent Limitations for Disinfection Byproducts

- 6.3.5.2.2. The Permittee shall implement activities according to the following schedule to achieve compliance with the effluent limitations for dichlorobromomethane, chlorodibromomethane, total trihalomethanes, and Haloacetic Acids established in Section 4.1.2 of this Order.

Table 5. Schedule for Compliance with Final Effluent Limitations for Disinfection Byproducts

Task	Task Description	Compliance Date
1	Complete full scale chloramine disinfection testing. - Install temporary chloramine disinfection system. - Conduct testing to verify chloramination represents a viable disinfection process.	October 31, 2023
2	Submit a technical report on full-scale chloramine disinfection study including monitoring results for disinfection byproducts and ammonia at monitoring locations EFF-001 and EFF-003; analysis of chloramine disinfection as viable disinfection process to comply with final effluent limitations for final ammonia impact ratio, total coliforms, dichlorobromomethane, chlorodibromomethane, total trihalomethanes, and haloacetic acids.	March 1, 2024
3	3.1 If the full-scale chloramine disinfection study concludes that chloramination will result in compliance with final effluent limitations, the Permittee shall submit full system design plans for full chloramine disinfection unit and follow compliance dates outlined in Tasks 4,5, and 7. 3.2 If the full-scale chloramine disinfection study concludes that chloramination will not result in compliance with final effluent limitations, then the Permittee shall submit, for Regional Water Board Executive Officer approval, a report identifying additional measures that the Permittee will evaluate in order to reduce the production of and achieve compliance with final effluent limitations	November 1, 2024

Task	Task Description	Compliance Date
	for chlorine disinfection by-products. Possible measures include evaluation of other disinfection methods such as ultraviolet light disinfection or conducting a study to evaluate the potential for dilution credits and a mixing zone to achieve compliance. The permittee shall also meet compliance dates outlined in Tasks 4,6, and 7.	
4	The Permittee shall submit an annual progress report on May 1 beginning in 2023.	May 1, 2023 May 1, 2024 May 1, 2025 May 1, 2026 May 1, 2027
5	Completion of construction and installation of chloramine disinfection unit (fully installed and operational).	July 1, 2025
6	Submit a work plan for alternate disinfection byproduct reduction strategy. Including sources of financing and specific milestone tasks to be completed.	November 1, 2025
7	Achieve compliance with final effluent limitations for dichlorobromomethane, chlorodibromomethane, total trihalomethanes, and haloacetic acids as required by this Order.	May 31, 2025 if Task 3.1 is implemented October 1, 2027 if Task 3.2 is implemented

7. COMPLIANCE DETERMINATION

Compliance with the effluent limitations contained in section 4 of this Order will be determined as specified below.

7.1. General

Compliance with effluent limitations for priority pollutants, when effluent limitations have been established, shall be determined using sample reporting protocols defined in the MRP and Attachment A of this Order. For purposes of reporting and administrative enforcement by the Regional and State Water Boards, the Permittee shall be deemed out of compliance with effluent limitations if the concentration of pollutant in the monitoring sample is greater than the effluent limitation and greater than or equal to the reported minimum level (ML).

7.2. Multiple Sample Data

When determining compliance with an AMEL for priority pollutants, and more than one sample result is available, the Permittee shall compute the arithmetic mean

unless the data set contains one or more reported determinations of “Detected, but Not Quantified” (DNQ) or “Not Detected” (ND). In those cases, the Permittee shall compute the median in place of the arithmetic mean in accordance with the following procedure.

- 7.2.1. The data set shall be ranked from low to high, ranking the reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
- 7.2.2. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two middle values, unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ and a value of zero shall be used for the ND or DNQ value in the median calculation for compliance purposes only. Using a value of zero for DNQ or ND samples does not apply when performing reasonable potential or antidegradation analyses.

7.3. **Average Monthly Effluent Limitation (AMEL)**

If the average (or when applicable, the median determined by subsection 7.2, above, for multiple sample data) of daily discharges over a calendar month exceeds the AMEL for a given parameter, this will represent a single violation, though the Permittee will be considered out of compliance for each day of that month for that parameter (e.g., resulting in 1 days of non-compliance in a 1-day month). If only a single sample is taken during the calendar month and the analytical result for that sample exceeds the AMEL, the Permittee will be considered out of compliance for that calendar month. The Permittee will only be considered out of compliance for days when the discharge occurs. If there are ND or DNQ results for a specific constituent in a calendar month, the Permittee shall calculate the median of all sample results within that month for compliance determination with the AMEL as described in section 7.2, above.

7.4. **Average Weekly Effluent Limitation (AWEL)**

If the average (or when applicable, the median determined by subsection 7.2, above, for multiple sample data) of daily discharges over a calendar week exceeds the AWEL for a given parameter, this will represent a single violation, though the Permittee will be considered out of compliance for each day of that week for that parameter, resulting in 7 days of non-compliance. If only a single sample is taken during the calendar week and the analytical result for that sample exceeds the AWEL, the Permittee will be considered out of compliance for that calendar week. The Permittee will only be considered out of compliance for days when the discharge occurs. If there are ND or DNQ results for a specific constituent in a calendar week, the Permittee shall calculate the median of all

sample results within that week for compliance determination with the AWEL as described in section 7.2, above.

7.5. Maximum Daily Effluent Limitation (MDEL)

If a daily discharge (or when applicable, the median determined by subsection 7.2, above, for multiple sample data of a daily discharge) exceeds the MDEL for a given parameter, the Permittee will be considered out of compliance for that parameter for that 1 day only within the reporting period.

7.6. Instantaneous Minimum Effluent Limitation

If the analytical result of a single grab sample is lower than the instantaneous minimum effluent limitation for a parameter, the Permittee will be considered out of compliance for that parameter for that single sample. Non-compliance for each sample will be considered separately (e.g., the results of two grab samples taken within a calendar day that both are lower than the instantaneous minimum effluent limitation would result in two instances of non-compliance with the instantaneous minimum effluent limitation).

If the Permittee monitors pH continuously, pursuant to 40 C.F.R. section 401.17, the Permittee shall be in compliance with the pH limitation specified herein provided that both of the following conditions are satisfied: (1) the total sum of time during which the pH values are outside the required range of pH values shall not exceed 7 hours and 26 minutes in any calendar month; and (2) no individual excursion from the range of pH values shall exceed 60 minutes.

7.7. Instantaneous Maximum Effluent Limitation

If the analytical result of a single grab sample is higher than the instantaneous maximum effluent limitation for a parameter, the Permittee will be considered out of compliance for that parameter for that single sample. Non-compliance for each sample will be considered separately (e.g., the results of two grab samples taken within a calendar day that both exceed the instantaneous maximum effluent limitation would result in two instances of non-compliance with the instantaneous maximum effluent limitation).

If the Permittee monitors pH continuously, pursuant to 40 C.F.R. section 401.17, the Permittee shall be in compliance with the pH limitation specified herein provided that both of the following conditions are satisfied: (1) the total sum of time during which the pH values are outside the required range of pH values shall not exceed 7 hours and 26 minutes in any calendar month; and (2) no individual excursion from the range of pH values shall exceed 60 minutes.

7.8. Bacteriological Limitations

- 7.8.1. Median (Total Coliform Bacteria).** The median is the central tendency concentration of the pollutant. The data set shall be ranked from low to high,

ranking any ND concentrations lowest, followed by quantified values. The median value is determined based on the number of data points in the set. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, the median is the average of the two middle values, unless one or both points are ND or DNQ, in which case the median value shall be the lower of the two middle data points. DNQ is lower than a detected value, and ND is lower than DNQ.

- 7.8.2. **Six-week Rolling Geometric Mean (*E. coli* bacteria).** The rolling geometric mean shall be calculated using at least 5 sample results over a 6-week period from a site using the following formula:

$$\text{Geometric Mean} = \sqrt[n]{(x_1)(x_2)(x_3) \dots (x_n)}$$

where x is the sample value and n is the number of samples taken.

A minimum of three samples over a six-week period is necessary to calculate the geometric mean. When less than three samples are taken in a six-week period, compliance with the *E. coli* receiving water objective shall be determined using the Statistical Threshold Value (STV). If the Permittee samples less than three times during a six-week period, compliance shall be assessed by comparing the single sample results to the STV.

- 7.8.3. **Statistical Threshold Value (*E. coli* bacteria).** (1) The data set shall be ranked from low to high, ranking any ND concentrations lowest, followed by quantified values. (2) The number of sample results should then be multiplied by 90 percent then rounded up to the nearest whole number. (3) Count the values in the data set starting from lowest to highest until the number indicated in step (2) is reached. (4) To be compliant with the statistical threshold value in Receiving Water Limitation 5.1.20, all sample results less than the point described in step 3 must be less than 100 MPN/100 mL.
- 7.8.4. **7-Day Median.** Compliance with the 7-day median will be determined as a rolling median using the bacteriological results of the last 7 days for which analyses have been completed
- 7.8.5. **Geometric Mean (GM).** The geometric mean is a type of mean or average that indicates the central tendency or typical value of a set of numbers by using the product of their values (as opposed to the arithmetic mean which uses their sum). The geometric mean shall be calculated using the 5 most recent samples from a site using the following formula:

$$\text{Geometric Mean} = \sqrt[n]{(x_1)(x_2)(x_3) \dots (x_n)}$$

Where x is the sample value and n is the number of samples taken.

7.9. Chronic Toxicity Requirements

- 7.9.1. If a chronic toxicity test exceeds the applicable chronic toxicity MDEL, as identified in sections 4.1.1.3.1 of this Order, the Permittee will be considered out of compliance for that single sample.
- 7.9.2. If chronic toxicity testing exceeds the chronic toxicity MMEL, as identified in section 4.1.1.3.2 of this Order, the Permittee will be considered out of compliance for that month. No more than one chronic toxicity test initiated in a calendar month¹ shall result in a “fail” at the IWC for any endpoint.
- 7.9.3. Compliance with chronic toxicity routine monitoring, compliance monitoring, and TRE provisions shall constitute compliance with the chronic toxicity requirements, as specified in the MRP (Attachment E, sections 5.1 and 5.2).

7.10. Average Dry Weather Flow

Compliance with Discharge Prohibition 3.8 will be determined once each calendar year by evaluating all equalized influent flow data collected at Monitoring Location INF-001 in a calendar year. The flow through the Facility, measured daily and averaged monthly, must be 1.4 mgd or less for the month with the lowest average monthly flow.

7.11. Peak Daily Wet Weather Flow

The peak daily wet weather flow is the maximum flow rate that occurs over a 24-hour period. Compliance with Discharge Prohibition 3.8 will be determined daily by measuring the daily average equalized influent flow at Monitoring Location INF-001. If the measured daily average flow exceeds 1.25 mgd, the Permittee is not in compliance with Prohibition 3.8 of this Order.

7.12. Percent Removal

Percent removal shall be determined from the monthly average value of influent wastewater concentration in comparison to the monthly average value of effluent concentration for the same constituent over the same time period as measured at Monitoring Locations INF-001 and EFF-001, respectively.

¹ For purposes of aquatic toxicity monitoring, a calendar month shall be defined as the period of time from a day of one month to the day before the corresponding day of the next month if the corresponding day exists, or if not to the last day of the next month (e.g., from January 1 to January 31, from June 15 to July 14, or from January 31 to February 28).

7.13. Six-Month Median Effluent Limitations

The six-month median effluent limitations shall apply as a moving median of daily values for any 180-day period in which daily values represent flow weighted average concentrations within a 24-hour period. For intermittent discharges, the daily value shall be considered to equal zero for days on which no discharge occurred.

7.14. Single Sample Maximum

All single sample results are compared to single sample maximum and median limitations. Single sample results are only compared to the geometric mean and six-week rolling geometric mean, and statistical threshold value when sampling is required at the frequency required to properly assess compliance, as further stated in 7.8.2. through 7.8.4, above. Compliance with a single annual sample is determined in comparison to single sample maximum limitations only. If single sample maximums are routinely exceeded, the Regional Water Board may require additional sampling to further assess the effluent and to determine impacts on the receiving water.

7.15. Chlorine Residual Effluent Limitations

- 7.15.1. Compliance with the chlorine residual effluent limitations in section 4.1.1, Table 2 shall be based on continuous monitoring at Monitoring Location EFF-001 in order to demonstrate that the discharge has been adequately dechlorinated. Continuous monitoring analyzers for chlorine residual or for dechlorination agent residual in the effluent are appropriate methods for compliance determination. A positive residual dechlorination agent in the effluent indicates that chlorine is not present in the discharge, which demonstrates compliance with the effluent limitations. This type of monitoring can also be used to prove that some chlorine residual exceedances are false positives. Continuous monitoring data showing either a positive dechlorination agent residual or a chlorine residual at or below the prescribed limit are sufficient to show compliance with the total residual chlorine effluent limitation prescribed in section 4.1.1, Table 2, provided that the instruments are maintained and calibrated in accordance with the manufacturer's recommendations.
- 7.15.2. The Permittee shall calibrate continuous analyzers (e.g., chlorine residual, bisulfite residual) against grab samples as frequently as necessary to maintain accurate and reliable operation.
- 7.15.3. The Permittee shall report from discrete readings of the continuous monitoring every hour on the hour. Compliance shall be based on an average of these discrete hourly readings on a daily basis. The Permittee shall retain continuous monitoring readings for at least three years. The Regional Water Board retains the right to use all continuous monitoring data for discretionary enforcement.

Any excursion above the chlorine residual effluent limitations specified in section 4.1.1, Table 2, of this Order is a violation. If the Permittee conducts continuous monitoring and the Permittee can demonstrate through data collected from a back-up monitoring system, that a chlorine spike recorded by the continuous monitor was not actually due to chlorine, then any excursion resulting from the recorded spike will not be considered an exceedance, but rather reported as a false positive. Records supporting validation of false positives shall be maintained in accordance with Attachment D, section 4 Standard Provisions.

ATTACHMENT A - DEFINITIONS

Arithmetic Mean (μ)

Also called the average, is the sum of measured values divided by the number of samples. For ambient water concentrations, the arithmetic mean is calculated as follows:

$$\text{Arithmetic mean } (\mu) = \frac{\sum x}{n}$$

where: $\sum x$ is the sum of the measured ambient water concentrations, and n is the number of samples.

Average Monthly Effluent Limitation (AMEL)

The highest allowable average of daily discharges over a calendar month, calculated as the sum of all daily discharges measured during a calendar month divided by the number of daily discharges measured during that month.

Average Weekly Effluent Limitation (AWEL)

The highest allowable average of daily discharges over a calendar week (Sunday through Saturday), calculated as the sum of all daily discharges measured during a calendar week divided by the number of daily discharges measured during that week.

Basin Plan

A Basin Plan is a water quality control plan that is specific to a Regional Water Quality Control Board (Regional Water Board), and serves as regulations that: (1) define and designate beneficial uses of surface and ground waters, (2) establish water quality objectives to protect the beneficial uses, and (3) provide implementation measures.

Bioaccumulative

Those substances taken up by an organism from its surrounding medium through gill membranes, epithelial tissue, or from food and subsequently concentrated and retained in the body of the organism.

California Integrated Water Quality System (CIWQS)

CIWQS is the State Water Board, statewide electronic reporting database that provides for electronic reporting of mandatory reports that are requirements of State and Regional Water Board-issued waste discharge requirements.

Carcinogenic

Pollutants are substances that are known to cause cancer in living organisms.

Chronic Toxicity

This parameter shall be used to measure the acceptability of waters for supporting a healthy marine biota until improved methods are developed to evaluate biological response. See also Test of Significant Toxicity.

Coefficient of Variation (CV)

CV is a measure of the data variability and is calculated as the estimated standard deviation divided by the arithmetic mean of the observed values.

Daily Discharge

Daily Discharge is defined as either: (1) the total mass of the constituent discharged over the calendar day (12:00 am through 11:59 pm) or any 24-hour period that reasonably represents a calendar day for purposes of sampling (as specified in the permit), for a constituent with limitations expressed in units of mass or; (2) the unweighted arithmetic mean measurement of the constituent over the day for a constituent with limitations expressed in other units of measurement (e.g., concentration).

The daily discharge may be determined by the analytical results of a composite sample taken over the course of one day (a calendar day or other 24-hour period defined as a day) or by the arithmetic mean of analytical results from one or more grab samples taken over the course of the day.

For composite sampling, if 1 day is defined as a 24-hour period other than a calendar day, the analytical result for the 24-hour period will be considered as the result for the calendar day in which the 24-hour period ends.

Detected, but Not Quantified (DNQ)

DNQ are those sample results less than the RL, but greater than or equal to the laboratory's MDL. Sample results reported as DNQ are estimated concentrations.

Dilution Credit

Dilution Credit is the amount of dilution granted to a discharge in the calculation of a water quality-based effluent limitation, based on the allowance of a specified mixing zone. It is calculated from the dilution ratio or determined through conducting a mixing zone study or modeling of the discharge and receiving water.

Effective Concentration (EC)

A point estimate of the toxicant concentration that would cause an adverse effect on a quantal, "all or nothing," response (such as death, immobilization, or serious incapacitation) in a given percent of the test organisms. If the effect is death or immobility, the term lethal concentration (LC) may be used. EC values may be calculated using point estimation techniques such as probit, logit, and Spearman-Kärber. EC25 is the concentration of toxicant (in percent effluent) that causes a response in 25 percent of the test organisms.

Effluent Concentration Allowance (ECA)

ECA is a value derived from the water quality criterion/objective, dilution credit, and ambient background concentration that is used, in conjunction with the coefficient of variation for the effluent monitoring data, to calculate a long-term average (LTA) discharge concentration. The ECA has the same meaning as wasteload allocation

(WLA) as used in U.S. EPA guidance (Technical Support Document For Water Quality-based Toxics Control, March 1991, second printing, EPA/505/2-90-001).

Enclosed Bays

Enclosed Bays means indentations along the coast that enclose an area of oceanic water within distinct headlands or harbor works. Enclosed bays include all bays where the narrowest distance between the headlands or outermost harbor works is less than 75 percent of the greatest dimension of the enclosed portion of the bay. Enclosed bays include, but are not limited to, Humboldt Bay, Bodega Harbor, Tomales Bay, Drake's Estero, San Francisco Bay, Morro Bay, Los Angeles-Long Beach Harbor, Upper and Lower Newport Bay, Mission Bay, and San Diego Bay. Enclosed bays do not include inland surface waters or ocean waters.

Estimated Chemical Concentration

The estimated chemical concentration that results from the confirmed detection of the substance by the analytical method below the ML value.

Estuaries

Estuaries means waters, including coastal lagoons, located at the mouths of streams that serve as areas of mixing for fresh and ocean waters. Coastal lagoons and mouths of streams that are temporarily separated from the ocean by sandbars shall be considered estuaries. Estuarine waters shall be considered to extend from a bay or the open ocean to a point upstream where there is no significant mixing of fresh water and seawater. Estuarine waters included, but are not limited to, the Sacramento-San Joaquin Delta, as defined in Water Code section 12220, Suisun Bay, Carquinez Strait downstream to the Carquinez Bridge, and appropriate areas of the Smith, Mad, Eel, Noyo, Russian, Klamath, San Diego, and Otay rivers. Estuaries do not include inland surface waters or ocean waters.

Inland Surface Waters

All surface waters of the state that do not include the ocean, enclosed bays, or estuaries.

Instantaneous Maximum Effluent Limitation

The highest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous maximum limitation).

Instantaneous Minimum Effluent Limitation

The lowest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous minimum limitation).

Maximum Daily Effluent Limitation (MDEL)

The highest allowable daily discharge of a pollutant, over a calendar day (or 24-hour period). For pollutants with limitations expressed in units of mass, the daily discharge is calculated as the total mass of the pollutant discharged over the day. For pollutants with limitations expressed in other units of measurement, the daily discharge is calculated as the arithmetic mean measurement of the pollutant over the day.

For the purposes of chronic and acute aquatic toxicity, an MDEL is an effluent limitation based on the outcome of the test of significant toxicity (TST) approach and the resulting percent effect at the IWC.

Median Monthly Effluent Limitation (MMEL):

For the purposes of chronic and acute aquatic toxicity, an MMEL is an effluent limitation based on a maximum of three independent toxicity tests, analyzed using the TST.

Method Detection Limit (MDL)

MDL is the minimum concentration of a substance that can be reported with 99 percent confidence that the measured concentration is distinguishable from method blank results, as defined in 40 C.F.R. part 136, Attachment B.

Minimum Level (ML)

ML is the concentration at which the entire analytical system must give a recognizable signal and acceptable calibration point. The ML is the concentration in a sample that is equivalent to the concentration of the lowest calibration standard analyzed by a specific analytical procedure, assuming that all the method specified sample weights, volumes, and processing steps have been followed.

Mixing Zone

Mixing Zone is a limited volume of receiving water that is allocated for mixing with a wastewater discharge where water quality criteria can be exceeded without causing adverse effects to the overall water body.

MMEL Compliance Tests

For the purposes of chronic and acute aquatic toxicity, MMEL compliance tests are a maximum of two tests that are used in addition to the routine monitoring test to determine compliance with the chronic and acute aquatic toxicity MMEL and MDEL.

Most Sensitive Species

The single species selected from an array of test species to be used in a single species laboratory test series to determine toxic effects of effluent or ambient water.

Not Detected (ND)

Sample results which are less than the laboratory's MDL.

Null Hypothesis

A statement used in statistical testing that has been put forward either because it is believed to be true or because it is to be used as a basis for argument but has not been proved.

Percent Effect

The value that denotes the difference in response between the test concentration and the control, divided by the mean control response, and multiplied by 100.

Permitting Authority

The State Water Board or a regional water board that issues a permit, waste discharge requirements, water quality certification, or other authorization for the discharge or proposed discharge of waste. To the extent that the action is delegable, the term “Permitting Authority” can include the Executive Officer or Executive Director.

Persistent Pollutants

Persistent pollutants are substances for which degradation or decomposition in the environment is nonexistent or very slow.

Pollutant Minimization Program (PMP)

PMP means waste minimization and pollution prevention actions that include, but are not limited to, product substitution, waste stream recycling, alternative waste management methods, and education of the public and businesses. The goal of the PMP shall be to reduce all potential sources of a priority pollutant(s) through pollutant minimization (control) strategies, including pollution prevention measures as appropriate, to maintain the effluent concentration at or below the water quality-based effluent limitation. Pollution prevention measures may be particularly appropriate for persistent bioaccumulative priority pollutants where there is evidence that beneficial uses are being impacted. The Regional Water Board may consider cost effectiveness when establishing the requirements of a PMP. The completion and implementation of a Pollution Prevention Plan, if required pursuant to Water Code section 13263.3(d), shall be considered to fulfill the PMP requirements.

Pollution Prevention

Pollution Prevention means any action that causes a net reduction in the use or generation of a hazardous substance or other pollutant that is discharged into water and includes, but is not limited to, input change, operational improvement, production process change, and product reformulation (as defined in Water Code section 13263.3). Pollution prevention does not include actions that merely shift a pollutant in wastewater from one environmental medium to another environmental medium, unless clear environmental benefits of such an approach are identified to the satisfaction of the State Water Resources Control Board (State Water Board) or Regional Water Board.

Publicly Owned Treatment Works (POTW)

A treatment works as defined in section 212 of the Clean Water Act (CWA), which is owned by a government agency as defined by section 502(4) of the CWA. Section 502(4) of the CWA defines a municipality as a city, town, borough, county, parish, district, association, or other public body created by or pursuant to state law and having jurisdiction over disposal of sewage, industrial wastes, or other wastes. This definition includes any devices and systems used in the storage, treatment, recycling, and recycling of municipal sewage or industrial wastes of a liquid nature. It also includes sewers, pipes and other conveyances only if they convey wastewater to a POTW Treatment Plant. The term also means the municipality as defined in section 502(4) of the CWA, which has jurisdiction over the Indirect Discharges to and the discharges from such a treatment works.

Reasonable Potential

A designation used for a waste discharge that is projected or calculated to cause or contribute to an excursion above a water quality standard.

Receiving Water

A receiving water is a water of the State that receives a discharge of waste.

Recycled Water

Water which, as a result of treatment of municipal wastewater, is suitable for a direct beneficial use or a controlled use that would not otherwise occur and is therefore considered a valuable resource (Water Code section 13050). The terms “recycled water” and “reclaimed water” have the same meaning (Water Code section 26).

Regulatory Management Decision (RMD)

The decision that represents the maximum allowable error rates and thresholds for toxicity and non-toxicity that would result in an acceptable risk to aquatic life.

Replicates

Two or more independent organism exposures of the same treatment (i.e., effluent concentration) within a toxicity test. Replicates are typically conducted with separate test chambers and test organisms, each having the same effluent concentration.

Reporting Level (RL)

The RL is the ML (and its associated analytical method) chosen by the Permittee for reporting and compliance determination from the MLs included in this Order, including an additional factor if applicable as discussed herein. The MLs included in this Order correspond to approved analytical methods for reporting a sample result that are selected by the Regional Water Board either from Appendix 4 of the SIP in accordance with section 2.4.2 of the SIP or established in accordance with section 2.4.3 of the SIP. The ML is based on the proper application of method-based analytical procedures for sample preparation and the absence of any matrix interferences. Other factors may be applied to the ML depending on the specific sample preparation steps employed. For example, the treatment typically applied in cases where there are matrix-effects is to dilute the sample or sample aliquot by a factor of ten. In such cases, this additional factor must be applied to the ML in the computation of the RL.

Satellite Collection System

The portion, if any, of a sanitary sewer system owned or operated by a different public agency than the agency that owns and operates the wastewater treatment facility that a sanitary sewer system is tributary to.

Septage

Defined as the liquid or solid material removed from a septic tank, cesspool, portable toilet, type III marine sanitation device, recreational vehicle’s sanitation tank, or similar storage or treatment works that receives domestic waste.

Six-Month Median Effluent Limitation

The highest allowable moving median of all daily discharges for any 180-day period.

Sludge and Biosolids

Sludge, as used in this Order, means the solid, semisolid, and liquid residues removed during primary, secondary, or secondary wastewater treatment processes. Solid waste refers to grit and screenings generated during preliminary treatment. Biosolids refers to sludge that has been treated, tested, and demonstrated to be capable of being beneficially and legally used pursuant to federal and state regulations as a soil amendment for agriculture, silviculture, horticulture, and land reclamation activities.

Source of Drinking Water

Any water designated as municipal or domestic supply (MUN) in a Regional Water Board Basin Plan.

Species Sensitivity Screening

An analysis to determine the single most sensitive species from an array of test species to be used in a single species laboratory test series.

Standard Deviation (σ)

Standard Deviation is a measure of variability that is calculated as follows:

$$\text{Standard Deviation } (\sigma) = \frac{\sum (X - \mu)^2}{(n - 1)^{0.5}}$$

where: x is the observed value; μ is the arithmetic mean of the observed values; and n is the number of samples.

Test of Significant Toxicity (TST)

The statistical approach described in National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document (EPA 833-R10-003, 2010). TST was developed by the U.S. Environmental Protection Agency (EPA) for analyzing whole effluent toxicity (WET) and ambient toxicity data. Using the TST approach, the sample is declared toxic if there is greater than or equal to a 25% effect in chronic tests, or if there is greater than or equal to a 20% effect in acute tests at the permitted instream waste concentration (IWC) (referred to as the toxic regulatory management decision (RMD)). The sample is declared non-toxic if there is less than or equal to a 10% effect at the IWC in acute or chronic tests (referred to as the non-toxic RMD).

Toxicity Identification Evaluation

Techniques used to identify the unexplained cause(s) of toxic event. A TIE involves selectively removing classes of chemicals through a series of sample manipulations, effectively reducing complex mixtures of chemicals in natural waters to simple components for analysis. Following each manipulation, the toxicity sample is assessed to see whether the toxicant class removed was responsible for the toxicity.

Toxicity Reduction Evaluation (TRE)

TRE is a study conducted in a stepwise process designed to identify the causative agents of effluent or ambient toxicity, isolate the sources of toxicity, evaluate the effectiveness of toxicity control options, and then confirm the reduction in toxicity. The

first steps of the TRE consist of the collection of data relevant to the toxicity, including additional toxicity testing, and an evaluation of facility operations and maintenance practices, and best management practices. A Toxicity Identification Evaluation (TIE) may be required as part of the TRE, if appropriate. (A TIE is a set of procedures to identify the specific chemical(s) responsible for toxicity. These procedures are performed in three phases (characterization, identification, and confirmation) using aquatic organism toxicity tests.)

Toxicity Provisions

Refers to Section III.B and Section IV.B of the Water Quality Control Plan for Inland Surface Waters, Enclosed Bays, and Estuaries of California.

Water Quality Objective

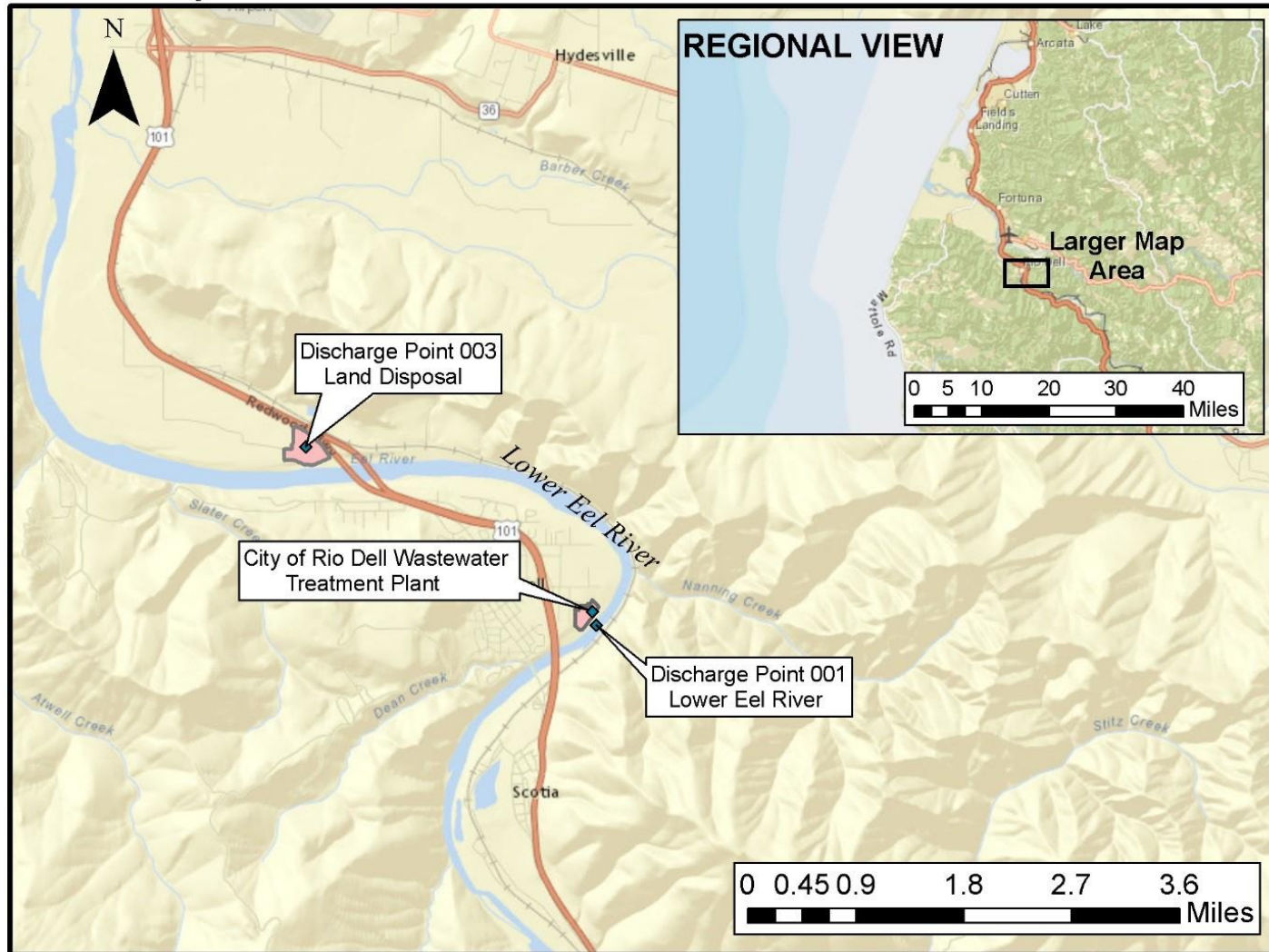
A water quality objective is the amount of pollutant or a parameter level which is established for the reasonable protection of beneficial uses of surface waters and groundwater, and the prevention of nuisance.

Water Recycling

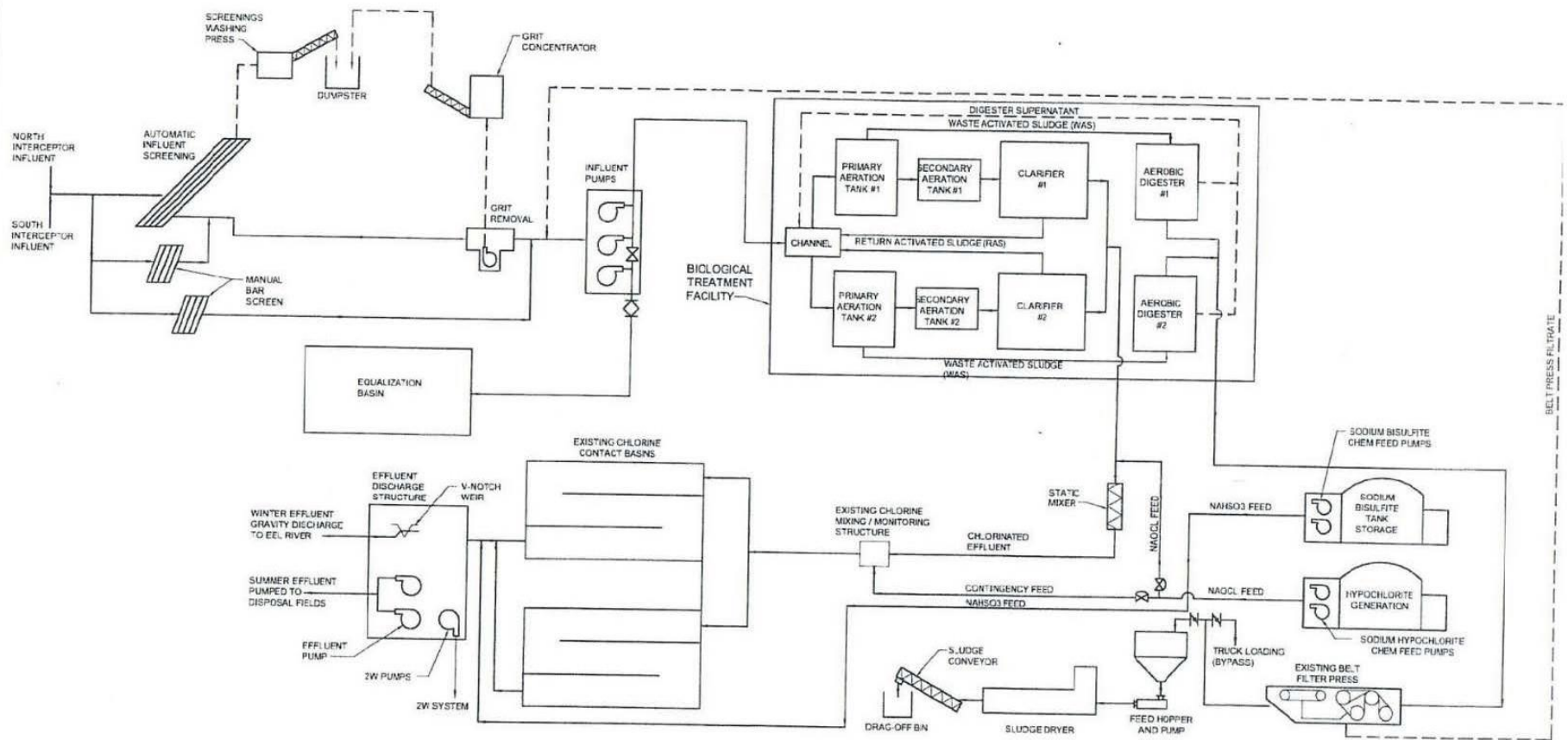
The treatment of wastewater to render it suitable for reuse, the transportation of treated wastewater to the place of use, and the actual use of treated wastewater for a direct beneficial use or controlled use that would not otherwise occur.

ATTACHMENT B - MAP

City of Rio Dell Wastewater Treatment Plant



ATTACHMENT C - FLOW SCHEMATIC



ATTACHMENT D - STANDARD PROVISIONS

1. STANDARD PROVISIONS – PERMIT COMPLIANCE

1.1. Duty to Comply

- 1.1.1. The Permittee must comply with all of the terms, requirements, and conditions of this Order. Any noncompliance constitutes a violation of the Clean Water Act (CWA) and the California Water Code and is grounds for enforcement action; permit termination, revocation and reissuance, or modification; denial of a permit renewal application; or a combination thereof. (40 C.F.R. § 122.41(a); Wat. Code, §§ 13261, 13263, 13265, 13268, 13000, 13001, 13304, 13350, 13385.)
- 1.1.2. The Permittee shall comply with effluent standards or prohibitions established under Section 307(a) of the CWA for toxic pollutants within the time provided in the regulations that establish these standards or prohibitions, even if this Order has not yet been modified to incorporate the requirement. (40 C.F.R. § 122.41(a)(1).)

1.2. Need to Halt or Reduce Activity Not a Defense

It shall not be a defense for a Permittee in an enforcement action that it would have been necessary to halt or reduce the permitted activity in order to maintain compliance with the conditions of this Order. (40 C.F.R. § 122.41(c).)

1.3. Duty to Mitigate

The Permittee shall take all reasonable steps to minimize or prevent any discharge in violation of this Order that has a reasonable likelihood of adversely affecting human health or the environment. (40 C.F.R. § 122.41(d).)

1.4. Proper Operation and Maintenance

The Permittee shall at all times properly operate and maintain all facilities and systems of treatment and control (and related appurtenances) which are installed or used by the Permittee to achieve compliance with the conditions of this Order. Proper operation and maintenance also include adequate laboratory controls and appropriate quality assurance procedures. This provision requires the operation of backup or auxiliary facilities or similar systems that are installed by a Permittee only when necessary to achieve compliance with the conditions of this Order. (40 C.F.R. § 122.41(e).)

1.5. Property Rights

- 1.5.1. This Order does not convey any property rights of any sort or any exclusive privileges. (40 C.F.R. § 122.41(g).)

- 1.5.2. The issuance of this Order does not authorize any injury to persons or property or invasion of other private rights, or any infringement of state or local law or regulations. (40 C.F.R. § 122.5(c).)

1.6. **Inspection and Entry**

The Permittee shall allow the Regional Water Board, State Water Board, U.S. EPA, and/or their authorized representatives (including an authorized contractor acting as their representative), upon the presentation of credentials and other documents, as may be required by law, to (33 U.S.C. § 118(a)(4)(B); 40 C.F.R. § 122.41(i); Wat. Code, §§ 13267, 13383):

- 1.6.1. Enter upon the Permittee's premises where a regulated facility or activity is located or conducted, or where records are kept under the conditions of this Order (33 U.S.C. § 118(a)(4)(B)(i); 40 C.F.R. § 122.41(i)(1); Wat. Code, §§ 13267, 13383);
- 1.6.2. Have access to and copy, at reasonable times, any records that must be kept under the conditions of this Order (33 U.S.C. § 118(a)(4)(B)(ii); 40 C.F.R. § 122.41(i)(2); Wat. Code, §§ 13267, 13383);
- 1.6.3. Inspect and photograph, at reasonable times, any facilities, equipment (including monitoring and control equipment), practices, or operations regulated or required under this Order (33 U.S.C. § 118(a)(4)(B)(ii); 40 C.F.R. § 122.41(i)(3); Wat. Code, §§ 13267, 13383); and
- 1.6.4. Sample or monitor, at reasonable times, for the purposes of assuring Order compliance or as otherwise authorized by the CWA or the Water Code, any substances or parameters at any location. (33 U.S.C. § 118(a)(4)(B); 40 C.F.R. § 122.41(i)(4); Wat. Code, §§ 13267, 13383.)

1.7. **Bypass**

1.7.1. **Definitions**

- 1.7.1.1. "Bypass" means the intentional diversion of waste streams from any portion of a treatment facility. (40 C.F.R. § 122.41(m)(1)(i).)
- 1.7.1.2. "Severe property damage" means substantial physical damage to property, damage to the treatment facilities, which causes them to become inoperable, or substantial and permanent loss of natural resources that can reasonably be expected to occur in the absence of a bypass. Severe property damage does not mean economic loss caused by delays in production. (40 C.F.R. § 122.41(m)(1)(ii).)
- 1.7.2. **Bypass not exceeding limitations.** The Permittee may allow any bypass to occur which does not cause exceedances of effluent limitations, but only if it is for essential maintenance to assure efficient operation. These bypasses are not

subject to the provisions listed in Standard Provisions – Permit Compliance 1.7.3, 1.7.4, and 1.7.5 below. (40 C.F.R. § 122.41(m)(2).)

1.7.3. **Prohibition of bypass.** Bypass is prohibited, and the Regional Water Board may take enforcement action against a Permittee for bypass, unless (40 C.F.R. § 122.41(m)(4)(i)):

1.7.3.1. Bypass was unavoidable to prevent loss of life, personal injury, or severe property damage (40 C.F.R. § 122.41(m)(4)(i)(A));

1.7.3.2. There were no feasible alternatives to the bypass, such as the use of auxiliary treatment facilities, retention of untreated wastes, or maintenance during normal periods of equipment downtime. This condition is not satisfied if adequate back up equipment should have been installed in the exercise of reasonable engineering judgment to prevent a bypass that occurred during normal periods of equipment downtime or preventive maintenance (40 C.F.R. § 122.41(m)(4)(i)(B)); and

1.7.3.3. The Permittee submitted notice to the Regional Water Board as required under Standard Provisions – Permit Compliance 1.7.5 below. (40 C.F.R. § 122.41(m)(4)(i)(C).)

1.7.4. The Regional Water Board may approve an anticipated bypass, after considering its adverse effects, if the Regional Water Board determines that it will meet the three conditions listed in Standard Provisions – Permit Compliance 1.7.3 above. (40 C.F.R. § 122.41(m)(4)(ii).)

1.7.5. **Notice**

1.7.5.1. **Anticipated bypass.** If the Permittee knows in advance of the need for a bypass, it shall submit prior notice, if possible, at least 10 days before the date of the bypass. The notice shall be sent to the Regional Water Board. As of December 21, 2025, a notice shall also be submitted electronically to the initial recipient defined in Standard Provisions – Reporting 5.10 below. Notices shall comply with 40 C.F.R. part 3, 40 C.F.R. section 122.22, and 40 C.F.R. part 127. (40 C.F.R. § 122.41(m)(3)(i).)

1.7.5.2. **Unanticipated bypass.** The Permittee shall submit a notice of an unanticipated bypass as required in Standard Provisions - Reporting 5.5 below (24-hour notice). Notices shall comply with 40 C.F.R. part 3, 40 C.F.R. section 122.22, and 40 C.F.R. part 127. (40 C.F.R. § 122.41(m)(3)(ii).)

1.8. **Upset**

Upset means an exceptional incident in which there is unintentional and temporary noncompliance with technology-based permit effluent limitations because of factors beyond the reasonable control of the Permittee. An upset does not include noncompliance to the extent caused by operational error, improperly designed

treatment facilities, inadequate treatment facilities, lack of preventive maintenance, or careless or improper operation. (40 C.F.R. § 122.41(n)(1).)

- 1.8.1. **Effect of an upset.** An upset constitutes an affirmative defense to an action brought for noncompliance with such technology-based permit effluent limitations if the requirements of Standard Provisions – Permit Compliance 1.8.2 below are met. No determination made during administrative review of claims that noncompliance was caused by upset, and before an action for noncompliance, is final administrative action subject to judicial review. (40 C.F.R. § 122.41(n)(2).)
- 1.8.2. **Conditions necessary for a demonstration of upset.** A Permittee who wishes to establish the affirmative defense of upset shall demonstrate, through properly signed, contemporaneous operating logs or other relevant evidence that (40 C.F.R. § 122.41(n)(3)):
 - 1.8.2.1. An upset occurred and that the Permittee can identify the cause(s) of the upset (40 C.F.R. § 122.41(n)(3)(i));
 - 1.8.2.2. The permitted facility was, at the time, being properly operated (40 C.F.R. § 122.41(n)(3)(ii));
 - 1.8.2.3. The Permittee submitted notice of the upset as required in Standard Provisions – Reporting 5.5.2.2 below (24-hour notice) (40 C.F.R. § 122.41(n)(3)(iii)); and
 - 1.8.2.4. The Permittee complied with any remedial measures required under Standard Provisions – Permit Compliance 1.3 above. (40 C.F.R. § 122.41(n)(3)(iv).)
- 1.8.3. **Burden of proof.** In any enforcement proceeding, the Permittee seeking to establish the occurrence of an upset has the burden of proof. (40 C.F.R. § 122.41(n)(4).)

2. STANDARD PROVISIONS – PERMIT ACTION

2.1. General

This Order may be modified, revoked and reissued, or terminated for cause. The filing of a request by the Permittee for modification, revocation and reissuance, or termination, or a notification of planned changes or anticipated noncompliance does not stay any Order condition. (40 C.F.R. § 122.41(f).)

2.2. Duty to Reapply

If the Permittee wishes to continue an activity regulated by this Order after the expiration date of this Order, the Permittee must apply for and obtain a new permit. (40 C.F.R. § 122.41(b).)

2.3. Transfers

This Order is not transferable to any person except after notice to the Regional Water Board. The Regional Water Board may require modification or revocation and reissuance of the Order to change the name of the Permittee and incorporate such other requirements as may be necessary under the CWA and the Water Code. (40 C.F.R. §§ 122.41(l)(3), 122.61.)

3. STANDARD PROVISIONS – MONITORING

- 3.1. Samples and measurements taken for the purpose of monitoring shall be representative of the monitored activity. (40 C.F.R. § 122.41(j)(1).)
- 3.2. Monitoring must be conducted according to test procedures approved under 40 C.F.R. part 136 for the analyses of pollutants unless another method is required under 40 C.F.R. chapter 1, subchapter N. Monitoring must be conducted according to sufficiently sensitive test methods approved under 40 C.F.R. part 136 for the analysis of pollutants or pollutant parameters or as required under 40 C.F.R. chapter 1, subchapter N. For the purposes of this paragraph, a method is sufficiently sensitive when:
 - 3.2.1. The method minimum level (ML) is at or below the level of the most stringent effluent limitation established in the permit for the measured pollutant or pollutant parameter, and either the method ML is at or below the level of the most stringent applicable water quality criterion for the measured pollutant or pollutant parameter or the method ML is above the applicable water quality criterion but the amount of the pollutant or pollutant parameter in the facility's discharge is high enough that the method detects and quantifies the level of the pollutant or pollutant parameter in the discharge; or
 - 3.2.2. The method has the lowest ML of the analytical methods approved under 40 C.F.R. part 136 or required under 40 C.F.R. chapter 1, subchapter N for the measured pollutant or pollutant parameter.

In the case of pollutants or pollutant parameters for which there are no approved methods under 40 C.F.R. part 136 or otherwise required under 40 C.F.R. chapter 1, subchapter N, monitoring must be conducted according to a test procedure specified in this Order for such pollutants or pollutant parameters. (40 C.F.R. § 122.21(e)(3), 122.41(j)(4), 122.44(i)(1)(iv).)

In the case of sludge use or disposal approved under 40 C.F.R. part 136, monitoring must be conducted according to test procedures in part 503 unless otherwise specified in 40 C.F.R. or other test procedures have been specified in this Order.

4. STANDARD PROVISIONS – RECORDS

- 4.1. Except for records of monitoring information required by this Order related to the Permittee's sewage sludge use and disposal activities, which shall be retained for a period of at least five years (or longer as required by 40 C.F.R. part 503), the Permittee shall retain records of all monitoring information, including all calibration and maintenance records and all original strip chart recordings for continuous monitoring instrumentation, copies of all reports required by this Order, and records of all data used to complete the application for this Order, for a period of at least three (3) years from the date of the sample, measurement, report or application. This period may be extended by request of the North Coast Regional Water Board Executive Officer at any time. (40 C.F.R. § 122.41(j)(2).)

4.2. Records of monitoring information shall include:

- 4.2.1. The date, exact place, and time of sampling or measurements (40 C.F.R. § 122.41(j)(3)(i));
- 4.2.2. The individual(s) who performed the sampling or measurements (40 C.F.R. § 122.41(j)(3)(ii));
- 4.2.3. The date(s) analyses were performed (40 C.F.R. § 122.41(j)(3)(iii));
- 4.2.4. The individual(s) who performed the analyses (40 C.F.R. § 122.41(j)(3)(iv));
- 4.2.5. The analytical techniques or methods used (40 C.F.R. § 122.41(j)(3)(v)); and
- 4.2.6. The results of such analyses. (40 C.F.R. § 122.41(j)(3)(vi).)

4.3. Claims of confidentiality for the following information will be denied (40 C.F.R. § 122.7(b)):

- 4.3.1. The name and address of any permit applicant or Permittee (40 C.F.R. § 122.7(b)(1)); and
- 4.3.2. Permit applications and attachments, permits and effluent data. (40 C.F.R. § 122.7(b)(2).)

5. STANDARD PROVISIONS – REPORTING

5.1. Duty to Provide Information

The Permittee shall furnish to the Regional Water Board, State Water Board, or U.S. EPA within a reasonable time, any information which the Regional Water Board, State Water Board, or U.S. EPA may request to determine whether cause exists for modifying, revoking and reissuing, or terminating this Order or to determine compliance with this Order. Upon request, the Permittee shall also furnish to the

Regional Water Board, State Water Board, or U.S. EPA copies of records required to be kept by this Order. (40 C.F.R. § 122.41(h); Wat. Code, §§ 13267, 13383.)

5.2. Signatory and Certification Requirements

- 5.2.1. All applications, reports, or information submitted to the Regional Water Board, State Water Board, and/or U.S. EPA shall be signed and certified in accordance with Standard Provisions – Reporting 5.2.2, 5.2.3, 5.2.4, 5.2.5, and 5.2.6 below. (40 C.F.R. § 122.41(k).)
- 5.2.2. All permit applications shall be signed by either a principal executive officer or ranking elected official. For purposes of this provision, a principal executive officer of a federal agency includes: (i) the chief executive officer of the agency, or (ii) a senior executive officer having responsibility for the overall operations of a principal geographic unit of the agency (e.g., Regional Administrators of U.S. EPA). (40 C.F.R. § 122.22(a)(3).)
- 5.2.3. All reports required by this Order and other information requested by the Regional Water Board, State Water Board, or U.S. EPA shall be signed by a person described in Standard Provisions – Reporting 5.2.2 above, or by a duly authorized representative of that person. A person is a duly authorized representative only if:
 - 5.2.3.1. The authorization is made in writing by a person described in Standard Provisions – Reporting 5.2.2 above (40 C.F.R. § 122.22(b)(1));
 - 5.2.3.2. The authorization specifies either an individual or a position having responsibility for the overall operation of the regulated facility or activity such as the position of facility manager, operator of a well or a well field, superintendent, position of equivalent responsibility, or an individual or position having overall responsibility for environmental matters for the company. (A duly authorized representative may thus be either a named individual or any individual occupying a named position.) (40 C.F.R. § 122.22(b)(2)); and
 - 5.2.3.3. The written authorization is submitted to the Regional Water Board and State Water Board. (40 C.F.R. § 122.22(b)(3).)
- 5.2.4. If an authorization under Standard Provisions – Reporting 5.2.3 above is no longer accurate because a different individual or position has responsibility for the overall operation of the facility, a new authorization satisfying the requirements of Standard Provisions – Reporting 5.2.3 above must be submitted to the Regional Water Board and State Water Board prior to or together with any reports, information, or applications, to be signed by an authorized representative. (40 C.F.R. § 122.22(c).)
- 5.2.5. Any person signing a document under Standard Provisions – Reporting 5.2.2 or 5.2.3 above shall make the following certification:

“I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.” (40 C.F.R. § 122.22(d).)

- 5.2.6. Any person providing the electronic signature for documents described in Standard Provisions – 5.2.1, 5.2.2, or 5.2.3 that are submitted electronically shall meet all relevant requirements of Standard Provisions – Reporting 5.2 and shall ensure that all relevant requirements of 40 C.F.R. part 3 (Cross-Media Electronic Reporting) and 40 C.F.R. part 127 (NPDES Electronic Reporting Requirements) are met for that submission. (40 C.F.R. § 122.22(e).)

5.3. Monitoring Reports

- 5.3.1. Monitoring results shall be reported at the intervals specified in the Monitoring and Reporting Program (Attachment E) in this Order. (40 C.F.R. § 122.41(l)(4).)
- 5.3.2. Monitoring results must be reported on a Discharge Monitoring Report (DMR) form or forms provided or specified by the Regional Water Board or State Water Board. As of December 21, 2016, all reports and forms must be submitted electronically to the initial recipient defined in Standard Provisions – Reporting 5.10 and comply with 40 C.F.R. part 3, 40 C.F.R. section 122.22, and 40 C.F.R. part 127. (40 C.F.R. § 122.41(l)(4)(i).)
- 5.3.3. If the Permittee monitors any pollutant more frequently than required by this Order using test procedures approved under 40 C.F.R. part 136, or another method required for an industry-specific waste stream under 40 C.F.R. chapter 1, subchapter N, the results of such monitoring shall be included in the calculation and reporting of the data submitted in the DMR or reporting form specified by the Regional Water Board or State Water Board. (40 C.F.R. § 122.41(l)(4)(ii).)
- 5.3.4. Calculations for all limitations, which require an averaging of measurements, shall utilize an arithmetic mean unless otherwise specified in this Order. (40 C.F.R. § 122.41(l)(4)(iii).)

5.4. Compliance Schedules

Reports of compliance or noncompliance with, or any progress reports on, interim and final requirements contained in any compliance schedule of this Order, shall be submitted no later than 14 days following each schedule date. (40 C.F.R. § 122.41(l)(5).)

5.5. Twenty-Four Hour Reporting

- 5.5.1. The Permittee shall report any noncompliance which may endanger health or the environment. Any information shall be provided orally within 24 hours from the time the Permittee becomes aware of the circumstances. A report shall also be provided within five (5) days of the time the Permittee becomes aware of the circumstances. The report shall contain a description of the noncompliance and its cause; the period of noncompliance, including exact dates and times, and if the noncompliance has not been corrected, the anticipated time it is expected to continue; and steps taken or planned to reduce, eliminate, and prevent reoccurrence of the noncompliance.

For noncompliance events related to combined sewer overflows, sanitary sewer overflows, or bypass events, these reports must include the data described above (with the exception of time of discovery) as well as the type of event (i.e., combined sewer overflow, sanitary sewer overflow, or bypass event), type of overflow structure (e.g., manhole, combined sewer overflow outfall), discharge volume untreated by the treatment works treating domestic sewage, types of human health and environmental impacts of the event, and whether the noncompliance was related to wet weather.

As of December 21, 2025, all reports related to combined sewer overflows, sanitary sewer overflows, or bypass events must be submitted to the initial recipient defined in Standard Provisions – Reporting 5.10. The reports shall comply with 40 C.F.R. part 3, 40 C.F.R. section 122.22, and 40 C.F.R. part 127. The Regional Water Board may also require the Permittee to electronically submit reports not related to combined sewer overflows, sanitary sewer overflows, or bypass events under this section. (40 C.F.R. § 122.41(l)(6)(i).)

- 5.5.2. The following shall be included as information that must be reported within 24 hours:

5.5.2.1. Any unanticipated bypass that exceeds any effluent limitation in this Order. (40 C.F.R. § 122.41(l)(6)(ii)(A).)

5.5.2.2. Any upset that exceeds any effluent limitation in this Order. (40 C.F.R. § 122.41(l)(6)(ii)(B).)

- 5.5.3. The Regional Water Board may waive the above required written report on a case-by-case basis if an oral report has been received within 24 hours. (40 C.F.R. § 122.41(l)(6)(ii)(B).)

5.6. Planned Changes

The Permittee shall give notice to the Regional Water Board as soon as possible of any planned physical alterations or additions to the permitted facility. Notice is required under this provision only when (40 C.F.R. § 122.41(l)(1)):

- 5.6.1. The alteration or addition to a permitted facility may meet one of the criteria for determining whether a facility is a new source in section 122.29(b) (40 C.F.R. § 122.41(l)(1)(i)); or
- 5.6.2. The alteration or addition could significantly change the nature or increase the quantity of pollutants discharged. This notification applies to pollutants that are not subject to the effluent limitations in this Order. (40 C.F.R. § 122.41(l)(1)(ii).)
- 5.6.3. The alteration or addition could significantly change the nature or increase the quantity of pollutants discharged. This notification applies to pollutants that are subject neither to effluent limitations in this Order nor to notification requirements under section 122.42(a)(1) (see Additional Provisions—Notification Levels 7.1.1). (40 C.F.R. § 122.41(l)(1)(ii).)

5.7. Anticipated Noncompliance

The Permittee shall give advance notice to the Regional Water Board of any planned changes in the permitted facility or activity that may result in noncompliance with this Order's requirements. (40 C.F.R. § 122.41(l)(2).)

5.8. Other Noncompliance

The Permittee shall report all instances of noncompliance not reported under Standard Provisions – Reporting 5.3, 5.4, and 5.5 above at the time monitoring reports are submitted. The reports shall contain the information listed in Standard Provision – Reporting 5.5 above. For noncompliance events related to combined sewer overflows, sanitary sewer overflows, or bypass events, these reports shall contain the information described in Standard Provision – Reporting 5.5 and the applicable required data in appendix A to 40 C.F.R. part 127. The Regional Water Board may also require the Permittee to electronically submit reports not related to combined sewer overflows, sanitary sewer overflows, or bypass events under this section. (40 C.F.R. § 122.41(l)(7).)

5.9. Other Information

When the Permittee becomes aware that it failed to submit any relevant facts in a permit application, or submitted incorrect information in a permit application or in any report to the Regional Water Board, State Water Board, or U.S. EPA, the Permittee shall promptly submit such facts or information. (40 C.F.R. § 122.41(l)(8).)

5.10. Initial Recipient for Electronic Reporting Data

The owner, operator, or the duly authorized representative is required to electronically submit NPDES information specified in appendix A to 40 C.F.R. part 127 to the initial recipient defined in 40 C.F.R. section 127.2(b). U.S. EPA will identify and publish the list of initial recipients on its website and in the Federal Register, by state and by NPDES data group [see 40 C.F.R. section 127.2(c)]. U.S. EPA will update and maintain this listing. (40 C.F.R. § 122.41(l)(9).)

6. STANDARD PROVISIONS – ENFORCEMENT

- 6.1. The Regional Water Board is authorized to enforce the terms of this permit under several provisions of the Water Code, including, but not limited to, sections 13268, 13350, 13385, 13386, and 13387.

7. ADDITIONAL PROVISIONS – NOTIFICATION LEVELS

7.1. Publicly Owned Treatment Works (POTWs)

All POTWs shall provide adequate notice to the Regional Water Board of the following (40 C.F.R. § 122.42(b)):

- 7.1.1. Any new introduction of pollutants into the POTW from an indirect Permittee that would be subject to sections 301 or 306 of the CWA if it were directly discharging those pollutants (40 C.F.R. § 122.42(b)(1)); and
- 7.1.2. Any substantial change in the volume or character of pollutants being introduced into that POTW by a source introducing pollutants into the POTW at the time of adoption of the Order. (40 C.F.R. § 122.42(b)(2).)
- 7.1.3. Adequate notice shall include information on the quality and quantity of effluent introduced into the POTW as well as any anticipated impact of the change on the quantity or quality of effluent to be discharged from the POTW. (40 C.F.R. § 122.42(b)(3)).

ATTACHMENT E - MONITORING AND REPORTING PROGRAM

TABLE OF CONTENTS

1.	General Monitoring Provisions	3
1.1.	Wastewater Monitoring Provision.	3
1.2.	Supplemental Monitoring Provision.	3
1.3.	Laboratory Certification.	3
1.4.	Instrumentation and Calibration Provision.	3
1.5.	Minimum Levels (ML) and Reporting Levels (RL).....	4
1.6.	Discharge Monitoring Report Quality Assurance (DMR-QA) Study.....	5
2.	Monitoring Locations	5
3.	Influent Monitoring Requirements	6
3.1.	Monitoring Location INF-001	6
4.	Effluent Monitoring Requirements	6
4.1.	Monitoring Location EFF-001	6
5.	Whole Effluent Toxicity Testing Requirements.....	11
5.1.	Chronic Toxicity Testing	11
5.2.	Toxicity Reduction Evaluation (TRE) Process	15
6.	Land Discharge Monitoring Requirements – Not Applicable	16
7.	Recycling Monitoring Requirements.....	16
7.1.	Monitoring Location EFF-003	16
8.	Receiving Water Monitoring Requirements.....	19
8.1.	Monitoring Location RSW-001.....	19
8.2.	Monitoring Location RSW-002.....	20
8.3.	Groundwater Monitoring	21
9.	Other Monitoring Requirements	22
9.1.	Disinfection Process Monitoring for the Chlorine Disinfection System.....	22
9.2.	Visual Monitoring (Monitoring Location EFF-001, RSW-001, and RSW-002)	23
9.3.	Sludge Monitoring (Monitoring Location BIO-001).....	23
10.	Reporting Requirements	23
10.1.	General Monitoring and Reporting Requirements.....	23
10.2.	Self-Monitoring Reports (SMRs)	24
10.3.	Discharge Monitoring Reports (DMRs)	27
10.4.	Other Reports	27
10.5.	Spill Notification	33

TABLE OF TABLES

Table E-1. Monitoring Station Locations.....	5
Table E-2. Influent Monitoring – Monitoring Location INF-001	6
Table E-3. Effluent Monitoring – Monitoring Location EFF-001	7
Table E-4. Recycled Water Monitoring Requirements – Monitoring Location EFF-003	17
Table E-5. Receiving Water Monitoring Requirements – Monitoring Location RSW-001	19
Table E-6. Receiving Water Monitoring Requirements – Monitoring Location RSW-002	20
Table E-7. Receiving Water Monitoring Requirements – Monitoring Location MW-001 through MW-004	21
Table E-8. Internal Effluent Monitoring - Monitoring Location INT-001	23
Table E-9. Monitoring Periods and Reporting Schedule.....	24
Table E-10. Reporting Requirements for Special Provisions Reports	28

ATTACHMENT E – MONITORING AND REPORTING PROGRAM (MRP)

Section 308 of the federal Clean Water Act (CWA) and sections 122.41(h), (j)-(l), 122.44(i), and 122.48 of title 40 of the Code of Federal Regulations (40 C.F.R.) require that all NPDES permits specify monitoring and reporting requirements. Water Code sections 13267 and 13383 also authorize the Regional Water Board to establish monitoring, inspection, entry, reporting, and recordkeeping requirements. This MRP establishes monitoring, reporting, and recordkeeping requirements that implement the federal and California laws and/or regulations.

1. GENERAL MONITORING PROVISIONS

1.1. Wastewater Monitoring Provision.

- 1.1.1. Composite samples may be taken by a proportional sampling device or by grab samples composited in proportion to flow. In compositing grab samples, the sampling interval shall not exceed 1 hour.

1.2. Supplemental Monitoring Provision.

- 1.2.1. If the Permittee monitors any pollutant more frequently than required by this Order, using test procedures approved by 40 C.F.R. part 136 or as specified in this Order, the results of such monitoring shall be included in the calculation and reporting of the data submitted in the quarterly and annual discharge monitoring reports.

1.3. Laboratory Certification.

- 1.3.1. Laboratories analyzing monitoring samples shall be certified by the State Water Resources Control Board (State Water Board), in accordance with the provision of Water Code section 1176, and must include quality assurance/quality control data with their reports.
- 1.3.2. The Permittee may analyze pollutants with short hold times (e.g., pH, chlorine residual, etc.) with field equipment or its on-site laboratory provided that the Permittee has standard operating procedures (SOPs) that identify quality assurance/quality control procedures to be followed to ensure accurate results. The Permittee must demonstrate sufficient capability to adequately perform these field tests (e.g., qualified and trained employees, properly calibrated and maintained field instruments). The program shall conform to U.S. EPA guidelines or other approved procedures.

1.4. Instrumentation and Calibration Provision.

- 1.4.1. All monitoring instruments and devices used by the Permittee to fulfill the prescribed monitoring program shall be properly maintained and calibrated as necessary to ensure their continued accuracy. All flow measurement devices shall be calibrated no less than the manufacturer's recommended intervals or

one-year intervals, (whichever comes first) to ensure continued accuracy of the devices. The Permittee shall calibrate continuous analyzers (e.g., chlorine residual, bisulfite residual) against grab samples as frequently as necessary to maintain accurate and reliable operation.

1.5. Minimum Levels (ML) and Reporting Levels (RL).

- 1.5.1. U.S. EPA published regulations for the Sufficiently Sensitive Methods Rule (SSM Rule) which became effective September 18, 2015. Unless otherwise specified by this MRP, all monitoring shall be conducted according to test procedures established at 40 C.F.R. 136, Guidelines Establishing Test Procedures for Analysis of Pollutants. All analyses shall be conducted using the lowest practical quantitation limit achievable using U.S. EPA approved methods. For the purposes of the NPDES program, when more than one test procedure is approved under 40 C.F.R., part 136 for the analysis of a pollutant or pollutant parameter, the test procedure must be sufficiently sensitive as defined at 40 C.F.R. 122.21(e)(3) and 122.44(i)(1)(iv).

A U.S. EPA-approved analytical method is sufficiently sensitive where:

- 1.5.2. The ML is at or below both the level of the applicable water quality criterion/objective and the permit limitation for the measured pollutant or pollutant parameter; or
- 1.5.3. In permit applications, the ML is above the applicable water quality criterion/objective, but the amount of the pollutant or pollutant parameter in a facility's discharge is high enough that the method detects and quantifies the level of the pollutant or pollutant parameter in the discharge; or
- 1.5.4. The method has the lowest ML of the U.S. EPA-approved analytical methods where none of the U.S. EPA-approved analytical methods for a pollutant can achieve the MLs necessary to assess the need for effluent limitations or to monitor compliance with a permit limitation.

Where effluent limitations are set below the lowest achievable quantitation limits, pollutants not detected at the lowest practical quantitation limits will be considered in compliance with effluent limitations. Analysis for toxics listed by the California Toxics Rule (CTR) shall also adhere to guidance and requirements contained in the *Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California* (2005) (SIP). However, there may be situations when analytical methods are published with MLs that are more sensitive than the MLs. For instance, U.S. EPA Method 1631E for mercury is not currently listed in SIP Appendix 4, but it is published with an ML of 0.5 ng/L that makes it a sufficiently sensitive analytical method. Similarly, U.S. EPA Method 245.7 for mercury is published with an ML of 5 ng/L.

1.6. Discharge Monitoring Report Quality Assurance (DMR-QA) Study.

- 1.6.1. The Permittee shall participate in the DMR-QA program and ensure that the results of the DMR-QA Study or the most recent Water Pollution Performance Evaluation Study from each laboratory providing testing services for the permit are submitted annually to the State Water Board at qualityassurance@waterboards.ca.gov. For more information on the DMR-QA Program, contact the State DMR-QA Coordinator at the aforementioned email address.

2. MONITORING LOCATIONS

The Permittee shall establish the following monitoring locations to demonstrate compliance with the effluent limitations, discharge specifications, and other requirements in this Order:

Table E-1. Monitoring Station Locations

Discharge Point Name	Monitoring Location Name	Monitoring Location Description
---	INF-001	Influent at the headworks of the wastewater treatment facility (WWTF) prior to treatment.
---	INT-001	Internal monitoring location for purposes of monitoring chlorine residual in chlorine treated wastewater within the contact chamber prior to dechlorination.
001	EFF-001	Treated effluent from the WWTF downstream of dechlorination, but prior to discharge to Lower Eel River.
002 (Historic)	EFF-002 (Historic)	Treated effluent from the WWTF downstream of disinfection processes before discharge to the percolation ponds. Discharges at Discharge Point 002 are not authorized by this Order and are shown here for information purposes only.
003	EFF-003	Disinfected secondary-23 recycled water downstream of the disinfection facilities and before effluent discharges to the irrigation site.
---	RSW-001	Lower Eel River surface water upstream and beyond the influence of the discharge.
---	RSW-002	Lower Eel River surface water that is at the point of discharge of Discharge Point 001.
---	MW-001 thru MW-004	Existing groundwater monitoring wells, located on the irrigation parcel.

Discharge Point Name	Monitoring Location Name	Monitoring Location Description
---	BIO-001	A representative sample of the sludge or biosolids generated when removed for disposal.

3. INFLUENT MONITORING REQUIREMENTS

3.1. Monitoring Location INF-001

3.1.1. The Permittee shall monitor influent to the facility at Monitoring Location INF-001 as follows:

Table E-2. Influent Monitoring – Monitoring Location INF-001

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method ¹
Influent Flow ²	mgd	Meter	Continuous	---
Biochemical Oxygen Demand 5 day @ 20°C (BOD ₅)	mg/L	24-hr Composite	Weekly	Standard Methods
Total Suspended Solids (TSS)	mg/L	24-hr Composite	Weekly	Standard Methods

Table Notes

1. In accordance with the current edition of Standard Methods for Examination of Water and Wastewater (American Public Health Administration) or current test procedures specified in 40 C.F.R. part 136.
2. The Permittee shall report the maximum daily, average daily, and average monthly flows

4. EFFLUENT MONITORING REQUIREMENTS

4.1. Monitoring Location EFF-001

4.1.1. The Permittee shall monitor secondary treated effluent prior to contact with the receiving water at Monitoring Location EFF-001 during periods of discharge to Lower Eel River as follows.

Table E-3. Effluent Monitoring – Monitoring Location EFF-001

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method¹
Effluent Flow ³	mgd	Meter	Continuous	--
Dilution Rate	% of stream flow	Calculation	Daily	--
Chlorine, Total Residual ⁴	mg/L	Grab/Meter	Continuous	Standard Methods ²
Biochemical Oxygen Demand 5-day @ 20°C (BOD ⁵)	mg/L	24-hr Composite	Weekly ⁵	Standard Methods ²
BOD ₅	% Removal	Calculate	Monthly	--
Total Suspended Solids (TSS)	mg/L	24-hr Composite	Weekly ⁵	Standard Methods ²
TSS	% Removal	Calculate	Monthly	--
pH	s.u.	Grab	Weekly ^{5,6}	Standard Methods ²
Temperature	°C	Grab	Weekly ⁶	Standard Methods ²
Settleable Solids	ml/L	Grab	Weekly ⁵	Standard Methods ²
Bromoform ¹⁸	µg/L	Grab	Monthly	Standard Methods ²
Chlorodibromomethane	µg/L	Grab	Monthly ⁷	Standard Methods ²
Chloroform ¹⁸	µg/L	Grab	Monthly	Standard Methods ²
Dichlorobromomethane	µg/L	Grab	Monthly ⁷	Standard Methods ²
Total Trihalomethanes ⁸	µg/L	Calculate	Monthly ⁷	--
Monochloroacetic Acid	µg/L	Grab	Quarterly ¹⁸	U.S. EPA Method 552
Dichloroacetic Acid	µg/L	Grab	Quarterly ¹⁸	U.S. EPA Method 552

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method¹
Trichloroacetic Acid	µg/L	Grab	Quarterly ¹⁸	U.S. EPA Method 552
Monobromoacetic Acid	µg/L	Grab	Quarterly ¹⁸	U.S. EPA Method 552
Dibromoacetic Acid	µg/L	Grab	Quarterly ¹⁸	U.S. EPA Method 552
Haloacetic Acids ^{9,17}	µg/L	Calculate	Quarterly ^{7,18}	U.S. EPA Method 552
Total Coliform Bacteria	MPN/100 mL	Grab	Weekly ⁵	Standard Methods ²
Ammonia Nitrogen, Total (as N)	mg/L	Grab	Monthly ⁷	Standard Methods ²
Ammonia Nitrogen, Unionized	mg/L	Calculation	Monthly	--
Ammonia Impact Ratio	Ratio	Calculation	Monthly	--
Nitrate Nitrogen, Total (as N)	mg/L	Grab	Monthly	Standard Methods ²
Nitrite Nitrogen, Total (as N)	mg/L	Grab	Monthly	Standard Methods ²
Organic Nitrogen, Total	mg/L	Grab	Monthly	Standard Methods ²
Nitrogen, Total (as N) ¹⁰	mg/L	Calculation	Monthly ⁷	Standard Methods ²
Phosphorus, Total (as P)	mg/L	Grab	Monthly	Standard Methods ²
Aluminum	µg/L	Grab	Quarterly	Standard Methods ²
CTR Priority Pollutants ¹¹	µg/L	24-hr Composite ¹²	Once per permit term ¹³	Standard Methods, ¹⁴
Chronic Toxicity ¹⁵	Pass or Fail, and % Effect	24-hr Composite	Annually ¹⁶	See Section 5 below
<i>E. coli</i> Bacteria ¹⁹	MPN/100 mL	Grab	Weekly ²⁰	Standard Methods

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method ¹
4,4-DDT	ug/L	Grab	Quarterly ^{7,21}	Standard Methods ²
Antimony, Total Recoverable	ug/L	Grab	Quarterly ^{7,21}	Standard Methods ²

Table Notes

1. Pollutants shall be analyzed using the analytical methods described in 40 C.F.R. part 136 or by methods approved by the Regional Water Board or State Water Board.
2. In accordance with the current edition of Standard Methods for Examination of Water and Wastewater (American Public Health Administration) or by methods approved by the Regional Water Board or State Water Board.
3. The Permittee shall report the maximum daily, average daily, and average monthly flows.
4. The Permittee shall monitor continuously to demonstrate that the discharge has been adequately dechlorinated to achieve chlorine residual effluent limitations specified in section 4.1.1, Table 2, at all times. The Permittee shall report from discrete readings of the continuous monitoring every hour on the hour and report the average of the hourly readings daily in accordance with Compliance Determination section 7.15 of this Order. The Permittee shall calibrate chlorine and bisulfite residual analyzers against grab samples as frequently as necessary to maintain accurate and reliable operation.
5. Accelerated Monitoring (weekly monitoring frequency). If two consecutive weekly test results exceed an effluent limitation, the Permittee shall take two samples each of the two weeks following receipt of the second sample result. During the intervening period, the Permittee shall take steps to identify the cause of the exceedance and take steps to return to compliance.
6. pH and temperature monitoring must coincide with monthly monitoring for ammonia.
7. Accelerated Monitoring (monthly and quarterly monitoring frequencies). If a test result exceeds an effluent limitation, the Permittee shall take two more samples, one within 7 days and one within 14 days following receipt of the initial sample result. During the intervening period, the Permittee shall take steps to identify the cause of the exceedance and take steps needed to return to compliance.
8. The sum of the concentrations of bromoform, chlorodibromomethane, chloroform, and dichlorobromomethane.
9. The sum of the concentrations of Monochloroacetic Acid, Dichloroacetic Acid, Trichloroacetic Acid, Monobromoacetic Acid, and Dibromoacetic Acid.

Table Notes

10. The sum of the concentrations of nitrate nitrogen, nitrite nitrogen, total organic nitrogen and ammonia.
11. Those pollutants identified by the California Toxics Rule at 40 C.F.R. section 131.38. The Permittee is not required to sample and analyze asbestos. Hardness shall be monitored concurrently with the priority pollutant sample.
12. CTR pollutant samples shall be collected using 24-hour composite sampling, except for pollutants that are volatile. Samples for volatile pollutants may be collected as grab samples.
13. CTR priority pollutant sampling shall be completed no later than **April 15, 2027** during a period of discharge to the Eel River. Effluent and receiving water monitoring shall occur concurrently.
14. Analytical methods must achieve the lowest minimum level (ML) specified in Appendix 4 of the SIP and, in accordance with section 2.4 of the SIP, the Permittee shall report the ML and MDL for each sample result.
15. Whole effluent chronic toxicity shall be monitored in accordance with the requirements of section 5 of this Monitoring and Reporting Program and shall be completed concurrently with the CTR priority pollutant monitoring.
16. Monitoring shall occur between January and April each year.
17. Based on the first full year of monitoring, the Regional Water Board Executive Officer may reduce the frequency of or eliminate monitoring for bromoform, chloroform, and Haloacetic Acids in the effluent if monitoring data demonstrates that the Permittee's discharge does not contain bromoform, chloroform or Haloacetic Acids at concentrations that pose reasonable potential to cause an exceedance of applicable water quality objectives for bromoform, chloroform or Haloacetic Acids.
18. Quarterly monitoring applies to Monitoring Locations EFF-001 and EFF-003. Sampling should be coordinated to ensure that two samples are collected during discharge to surface waters and two samples are collected during discharge to the irrigation site.
19. The Permittee may use any *E. coli* method specified in 40 CFR 136 for compliance monitoring.
20. The Permittee shall complete weekly monitoring for a minimum of three distinct 6-week periods, each discharge season (between October 1 and May 14). Monitoring shall be concurrent with *E. coli* monitoring at monitoring location RSW-001 and RSW-002.
21. The monitoring frequency for 4,4-DDT and Antimony may be reduced to annually if the first 12 months of monitoring data collected after the effective date of this Order demonstrate that concentrations of 4,4-DDT and Antimony are consistently lower than water quality objectives for protection of surface water.

5. WHOLE EFFLUENT TOXICITY TESTING REQUIREMENTS

5.1. Chronic Toxicity Testing

The Permittee shall conduct chronic toxicity testing in accordance with the *Water Quality Control Plan for Inland Surface Waters, Enclosed Bays, and Estuaries of California* (Toxicity Provisions), adopted on December 1, 2021. The following chronic toxicity testing requirements have been identified as applicable to this Order:

- 5.1.1. **Toxicity Testing Sample and Location.** The effluent sample shall be collected from Monitoring Location EFF-001. Dilution water and control water shall be prepared and used as specified by the test methods.
- 5.1.2. **In-stream Waste Concentration (IWC) for Chronic Toxicity.** The chronic toxicity IWC for this discharge is 100 percent effluent.
- 5.1.3. **Toxicity Test Methods.** Chronic aquatic toxicity tests shall be conducted using one or more of the test species listed below and selected by the Regional Water Board in accordance with the Toxicity Provisions, and shall follow methods identified in the Code of Federal Regulations, title 40, part 136, or other U.S. EPA-approved methods, or included in the following U.S. EPA method manuals: Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms, Fourth Edition (EPA-821-R-02-013).
 - 5.1.3.1. A 7-day static renewal toxicity test with a vertebrate, the fathead minnow, *Pimephales promelas* (Larval Survival and Growth Test Method 1000.0).
 - 5.1.3.2. A static renewal toxicity test with an invertebrate, the water flea, *Ceriodaphnia dubia* (Survival and Reproduction Test Method 1002.0).
 - 5.1.3.3. A 96-hour static non-renewal toxicity test with a plant, the green algae, *Selenastrum capricornutum* (also named *Raphidocelis subcapitata*) (Growth Test Method 1003.0).

Test results shall be analyzed using the TST as described below. To the extent that U.S. EPA-approved methods require that observations be made of organisms' response in multiple concentrations of effluent or receiving water, the instream waste concentration (IWC) shall be included as one of the selected concentrations, and the TST shall be conducted using the IWC and control as described in Section 5.1.4. below.

- 5.1.4. **Test of Significant Toxicity.** Aquatic toxicity test data shall be analyzed using the test of significant toxicity (TST) as described in Steps 1 through 7, within section IV.B.1.c of the Toxicity Provisions (Steps). For any chronic aquatic toxicity test method with both lethal and sub-lethal endpoints, the sub-lethal endpoint data shall be in these Steps. For any chronic aquatic toxicity test

method with more than one sub-lethal endpoint, the data for each sub-lethal endpoint shall be independently analyzed using these Steps. The TST is applicable for a data analysis of an IWC compared to a control. For assessing whether ambient water meets the water quality objectives, the undiluted ambient water shall be used as the IWC for purposes of the data analysis as described in the Toxicity Provisions.

- 5.1.5. **Percent Effect.** The percent effect at the IWC shall be calculated for each endpoint in an aquatic toxicity test, using untransformed data and the following equation:

$$\text{Percent Effect at the IWC} = \frac{\text{Mean Control Response} - \text{Mean IWC Response}}{\text{Mean Control Response}} \cdot 100$$

- 5.1.6. **Species Sensitivity Screening.** Species sensitivity screening shall be conducted within 18 months of the Order's adoption. The Permittee shall collect a single effluent sample and concurrently conduct three chronic toxicity tests using the fish, the invertebrate, and the algae species identified in section 5.1.3, above. This sample shall also be analyzed for the parameters required for the discharge at EFF-001. The species that exhibits the highest "Percent (%) Effect" at the discharge IWC during species sensitivity screening shall be used for routine monitoring during the permit term.

Species sensitivity screening conducted prior to an Order's adoption may be considered by the Regional Water Board if that species sensitivity screening data was generated within the last 10 years, remains representative of the Permittee's discharge, and fulfills the species sensitivity screening requirement. The Regional Water Board has determined that species sensitivity screening conducted between February 13 and 17, 2017 remains representative of the Permittee's effluent and the species used for chronic toxicity monitoring shall be *Ceriodaphnia dubia*, until the Order is modified to reflect a new most sensitive species, as identified by the required species sensitivity screening.

- 5.1.7. **Routine Monitoring Requirements.** The Permittee shall conduct at least one chronic aquatic toxicity test annually during which there is expected to be at least 15 days of discharge in the quarter. Initiation of the routine monitoring test shall be at a time that would allow any required MMEL compliance tests to be initiated within the same calendar month¹ as the routine monitoring test.

¹ For purposes of aquatic toxicity monitoring, a calendar month shall be defined as the period of time from a day of one month to the day before the corresponding day of the next month if the corresponding day exists, or if not to the last day of the next month

To the extent feasible, routine monitoring tests shall be evenly distributed across the calendar year or period of seasonal or intermittent discharge.

- 5.1.8. **Additional Routine Monitoring Requirement.** An additional routine monitoring test shall be required when there is one violation of the MDEL or MMEL, but not two violations in a single calendar month. This additional routine monitoring test is not required if the Permittee is already conducting a TRE, or if the Permittee is required to conduct routine monitoring at or more frequently than a monthly frequency.

This additional routine monitoring test shall be initiated within two weeks after the calendar month in which the MMEL or MDEL violation occurred. The calendar month of the violation and the calendar month of the additional routine monitoring shall be considered “successive calendar months” for purposes of determining whether a TRE is required under section 5.2.2, below. This additional routine monitoring test is used to determine if a TRE is necessary. This additional routine monitoring test is also used for compliance purposes and could result in the need to conduct MMEL compliance tests.

- 5.1.9. **Compliance Monitoring Requirements.** If a chronic aquatic toxicity routine monitoring test results in a “fail” at the IWC, then the Permittee shall complete a maximum of two MMEL compliance tests. The MMEL compliance tests shall be initiated within the same calendar month that the first routine monitoring test was initiated that resulted in the “fail” at the IWC. If the first chronic MMEL compliance tests results in a “fail” at the IWC, then the second MMEL compliance test is waived because the first chronic MMEL compliance test that results in a “fail” constitutes a violation and so the second MMEL compliance test is not required.

5.1.10. **Other Requirements.**

- 5.1.10.1. When a required toxicity test for routine monitoring or MMEL compliance tests is not completed, a new toxicity test to replace the toxicity test that was not completed shall be initiated as soon as possible. The new toxicity test shall replace the routine monitoring or MMEL compliance tests, as applicable, for the calendar month in which the toxicity test that was not completed was required to be initiated, even if the new toxicity test is initiated in a subsequent month. The new toxicity test for routine monitoring or MMEL compliance tests, as applicable, and any MMEL compliance tests required to be conducted due to the results of the new toxicity test shall be used to determine compliance with the effluent limitations for the calendar month in which the toxicity test that was not completed and was required to

(e.g., from January 1 to January 31, from June 15 to July 14, or from January 31 to February 28).

be initiated. The new toxicity test and any MMEL compliance tests required to be conducted due to the results of the new toxicity test shall not be used to substitute for any other required toxicity tests.

When any monitoring test is not initiated in the required time period due to circumstances outside of the Permittee's control that were not preventable with the reasonable exercise of care, and the Permittee promptly initiates, and ultimately completes a replacement test, the Regional Water Board may determine that the replacement monitoring test was not required to be initiated in the required time period.

- 5.1.10.2. When there is no discharge to surface waters and therefore no effluent available to complete a routine monitoring test or MMEL compliance test, the test shall not be required, and routine monitoring continues at the frequency specified in the permit.
- 5.1.11. **Reporting.** Results obtained from toxicity tests shall be reported to the Regional Water Board in the Permittee's quarterly Self-Monitoring Report (SMR), as either a "pass" or a "fail," and the percent effect at the IWC for each endpoint. The SMR shall include a full laboratory report for each toxicity test that was performed (WET report).
 - 5.1.11.1. WET reports shall include the contracting laboratory's complete report provided to the Permittee and shall be consistent with the appropriate "Report Preparation and Test Review" sections of the methods manual and this MRP. The WET test reports shall contain a narrative report that includes details about WET test procedures and results, including the following:
 - 5.1.11.1.1. Receipt and handling of the effluent sample that includes a tabular summary of initial water quality characteristics (e.g., pH, dissolved oxygen, temperature, conductivity, hardness, salinity, chlorine, ammonia);
 - 5.1.11.1.2. Toxic substances (such as chlorine, ammonia, and other chemicals) shall not be removed from the effluent sample prior to conducting the toxicity analysis;
 - 5.1.11.1.3. Any manipulations done to lab control/diluent and effluent such as filtration, nutrient addition, etc.;
 - 5.1.11.1.4. Tabular summary of test results for control water and each effluent dilution;
 - 5.1.11.1.5. The toxicity test results reported as either a "Pass" or "Fail", and the "Percent Effect" at the IWC for each endpoint;
 - 5.1.11.1.6. Identification of any anomalies or nuances in the test procedures or results.

- 5.1.12. **Notification.** All toxicity tests at the IWC shall be used for determining compliance with any toxicity MDEL or MMEL contained in this Order. The Permittee shall notify the Regional Water Board of a violation of a toxicity MDEL or MMEL as soon as the discharger learns of the violation, but no later than 24 hours of the discharger receiving the monitoring results.

5.2. Toxicity Reduction Evaluation (TRE) Process

- 5.2.1. **Generic TRE Work Plan.** The Permittee shall prepare and submit to the Regional Water Board Executive Officer a TRE workplan **within 180 days of the effective date of this Order.** This plan shall be reviewed and updated as necessary in order to remain current and applicable to the discharge and discharge facilities. The workplan shall describe the steps the Permittee intends to follow if toxicity is detected, and should include at a minimum the following items:
- 5.2.1.1. A description of the investigation and evaluation techniques that would be used to identify potential causes and sources of toxicity, effluent variability, and treatment system efficiency.
 - 5.2.1.2. A description of the facility's methods of maximizing in-house treatment efficiency, good housekeeping practices, and a list of all chemicals used in the operation of this Facility.
 - 5.2.1.3. If a toxicity identification evaluation (TIE) is necessary, an indication of the person who would conduct the TIEs (i.e., an in-house expert or an outside contractor).
- 5.2.2. **TRE Work Plan.** A TRE Work Plan is required to be submitted and implemented when a Permittee does not meet any combination of two or more MDEL or MMEL within a single calendar month or within two successive calendar months. In addition, if other information indicates toxicity (e.g., results of additional monitoring, results of monitoring at a higher concentration than the IWC, fish kills, intermittent recurring toxicity), then the Regional Water Board may also require a TRE. A TRE may also be required when there is no effluent available to complete a routine monitoring test or MMEL test. Routine Monitoring shall continue during a TRE.

The TRE Work Plan shall be submitted for Regional Water Board approval within 30 days from receipt of the chronic toxicity monitoring result, or other toxicity event, that initiated the TRE requirement. The TRE Work Plan shall follow the generic TRE Work Plan and be revised as appropriate for the initiating toxicity events.

The TRE shall be conducted according to the EPA manual Generalized Methodology for Conducting Industrial Toxicity Reduction Evaluations (EPA/600/2-88/070, 1989). The TRE Work Plan shall include the following

information, and comply with additional conditions set by the Regional Water Board Executive Officer:

- 5.2.2.1. Further actions by the Permittee to investigate, identify, and correct causes of toxicity.
- 5.2.2.2. Actions the Permittee will take to mitigate effects of the discharge and prevent the recurrence of toxicity.
- 5.2.2.3. A schedule for these actions, progress reports, and the final report.
- 5.2.3. **TIE Implementation.** The Permittee may initiate a TIE as part of a TRE to identify the causes of toxicity using the same species and test methods and, as guidance, EPA manuals: Methods for Aquatic Toxicity Identification Evaluations: Phase I Toxicity Characterization Procedures (EPA/600/6-91/003, 1991); Methods for Aquatic Toxicity Identification Evaluations, Phase II Toxicity Identification Procedures for Samples Exhibiting Acute and Chronic Toxicity (EPA/600/R-92/080, 1993); Methods for Aquatic Toxicity Identification Evaluations, Phase III Toxicity Confirmation Procedures for Samples Exhibiting Acute and Chronic Toxicity (EPA/600/R-92/081, 1993). The TIE should be conducted on the species demonstrating the most sensitive toxicity response.
- 5.2.4. Many recommended TRE elements parallel required or recommended efforts for source control, pollution prevention, and storm water control programs. TRE efforts should be coordinated with such efforts. As toxic substances are identified or characterized, the Permittee shall continue the TRE by determining the sources and evaluating alternative strategies for reducing or eliminating the substances from the discharge. All reasonable steps shall be taken to reduce toxicity to levels consistent with toxicity evaluation parameters.
- 5.2.5. The Regional Water Board recognizes that toxicity may be episodic and identification of the causes and reduction of sources of toxicity may not be successful in all cases. The TRE may be ended at any stage if monitoring finds there is no longer toxicity.

6. LAND DISCHARGE MONITORING REQUIREMENTS – NOT APPLICABLE

This Order does not authorize discharges to land.

7. RECYCLING MONITORING REQUIREMENTS

7.1. Monitoring Location EFF-003

- 7.1.1. The Permittee shall monitor secondary treated wastewater to be recycled at Monitoring Location EFF-003 during periods of discharge to the irrigation site as follows:

Table E-4. Recycled Water Monitoring Requirements – Monitoring Location EFF-003

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method ¹
Effluent Flow ³	mgd	Meter	Continuous	---
Biochemical Oxygen Demand 5 day @ 20°C (BOD ₅)	mg/L	Grab	Monthly	Standard Methods ²
Total Suspended Solids (TSS)	mg/L	Grab	Monthly	Standard Methods ²
pH	standard units	Grab	Weekly	Standard Methods ²
Total Coliform Bacteria	MPN/100 mL	Grab	Weekly	Standard Methods ²
Nitrate Nitrogen, Total (as N)	mg/L	Grab	June and September	Standard Methods ²
Nitrogen, Total (as N) ⁴	mg/L	Grab	June and September	Standard Methods ²
Total Dissolved Solids	mg/L	Grab	June and September	Standard Methods ²
Chloroform ⁵	µg/L	Grab	June and September	Standard Methods ²
Monochloroacetic Acid	µg/L	Grab	Quarterly ⁷	U.S. EPA Method 552
Dichloroacetic Acid	µg/L	Grab	Quarterly ⁷	U.S. EPA Method 552
Trichloroacetic Acid	µg/L	Grab	Quarterly ⁷	U.S. EPA Method 552
Monobromoacetic Acid	µg/L	Grab	Quarterly ⁷	U.S. EPA Method 552
Dibromoacetic Acid	µg/L	Grab	Quarterly ⁷	U.S. EPA Method 552
Haloacetic Acids ^{5,6}	µg/L	Calculate	Quarterly ⁷	U.S. EPA Method 552
4,4-DDT	ug/L	Grab	Quarterly ⁸	Standard Methods ²

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method ¹
Antimony, Total Recoverable	ug/L	Grab	Quarterly ⁸	Standard Methods ²

Table Notes

1. Pollutants shall be analyzed using the analytical methods described in 40 C.F.R. part 136 or by methods approved by the Regional Water Board or State Water Board.
2. In accordance with the current edition of *Standard Methods for Examination of Water and Wastewater* (American Public Health Administration) or by methods approved by the Regional Water Board or State Water Board.
3. The Permittee shall report the maximum daily, average daily, and average monthly flows.
4. The sum of the concentrations of nitrate nitrogen, nitrite nitrogen, total organic nitrogen and ammonia.
5. Based on the first full year of monitoring, the Regional Water Board Executive Officer may reduce the frequency of or eliminate monitoring for chloroform and Haloacetic Acids in the effluent if monitoring data demonstrates that the Permittee's discharge does not contain chloroform or Haloacetic Acids at concentrations that pose reasonable potential to cause an exceedance of applicable water quality objectives for chloroform or Haloacetic Acids.
6. The sum of the concentrations of Monochloroacetic Acid, Dichloroacetic Acid, Trichloroacetic Acid, Monobromoacetic Acid, and Dibromoacetic Acid.
7. Quarterly monitoring applies to Monitoring Locations EFF-001 and EFF-003. Sampling shall be coordinated to ensure that two samples are collected during discharge to surface waters and two samples are collected during discharge to the irrigation site.
8. The monitoring frequency for 4,4-DDT and Antimony may be reduced to annually if the first 12 months of monitoring data collected after the effective date of this Order demonstrate that concentrations of 4,4-DDT and Antimony are consistently lower than water quality objectives for protection of ground water.

- 7.1.2. The irrigation system shall be visually monitored on a daily basis during periods of use to ensure compliance with Water Recycling Requirements in section 4.3.2. of the Order. Visual observations shall include a record of any odors, evidence of surface runoff or spray mist leaving the property boundaries, or other signs of malfunction or improper operation of the irrigation system. A summary of these observations shall be submitted with each quarterly monitoring report.

- 7.1.3. Operating records for the irrigation system shall be maintained by the Permittee and shall include analytical results, specified above; records of operational problems with the irrigation system, plant and equipment breakdowns; and all corrective and preventative actions taken relative to the irrigation system. A summary of these operating records shall be maintained at the Facility.

8. RECEIVING WATER MONITORING REQUIREMENTS

8.1. Monitoring Location RSW-001

- 8.1.1. The Permittee shall monitor ambient conditions in Lower Eel River at Monitoring Location RSW-001 during periods of discharge to Lower Eel River as follows:

Table E-5. Receiving Water Monitoring Requirements – Monitoring Location RSW-001

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Flow ¹	mgd	Meter ¹	Daily	---
pH	standard units	Grab	Monthly ^{4,9}	Standard Methods ²
Hardness, Total (as CaCO ₃)	mg/L	Grab	Monthly ⁹	Standard Methods ²
Dissolved Oxygen	mg/L	Grab	Monthly	Standard Methods ²
Temperature	°C	Grab	Monthly ⁴	Standard Methods ²
Turbidity	NTU	Grab	Monthly	Standard Methods ²
Total Dissolved Solids	mg/L	Grab	Monthly	Standard Methods
CTR Priority Pollutants ^{5,6,7}	µg/L	Grab	Once per permit term	Standard Methods ³
Electrical Conductivity @ 25°C	µmhos/cm	Grab	Monthly	Standard Methods
<i>E. coli</i> Bacteria ⁸	MPN/100 mL	Grab	10	Standard Methods ²
Aluminum, Total Recoverable	µg/L	Grab	Quarterly ⁹	Standard Methods ²
Dissolved Organic Carbon	mg/L	Grab	Quarterly ⁹	Standard Methods ²

Table Notes

1. The flow rate shall be determined using the flow at United States Geological Survey (USGS) Gage No. 11477000 in the Lower Eel River near Scotia.
2. In accordance with the current edition of *Standard Methods for Examination of Water and Wastewater* (American Public Health Administration) or by methods approved by the Regional Water Board or State Water Board.
3. Pollutants shall be analyzed using the analytical methods described in 40 C.F.R. part 136 or by methods approved by the Regional Water Board or State Water Board.
4. pH and temperature monitoring must coincide with monthly effluent monitoring for ammonia.
5. Those pollutants identified by the California Toxics Rule at 40 C.F.R. section 131.38. The Permittee is not required to sample and analyze for asbestos. Hardness shall be monitored concurrently with the priority pollutant sample. Monitoring shall occur simultaneously with effluent monitoring for CTR priority pollutants required by section 4.1 of this MRP.
6. CTR priority pollutant sampling shall be completed no later than **April 15, 2027** during a period of discharge to the Eel River. Effluent and receiving water monitoring shall occur concurrently.
7. Analytical methods must achieve the lowest minimum level (ML) specified in Appendix 4 of the SIP and, in accordance with section 2.4 of the SIP, the Permittee shall report the ML and MDL for each sample result.
8. The Permittee may use any *E. coli* method specified in 40 CFR 136 for compliance monitoring.
9. Receiving water monitoring for aluminum, pH, dissolved organic carbon, and hardness shall be conducted concurrently with effluent monitoring for aluminum.
10. The Permittee shall monitor for *E. coli* at least once per 6-week monitoring period, concurrently with *E. coli* monitoring at monitoring location EFF-001 and RSW-002.

8.2. Monitoring Location RSW-002

- 8.2.1. The Permittee shall monitor the Lower Eel River at Monitoring Location RSW-002 during periods of discharge to Lower Eel River as follows:

Table E-6. Receiving Water Monitoring Requirements – Monitoring Location RSW-002

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
pH	standard units	Grab	Monthly ²	Standard Methods ¹

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Dissolved Oxygen	mg/L	Grab	Monthly ²	Standard Methods ¹
Temperature	°C	Grab	Monthly	Standard Methods ¹
Turbidity	NTU	Grab	Monthly	Standard Methods ¹
Total Dissolved Solids	mg/L	Grab	Monthly	Standard Methods ¹
Electrical Conductivity @ 25°C	µmhos/cm	Grab	Monthly	Standard Methods ¹
<i>E. coli</i> Bacteria ³	MPN/100 mL	Grab	4	Standard Methods ¹

Table Notes

1. In accordance with the current edition of *Standard Methods for Examination of Water and Wastewater* (American Public Health Administration) or by methods approved by the Regional Water Board or State Water Board.
2. pH and temperature monitoring must coincide with monthly effluent monitoring for ammonia.
3. The Permittee may use any *E. coli* method specified in 40 CFR 136 for compliance monitoring.
4. The Permittee shall monitor for *E. coli* at least once per 6-week monitoring period, concurrently with *E. coli* monitoring at monitoring location EFF-001 and RSW-001.

8.3. Groundwater Monitoring

Monitoring Location MW-001, MW-002, MW-003, and MW-004

The Permittee shall monitor groundwater at Monitoring Location MW-001, MW-002, MW-003, and MW-004 during periods of discharge of recycled water at Discharge Point 003 as follows:

Table E-7. Receiving Water Monitoring Requirements – Monitoring Location MW-001 through MW-004

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method ¹
Depth to Groundwater	0.01 feet	Grab	Semiannually	Measurement

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method¹
Nitrate Nitrogen, Total (as N)	mg/L	Grab	Semiannually	Part 136 ¹
Total Coliform Bacteria	MPN/100 mL	Grab	Semiannually	Part 136 ¹
Total Dissolved Solids	mg/L	Grab	Semiannually	Standard Methods ²
Chloroform ³	µg/L	Grab	Annually	Part 136 ¹
4,4-DDT	ug/L	Grab	Semiannually ⁴	Standard Methods ²
Antimony, Total Recoverable	ug/L	Grab	Semiannually ⁴	Standard Methods ²

Table Notes

1. Pollutants shall be analyzed using the analytical methods described in 40 C.F.R. part 136 or by methods approved by the Regional Water Board or State Water Board.
2. In accordance with the current edition of Standard Methods for Examination of Water and Wastewater (American Public Health Administration) or by methods approved by the Regional Water Board or State Water Board.
3. The Regional Water Board Executive Officer may reduce the frequency of or eliminate monitoring for chloroform in the effluent if monitoring data demonstrates that the Permittee's discharge does not result in the presence of chloroform in groundwater at concentrations that pose reasonable potential to cause an exceedance of applicable water quality objectives for chloroform.
4. The monitoring frequency for 4,4-DDT and Antimony may be reduced to annually if the first 12 months of monitoring data collected after the effective date of this Order demonstrate that concentrations of 4,4-DDT and Antimony are consistently lower than water quality objectives for protection of ground water.

9. OTHER MONITORING REQUIREMENTS

9.1. Disinfection Process Monitoring for the Chlorine Disinfection System

9.1.1. Monitoring Locations INT-001

- 9.1.2. The Permittee shall monitor the discharge from the chlorine contact chamber prior to dechlorinating at Monitoring Location INT-001 as follows:

Table E-8. Internal Effluent Monitoring - Monitoring Location INT-001

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Chlorine, Total Residual ²	mg/L	Grab/Meter	Daily/Continuous	Part 136 ¹

Table Notes

1. Pollutants shall be analyzed using the analytical methods described in 40 C.F.R. part 136 or by methods approved by the Regional Water Board or State Water Board.
2. The Permittee shall monitor continuously to demonstrate that the appropriate chlorine residual concentration is maintained in the effluent at INT-001 at all times. At a minimum, the Permittee shall record readings of the continuous monitoring every hour on the hour and report the maximum-recorded daily chlorine residual. The Permittee shall calibrate chlorine residual analyzers against grab samples as frequently as necessary to maintain accurate and reliable operation. The Permittee may measure residual de-chlorination compounds to determine compliance with residual chlorine limitations.

9.2. Visual Monitoring (Monitoring Location EFF-001, RSW-001, and RSW-002)

Visual observations of the discharge and receiving water shall be recorded monthly and on the first day of each intermittent discharge. Visual monitoring shall include, but not be limited to, observations for floating materials, coloration, objectionable aquatic growths, oil and grease films, and odors. Visual observations shall be recorded and included in the Permittee's quarterly SMRs.

9.3. Sludge Monitoring (Monitoring Location BIO-001)

- 9.3.1. Sludge sampling shall be conducted according to the requirements specified by the location and type of disposal activities undertaken.
- 9.3.2. Sampling records shall be retained for a minimum of 5 years. A log shall be maintained for sludge quantities generated and of handling and disposal activities. The frequency of entries is discretionary; however, the log must be complete enough to serve as a basis for developing the Sludge Handling and Disposal report that is required as part of the Annual Report.

10. REPORTING REQUIREMENTS**10.1. General Monitoring and Reporting Requirements**

- 10.1.1. The Permittee shall comply with all Standard Provisions (Attachment D) related to monitoring, reporting, and recordkeeping.

10.2. Self-Monitoring Reports (SMRs)

- 10.2.1. The Permittee shall electronically submit SMRs using the State Water Board's [California Integrated Water Quality System \(CIWQS\) Program website](http://www.waterboards.ca.gov/water_issues/programs/ciwqs/) (http://www.waterboards.ca.gov/water_issues/programs/ciwqs/). The CIWQS website will provide additional information for SMR submittal in the event there will be a planned service interruption for electronic submittal. The Permittee shall maintain sufficient staffing and resources to ensure it submits eSMRs that are complete and timely. This includes provisions of training and supervision of individuals (e.g., Permittee personnel or consultant) on how to prepare and submit eSMRs.
- 10.2.2. The Permittee shall report in the SMR the results for all monitoring specified in this MRP under sections 3 through 9. The Permittee shall submit quarterly SMRs including the results of all required monitoring using U.S. EPA-approved test methods or other test methods specified in this Order. SMRs are to include all new monitoring results obtained since the last SMR was submitted. If the Permittee monitors any pollutant more frequently than required by this Order, the results of this monitoring shall be included in the calculations and reporting of the data submitted in the SMR.
- 10.2.3. All monitoring results reported shall be supported by the inclusion of the complete analytical report from the laboratory that conducted the analyses.
- 10.2.4. Monitoring periods and reporting for all required monitoring shall be completed according to the following schedule:

Table E-9. Monitoring Periods and Reporting Schedule

Sampling Frequency	Monitoring Period Begins On...	Monitoring Period	SMR Due Date
Continuous	Permit effective date	All	First day of second calendar month following the month of sampling
Daily	Permit effective date	(Midnight through 11:59 PM) or any 24-hour period that reasonably represents a calendar day for purposes of sampling	First day of second calendar month following the month of sampling

Sampling Frequency	Monitoring Period Begins On...	Monitoring Period	SMR Due Date
Weekly	Sunday following permit effective date or on permit effective date if on a Sunday	Sunday through Saturday	First day of second calendar month following the month of sampling
Monthly	First day of calendar month following permit effective date or on permit effective date if that date is first day of the month	1st day of calendar month through last day of calendar month	First day of second calendar month following the month of sampling
Quarterly	Closest of January 1, April 1, July 1, or October 1 following (or on) permit effective date	January 1 through March 1 April 1 through June 30 July 1 through September 30 October 1 through December 1	First day of second calendar month following the month of sampling
Annually	January 1 following (or on) permit effective date	January 1 through December 1	March 1, each year (with annual report)
Semiannually	Closest of January 1 or July 1 following (or on) permit effective date.	January 1 through June 30 July 1 through December 1	September 1, each year March 1, each year
Once per Permit Term	Permit Effective Date	All	March 1 following the year that monitoring is completed (with annual report) with last data to be submitted at least 180 days prior to permit expiration

10.2.5. Reporting Protocols. The Permittee shall report with each sample result the applicable Reporting Level (RL) and the current Method Detection Limit (MDL), as determined by the procedure in 40 C.F.R. part 136.

The Permittee shall report the results of analytical determinations for the presence of chemical constituents in a sample using the following reporting protocols:

- 10.2.5.1. Sample results greater than or equal to the RL shall be reported as measured by the laboratory (i.e., the measured chemical concentration in the sample).
- 10.2.5.2. Sample results less than the RL, but greater than or equal to the laboratory's MDL, shall be reported as "Detected, but Not Quantified," or DNQ. The estimated chemical concentration of the sample shall also be reported.

For the purposes of data collection, the laboratory shall write the estimated chemical concentration next to DNQ as well as the words "Estimated Concentration" (may be shortened to Est. Conc.).

The laboratory may, if such information is available, include numerical estimates of the data quality for the reported result. Numerical estimates of data quality may be percent accuracy ($\pm$ a percentage of the reported value), numerical ranges (low to high), or any other means considered appropriate by the laboratory.

- 10.2.5.3. Sample results less than the laboratory's MDL shall be reported as "Not Detected," or ND.
 - 10.2.5.4. The Permittee is to instruct laboratories to establish calibration standards so that the ML value (or its equivalent if there is differential treatment of samples relative to calibration standards) is the lowest calibration standard. At no time is the Permittee to use analytical data derived from extrapolation beyond the lowest point of the calibration curve.
- 10.2.6. **Self-Monitoring Reports.** The Permittee shall submit quarterly SMRs in accordance with the following requirements:
- 10.2.6.1. The Permittee shall arrange all reported data in a tabular format. The data shall be summarized to clearly illustrate whether the facility is operating in compliance with interim and/or final effluent limitations. The Permittee is not required to duplicate the submittal of data that is entered in a tabular format within CIWQS. When electronic submittal of data is required and CIWQS does not provide for entry into a tabular format within the system, the Permittee shall electronically submit the data in a tabular format as an attachment.
 - 10.2.6.2. The Permittee shall attach a cover letter to the SMR. The information contained in the cover letter shall clearly identify:
 - 10.2.6.2.1. Facility name and address;

- 10.2.6.2.2. WDID number;
 - 10.2.6.2.3. Applicable period of monitoring and reporting;
 - 10.2.6.2.4. Violations of the WDRs (identified violations must include a description of the requirement that was violated and a description of the violation);
 - 10.2.6.2.5. Corrective actions taken or planned; and
 - 10.2.6.2.6. The proposed time schedule for corrective actions.
- 10.2.6.3. SMRs must be submitted to the Regional Water Board, signed and certified as required by the Standard Provisions (Attachment D), to the [CIWQS Program Website](http://www.waterboards.ca.gov/ciwqs/index.html) (<http://www.waterboards.ca.gov/ciwqs/index.html>).
 - 10.2.6.4. In the event that an alternate method for submittal of SMRs is required, the Permittee shall submit the SMR electronically via e-mail to NorthCoast@waterboards.ca.gov or on disk (CD or DVD) in Portable Document Format (PDF) file in lieu of paper-sourced documents. The guidelines for electronic submittal of documents can be found on the [Regional Water Board website](http://waterboards.ca.gov/northcoast/) (<http://waterboards.ca.gov/northcoast/>).
 - 10.2.6.5. At any time during the term of this permit, the Regional Water Board may notify the Permittee to electronically submit both technical and Self-Monitoring Reports (SMRs) to the State Water Board's GeoTracker database in searchable Portable Document Format (pdf). In addition, analytical data will be required to be uploaded to the GeoTracker database under a site-specific global identification number that will be assigned to the Permittee. Information on the GeoTracker database is provided on the [State Water Board website](https://www.waterboards.ca.gov/resources/data_databases/groundwater.html). (https://www.waterboards.ca.gov/resources/data_databases/groundwater.html).

10.3. Discharge Monitoring Reports (DMRs)

- 10.3.1. DMRs are U.S. EPA reporting requirements. The Permittee shall electronically certify and submit DMRs together with SMRs using Electronic Self-Monitoring Reports module eSMR 2.5 or any upgraded version. Electronic DMR submittal shall be in addition to electronic SMR submittal. DMRs shall be submitted quarterly on the first day of the second calendar month following the end of each quarter (February 1, May 1, August 1, November 1). Information about electronic DMR submittal is available at the [DMR website](http://www.waterboards.ca.gov/water_issues/programs/discharge_monitoring): (http://www.waterboards.ca.gov/water_issues/programs/discharge_monitoring).

10.4. Other Reports

- 10.4.1. Special Study Reports and Progress Reports. As specified in the Special Provisions contained in section 6.3. of the Order, special study and progress

reports shall be submitted in accordance with the following reporting requirements.

Table E-10. Reporting Requirements for Special Provisions Reports

Order Section	Special Provision Requirement	Reporting Requirement
Special Provision 6.3.2.1	Pollutant Minimization Program	If required by the Regional Water Board Executive Officer
Special Provision 6.3.2.2.5	Pollutant Minimization Program, Annual Facility Report	March 1, annually, following development of Pollutant Minimization Program
Special Provision 6.3.4.2.1	Source Control Provisions, Annual Report	March 1, annually
Special Provision 6.3.4.6	Adequate Capacity, Technical Report	Within 120 days of notification that the Facility will reach capacity within 4 years
MRP General Monitoring Provision 1.6	DMR-QA Study Report	Annually, per State Water Board instructions
MRP Effluent Monitoring Requirement 5.1.6	Verbal and written notification of chronic toxicity fail result	Within 24 hours after receipt of a fail result.
MRP Effluent Monitoring Requirement 5.2.1	Generic TRE Work Plan review and update	Review by July 1, 2024 Update as necessary
MRP Effluent Monitoring Requirement 5.2.2	TRE Workplan	No later than 30 days receipt of the chronic toxicity monitoring result, or other toxicity event, that initiated the TRE requirement.
MRP Reporting Requirement 10.4.2	Disaster Preparedness Assessment Report and Action Plan	September 1, 2025
MRP Reporting Requirement 10.4.3	Annual Report	March 1, annually
MRP Reporting Requirement 10.4.4	Annual Volumetric Report	April 30, annually
MRP Reporting Requirement 10.5.1	Notification of spills and unauthorized discharges.	Oral reporting within 24 hours and written report within 5 days

10.4.2. **Disaster Preparedness Assessment Report and Action Plan.** Natural disasters, extreme weather events, sea level rise, and shifting precipitation patterns, some of which are projected to intensify due to climate change, have significant implications for wastewater treatment and operations. Some natural disasters are expected to become more frequent and extreme according to the current science on climate change. In order to ensure that Facility operations are not disrupted, compliance with conditions of this Order are achieved, and receiving waters are not adversely impacted by permitted and unpermitted discharges, the Permittee shall submit a Disaster Preparedness Assessment Report and Action Plan to the Regional Water Board by **September 1, 2025**, for Executive Officer review and approval.

10.4.2.1. The Permittee shall:

10.4.2.1.1. Conduct an assessment of the wastewater treatment facility, operations, collection, and discharge systems (including the discharge outfall) to determine areas of short- and long-term vulnerabilities related to natural disasters and extreme weather, including sea level rise and other conditions projected by climate change science, if applicable; the assessment shall consider, as applicable, impacts to facility operations due to changing influent and receiving water quality, rising sea level, storm surges, fires, floods, earthquakes, tsunamis, back-to-back severe storms, and other extreme conditions that pose a risk to facility operations and water quality;

10.4.2.1.2. Identify control measures needed to protect, improve, and maintain wastewater infrastructure, waste discharge compliance, and receiving water quality in the event of a natural disaster or, if applicable, under conditions resulting from climate change;

10.4.2.1.3. Develop a schedule to implement necessary control measures. Control measures shall include, but are not limited to, emergency procedures, contingency plans, alarm/notification systems, training, backup power and equipment, and the need for planned mitigations to ameliorate potential risks associated with extreme weather events and changing conditions resulting from climate change; and

10.4.2.1.4. Implement the necessary control measures per the approved schedule of implementation.

10.4.3. **Annual Report.** The Permittee shall submit an annual report to the Regional Water Board for each calendar year through the CIWQS Program Web site. In the event that a paper copy of the annual report is required, the Permittee shall submit the report to the email address in section 10.2.6.3, above. The report shall be submitted by **March 1st** of the following year. The report shall, at a minimum, include the following:

- 10.4.3.1. Where appropriate, tabular and/or graphical summaries of the monitoring data and disposal records from the previous year. If the Permittee monitors any pollutant more frequently than required by this Order, using test procedures approved under 40 C.F.R. part 136 or as specified in this Order, the results of this monitoring shall be included in the calculation and report of the data submitted in the SMR.
- 10.4.3.2. A comprehensive discussion of the Facility's compliance (or lack thereof) with all effluent limitations and other WDRs, and the corrective actions taken or planned, which may be needed to bring the discharge into full compliance with the Order.
- 10.4.3.3. The names and general responsibilities of all persons employed at the Facility;
- 10.4.3.4. The names and telephone numbers of persons to contact regarding the Facility for emergency and routine situations; and
- 10.4.3.5. A statement certifying when the flow meter(s) and other monitoring instruments and devices were last calibrated, including identification of who performed the calibration.
- 10.4.3.6. **Source Control Activity Reporting.** The Permittee shall submit, as part of its Annual Report to the Regional Water Board, a description of the Permittee's source control activities, as required by Special Provision 6.3.4.2, during the past year. This annual report is due on March 1st of each year, and shall contain:
 - 10.4.3.6.1. A copy of the source control standards, including a table presenting local limits.
 - 10.4.3.6.2. A description of the waste hauler permit system; if applicable.
 - 10.4.3.6.3. A summary of the compliance and enforcement activities taken by the Permittee during the past year, which ensures industrial user compliance. The summary shall include the names and addresses of any industrial or commercial users under surveillance by the Permittee, an explanation of whether they were inspected, sampled, or both, the frequency of these activities at each user, and the conclusions or results from the inspection or sampling of each user.
 - 10.4.3.6.4. An updated list of industrial users (by North American Industrial Classification/Standard Industrial Classification categories) which were issued permits and/or enforcement orders, and a status of compliance for each user.
 - 10.4.3.6.5. The name and address of each user that received a discharge limit.

- 10.4.3.6.6. A summary of any industrial waste survey results.
- 10.4.3.6.7. A summary of public outreach activities to educate industrial, commercial, and residential users about the importance of preventing discharges of industrial and toxic wastes to the Facility.
- 10.4.3.7. **Sludge Handling and Disposal Activity Reporting.** The Permittee shall submit, as part of its annual report to the Regional Water Board, a description of the Permittee's solids handling, disposal and reuse activities over the previous 12 months. At a minimum, the report shall contain:
 - 10.4.3.7.1. Annual sludge production, in dry tons and percent solids;
 - 10.4.3.7.2. Sludge monitoring results;
 - 10.4.3.7.3. A schematic diagram showing sludge handling facilities (e.g., digesters, thickeners, drying beds, etc.), if any and a solids flow diagram;
 - 10.4.3.7.4. Methods of final disposal of sludge:
 - 10.4.3.7.4.1. For any portion of sludge discharged to a sanitary landfill, the Permittee shall provide the volume of sludge transported to the landfill, the names and locations of the facilities receiving sludge, the Regional Water Board's WDRs Order number for the regulated landfill, and the landfill classification.
 - 10.4.3.7.4.2. For any portion of sludge discharged through land application, the Permittee shall provide the volume of biosolids applied, the date and locations where biosolids were applied, the Regional Water Board's WDRs Order number for the regulated discharge, a demonstration that the discharge was conducted in compliance with applicable permits and regulations, and, if applicable, corrective actions taken or planned to bring the discharge into compliance with WDRs.
 - 10.4.3.7.4.3. For any portion of sludge further treated through composting, the Permittee shall provide a summary of the composting process, the volume of sludge composted, and a demonstration and signed certification statement that the composting process and final product met all requirements for Class A biosolids.
 - 10.4.3.7.5. Results of internal or external third-party audits of the Biosolids Management System, including reported program deficiencies and recommendations, required corrective actions, and a schedule to complete corrective actions.
- 10.4.3.8. **Storm Water Reporting.** The Permittee shall submit, as part of its annual report to the Regional Water Board, an evaluation of the effectiveness of the Permittee's best management practices (BMPs) to control the run-on of

storm water to the Facility site, as well as activities to maintain and upgrade these BMPs.

- 10.4.3.9. **Chlorine Usage.** The Permittee is required to update the SOPs as needed and report any changes to the SOPs or change in chlorine usage. If there have been no changes to the SOPs or chlorine usage, the Permittee shall report that no changes have been made in the annual report.
- 10.4.4. **Annual Volumetric Reporting.** The Permittee shall electronically certify and submit an annual volumetric report, containing monthly data in electronic format, to State Water Board's GeoTracker system by April 30 of the following year. Required data shall be submitted to the GeoTracker database under a site-specific global identification number. The Permittee shall report in accordance with each of the items in Section 3 of the Policy for Water Quality Control for Recycled Water (Recycled Water Policy) as described below:
 - 10.4.4.1. **Influent.** Monthly volume of wastewater collected and treated by the Facility.
 - 10.4.4.2. **Production.** Monthly volume of waster treated, specifying level of treatment.
 - 10.4.4.3. **Discharge.** Monthly volume of treated wastewater discharged to each of the following, specifying level of treatment:
 - 10.4.4.3.1. Inland surface waters, specifying volume required to maintain minimum instream flow, if any; and
 - 10.4.4.3.2. Land, where beneficial use is not taking place, including evaporation or percolation ponds, overland flow, or spray irrigation disposal, excluding pasture or fields with harvested crops.
 - 10.4.4.4. **Reuse.**
 - 10.4.4.4.1. Monthly Volume of treated wastewater distributed.
 - 10.4.4.4.2. Annual volume of treated wastewater distributed for beneficial use in compliance with California Code of Regulations, title 22 in each of the use categories listed below:
 - 10.4.4.4.2.1. **Agricultural irrigation:** pasture or crop irrigation.
 - 10.4.4.4.2.2. **Other non-potable uses:** including but not limited to landscape irrigation, commercial application, industrial application, dust control, flushing sewers, fire protection, fill stations, snow making, and recreational impoundments.

10.5. Spill Notification

- 10.5.1. **Spills and Unauthorized Discharges.** Information regarding all spills and unauthorized discharges (except SSOs) that may endanger health, or the environment shall be provided orally to the Regional Water Board ² **within 24 hours** from the time the Permittee becomes aware of the circumstances and a written report shall also be provided **within five days** of the time the Permittee becomes aware of the circumstances, in accordance with section 5.5 of Attachment D.

Information to be provided verbally to the Regional Water Board includes:

- 10.5.1.1. Name and contact information of caller;
 - 10.5.1.2. Date, time, and location of spill occurrence;
 - 10.5.1.3. Estimates of spill volume, rate of flow, and spill duration, if available and reasonably accurate;
 - 10.5.1.4. Surface water bodies impacted, if any;
 - 10.5.1.5. Cause of spill, if known at the time of the notification;
 - 10.5.1.6. Cleanup actions taken or repairs made at the time of the notification; and
 - 10.5.1.7. Responding agencies.
- 10.5.2. **Sanitary Sewer Overflows.** Notification and reporting of sanitary sewer overflows are conducted in accordance with the requirements of Order No. 2022-0103-DWQ (Statewide General WDRs for Sanitary Sewer Systems), which is not incorporated herein by reference, and any revisions thereto.
- 10.5.3. **Recycled Water Spills.** Notification and reporting of spills and unauthorized discharges of recycled water discharged in or on any waters of the state, as defined in Water Code section 13050, shall be conducted in accordance with the following:
- 10.5.4. **Secondary Recycled Water**

² The contact number of the Regional Water Board during normal business hours is (707) 576-2220. After normal business hours, spill reporting to the California Governor's Office of Emergency Services Warning Center (CalOES) will satisfy the 24-hour spill reporting requirement for the Regional Water Board. The contact number for spill reporting for the CalOES is (800) 852-7550.

- 10.5.4.1. For unauthorized discharges of 50,000 gallons or more of secondary recycled water, the Permittee shall immediately notify the Regional Water Board as soon as (a) the Permittee has knowledge of the discharge or probable discharge, (b) notification is possible, and (c) notification can be provided without substantially impeding cleanup or other emergency measures.
- 10.5.4.2. For unauthorized discharges of more than 1,000 gallons, but less than 50,000 gallons of secondary recycled water, the Permittee shall notify the Regional Water Board as soon as possible, but no longer than 3 days after becoming aware of the discharge.

ATTACHMENT F - FACT SHEET

TABLE OF CONTENTS

1.	Permit Information	F-3
2.	Facility Description	F-4
2.1.	Description of Wastewater and Biosolids Treatment and Controls	F-5
2.2.	Discharge Points and Receiving Waters	F-6
2.3.	Summary of Existing Requirements and SMR Data	F-7
2.4.	Compliance Summary	F-9
2.5.	Planned Changes	F-11
3.	Applicable Plans, Policies, and Regulations.....	F-12
3.1.	Legal Authorities.....	F-12
3.2.	California Environmental Quality Act (CEQA).....	F-12
3.3.	State and Federal Laws, Regulations, Policies, and Plans.....	F-12
3.4.	Impaired Water Bodies on the CWA section 303(d) List.....	F-16
3.5.	Other Plans, Policies and Regulations	F-17
4.	Rationale for Effluent Limitations and Discharge Specifications.....	F-19
4.1.	Discharge Prohibitions.....	F-19
4.2.	Technology-Based Effluent Limitations	F-23
4.3.	Water Quality-Based Effluent Limitations (WQBELs)	F-26
4.4.	Final Effluent Limitation Considerations	F-45
4.5.	Interim Effluent Limitations	F-49
4.6.	Land Discharge Specifications	F-50
4.7.	Recycling Specifications	F-50
5.	Rationale for Receiving Water Limitations.....	F-54
5.1.	Surface Water.....	F-54
5.2.	Groundwater.....	F-54
6.	Rationale for Provisions	F-54
6.1.	Standard Provisions	F-54
6.2.	Special Provisions	F-55
7.	Rationale for Monitoring and Reporting Requirements	F-61
7.1.	Influent Monitoring	F-61
7.2.	Effluent Monitoring.....	F-61
7.3.	Toxicity Testing Requirements	F-63
7.4.	Recycled Water Monitoring Requirements	F-63
7.5.	Receiving Water Monitoring.....	F-63
7.6.	Other Monitoring Requirements.....	F-64
8.	Public Participation.....	F-66

8.1. Notification of Interested Parties.....	F-66
8.2. Written Comments.....	F-66
8.3. Public Hearing	F-67
8.4. Reconsideration of Waste Discharge Requirements	F-67
8.5. Information and Copying	F-68
8.6. Register of Interested Persons	F-68
8.7. Additional Information.....	F-68

TABLE OF TABLES

Table F-1. Facility Information.....	F-3
Table F-2. Historic Effluent Limitations and Monitoring Data ¹	F-7
Table F-3. Basin Plan Beneficial Uses	F-13
Table F-4. Summary of Technology Based Effluent Limitations – Discharge Point 001 (Monitoring Location EFF-001).....	F-25
Table F-5. Effluent Trihalomethane Concentrations on March 18, 2020 Sample	F-35
Table F-6. Effluent Haloacetic Acid Concentrations on November 23, 2021 Sample.....	F-35
Table F-7. Summary of Reasonable Potential Analysis Results for Priority Pollutants, Ammonia, and Title 22 Pollutants.....	F-37
Table F-8. Determination of Final WQBELs Based on Human Health Criteria.....	F-41
Table F-9. Summary of Acute Toxicity Results (Water Flea).....	F-44
Table F-10. Summary of Chronic Toxicity Results (Water Flea)	F-44
Table F-11. Summary of Effluent Limitations – Discharge Point 001 (Monitoring Location EFF-001)	F-47
Table F-12. Wastewater Treatment Facility RPA Summary	F-69

ATTACHMENT F - FACT SHEET

As described in section 2.2 of this Order, the Regional Water Board incorporates this Fact Sheet as findings of the Regional Water Board supporting the issuance of this Order. This Fact Sheet includes the legal requirements and technical rationale that serve as the basis for the requirements of this Order.

This Order has been prepared under a standardized format to accommodate a broad range of discharge requirements for Permittees in California. Only those sections or subsections of this Order that are specifically identified as “not applicable” have been determined not to apply to this Permittee. Sections or subsections of this Order not specifically identified as “not applicable” are fully applicable to this Permittee.

1. PERMIT INFORMATION

The following table summarizes administrative information related to the facility.

Table F-1. Facility Information

WDID	1B8314OHUM
Permittee	City of Rio Dell
Name of Facility	City of Rio Dell Wastewater Treatment Facility
Facility Address	475 Hilltop Drive Rio Dell, CA 95562 Humboldt County
Facility Contact, Title and Phone	Derek Taylor, Wastewater Superintendent, (707) 764-5754
Authorized Person to Sign and Submit Reports	Kyle Knopp, City Manager, (707) 764-3532
Mailing Address	675 Wildwood Ave, Rio Dell, CA 95562
Billing Address	Same as mailing address
Type of Facility	Publicly Owned Treatment Works (POTW)
Major or Minor Facility	Minor
Threat to Water Quality	2
Complexity	A
Pretreatment Program	Not Applicable
Recycling Requirements	Producer, Distributer, User

Facility Permitted Flow	0.40 mgd (average dry weather flow capacity) 1.25 mgd (average monthly wet-weather flow capacity) 2.51 mgd (peak wet weather flow capacity)
Facility Design Flow	0.40 mgd (average dry weather flow capacity) 1.25 mgd (average monthly wet-weather flow capacity) 2.51 mgd (peak wet weather flow capacity)
Watershed	Eel River Hydrologic Unit, Ferndale Hydrologic Subarea
Receiving Water	Lower Eel River
Receiving Water Type	Inland Surface Waters

- 1.1.** The City of Rio Dell (hereinafter Permittee) is the owner and operator of the City of Rio Dell Wastewater Treatment, Recycling and Disposal Facility (hereinafter Facility), a POTW.

For the purposes of this Order, references to the “discharger” or “permittee” in applicable federal and state laws, regulations, plans, or policy are held to be equivalent to references to the Discharger herein.

- 1.2.** The Facility discharges secondary treated wastewater to Lower Eel River, a water of the United States. The Permittee was previously regulated by Order No. R1-2017-0007 and National Pollutant Discharge Elimination System (NPDES) Permit No. CA0022748 adopted on November 1, 2017 and expired on October 1, 2022. Attachment B provides a map of the area around the Facility. Attachment C provides a flow schematic of the Facility.
- 1.3.** The Permittee filed a report of waste discharge (ROWD) and submitted an application for reissuance of its WDRs and NPDES permit on October 28, 2021. The application was deemed complete on September 26, 2022.
- 1.4.** Regulations at 40 C.F.R. section 122.46 limit the duration of NPDES permits to a fixed term not to exceed five years. However, pursuant to California Code of Regulations, title 23, section 2235.4, the terms and conditions of an expired permit are automatically continued pending reissuance of the permit if the Permittee complies with all federal NPDES requirements for continuation of expired permits.

2. FACILITY DESCRIPTION

The Permittee owns and operates a municipal wastewater treatment facility (WWTF) and associated wastewater collection and disposal facilities that serve a population of 3,900 residential users in the Community of Rio Dell. The Facility discharges secondary treated wastewater from Discharge Point 001 to the Lower Eel River, just downstream of Highway 101 Bridge. The Facility also discharges disinfected

secondary-23 recycled water to an irrigation site located northwest of the Facility and west of the southbound Highway 101 Bridge, via Discharge Point 003.

2.1. Description of Wastewater and Biosolids Treatment and Controls

The Facility is located in the Ferndale Hydrologic Subarea of the Lower Eel River Hydrologic Area. The Facility is designed to treat an average dry weather flow of 0.40 mgd, an average monthly wet-weather flow of 1.25 mgd, and a peak wet weather flow of 2.51 mgd.

2.1.1. Collection System

The collection system includes two lift stations with dual pump systems and generators at each. The lift stations convey the wastewater to the headworks building via a North and South mainline.

2.1.2. Wastewater Treatment Facility

The raw sewage flows through a quarter inch opening automatic bar screen and grit removal system then discharges into the influent pump station wet well. A manual bar screen is also available as a backup.

The treatment system consists of headworks, Aero-Mod secondary treatment and solids stabilization system, chlorine disinfection in two chlorine contact tanks, and dechlorination using sodium bisulfite.

2.1.3. Recycled Water

The Permittee produces disinfected secondary recycled water. May 15 through September 30, the recycled water is used to irrigate hay grass and alfalfa at the Irrigation Site. The hay grass and alfalfa are harvested as fodder for beef cattle.

The use of the Irrigation Site commenced in 2014. The irrigation site was previously regulated as discharge to land under NPDES permit No. R1-2017-0007. In April 2022 discharge to Discharge Point 003 was reclassified as recycled water when the Division of Drinking Water approved the Rio Dell Title 22 Engineering Report. Operations at the Irrigation Site will not undergo substantial change due to reclassification.

2.1.4. Biosolids

Solids removed from the wastewater are stored and thickened in two aerated digesters and subsequently dewatered with a belt filter press. The dewatered biosolids are further processed through an indirect sludge dryer, which produces Class A biosolids that the Permittee gives away to a local farmer for use as a soil amendment.

2.2. Discharge Points and Receiving Waters

- 2.2.1. The Facility is located within the Ferndale Hydrologic Subarea of the Lower Eel River Hydrologic Area. The main tributaries to the Lower Eel River are the Van Duzen River, Yager Creek, Larabee Creek, Bull Creek, and Salmon Creek. The upper watershed is mountainous and vegetated by redwood and Douglas fir, interspersed with some hardwoods and meadows. Towards the coast, the river spreads out on a coastal plain where the Salt River joins the Eel River Estuary. The Eel River is designated a Critical Coastal Area by the Statewide Critical Coastal Areas Committee.

The Eel River Watershed Management Area (WMA) encompasses roughly 3,684 square miles in highly erodible soils in the steep coastal mountains of the Region, supporting a variety of water uses including municipal and agricultural supply systems, salmonid fisheries, and recreation. The Eel River WMA is a prime recreational area boasting numerous state and private campgrounds along its length with both water contact and non- contact uses such as boating and swimming. The Eel River is the third largest producer of salmon and steelhead in California and supports a large recreational fishing industry. The erodible soils, steep terrain, and other contributing factors evoke a high level of concern for the anadromous fishery resource. Coho salmon, a native species of the Eel River watershed, were listed as endangered under the federal Endangered Species Act in 1997.

- 2.2.2. During October 1 through May 14, effluent may be discharged to the Lower Eel River at Discharge Point 001 at 40° 29' 48.7" N latitude and 124° 5' 42.2" W longitude. The Lower Eel River is a water of the United States.
- 2.2.3. The City's recycled water use is identified in Title 22, Section 60304(d) as surface irrigation of "fodder and fiber crops and pasture for animals not producing milk for human consumption." The minimum recycled water quality allowed for this use is "undisinfected secondary recycled water" defined in Title 22, Section 60301.900. However, the Permittee discharges disinfected secondary-23 recycled water in practice which is treated to a level above what is required by Title 22.
- 2.2.4. The disinfected recycled water is discharged to a 23-acre irrigation site located northwest of the City and west of the southbound Highway 101 Bridge from May 15 to September 30 at Discharge Point 003 at 40° 30' 47.79" N latitude and 124° 7' 54.54" W longitude. Operation of the Irrigation Site was previously regulated as land disposal under NPDES permit Order No. R1-2017-0007.
- 2.2.5. The Permittee previously discharged treated wastewater to a percolation pond adjacent to the Lower Eel River at Discharge Point 002 during the period of May 15 to September 30. The percolation pond was seasonally constructed within the gravels of the active channel of the Lower Eel River. The Permittee has discontinued discharges to the percolation pond at Discharge Point 002.

This Order does not authorize discharges at Discharge Point 002.

2.3. Summary of Existing Requirements and SMR Data

Effluent limitations contained in Order No. R1-2017-0007 for discharges from Discharge Point 001 (Monitoring Location EFF-001) and representative monitoring data from the term of Order No. R1-2017-0007 are as follows:

Table F-2. Historic Effluent Limitations and Monitoring Data¹

Parameter	Units	Average Monthly	Average Weekly	Maximum Daily	Highest Average Monthly Discharge	Highest Average Weekly Discharge	Highest Daily Discharge
Biochemical Oxygen Demand 5-day @ 20°C (BOD ₅)	mg/L	30	45	--	7.2	11	11
BOD ₅	% Removal	85	--	--	94.4 ²	--	94.4 ²
Total Suspended Solids (TSS)	mg/L	30	45	--	4	8	8
TSS	% Removal	85	--	--	95.9 ²	--	95.9 ²
pH	s.u.	--	--	6.5 – 8.5	--	--	6.40-7.93
Settleable Solids	ml/L	0.1	--	0.2	<0.1	--	<0.1
Chlorine, Total Residual ²	mg/L	0.01	--	0.02	<0.01	--	<0.01
Chlorodibromomethane	µg/L	0.41	--	0.80	2.6	--	2.6
Dichlorobromomethane	µg/L	0.56	--	1.1	16.3	--	16.3
Total Trihalomethanes	µg/L	80	--	--	101.39	--	101.39
Haloacetic Acids	µg/L	60	--	--	109.1	--	109.1

Parameter	Units	Average Monthly	Average Weekly	Maximum Daily	Highest Average Monthly Discharge	Highest Average Weekly Discharge	Highest Daily Discharge
Nitrogen, Total (as N)	mg/L	10	--	--	0.14	--	0.14
Total Coliform Bacteria	MPN/100 mL	23 ³	--	240	66.5 ³	--	540
Acute Toxicity	% Survival	70 ⁴ /90 ⁵	--	--	80 ⁶	--	80

Table Notes

1. Data represented in the table includes all relevant monitoring results from November 1, 2017 – August 30, 2022 at both EFF-001 and EFF-003.
2. Represents the minimum observed monthly average percent removal value.
3. Represents the median concentrations Most Probable Number (MPN) per 100 milliliters, using the bacteriological results of the last 30 calendar days for which analyses have been completed.
4. Minimum for any one bioassay.
5. Median for any three or more consecutive bioassays.
6. Represents the minimum observed percent survival.

2.4. Compliance Summary

2.4.1. The Permittee reported 17 sanitary sewer overflows from its collection system since January of 2018, as follows:

- 2.4.1.1. On January 24, 2018, during a period of heavy rainfall, 19,200 gallons of raw sewage was discharged from a sewer manhole on Painter Street and to the Eel River.
- 2.4.1.2. On March 19, 2018, during a period of heavy rainfall, 86,400 gallons of raw sewage was discharged from a sewer manhole on 780 Rigby Ave. and to the Eel River.
- 2.4.1.3. January 16, 2019, during a period of heavy rainfall, 4,600 gallons of raw sewage was discharged from a sewer manhole on Painter Street and to the Eel River. An estimated 750 gallons were recovered and treated at the Facility.
- 2.4.1.4. February 25, 2019, during a period of heavy rainfall, 28,000 gallons of raw sewage was discharged from a sewer manhole on Painter Street and to the Eel River. An estimated 17,400 gallons were recovered and treated at the Facility.
- 2.4.1.5. February 26, 2019, during a period of heavy rainfall, 2,100 gallons of raw sewage was discharged to land from a sewer manhole on Painter Street. An estimated 2,100 gallons were recovered and treated at the Facility.
- 2.4.1.6. On January 16, 2020, during a period of heavy rainfall, 9,300 gallons of raw sewage was discharged to land from a sewer manhole on Painter Street. An estimated 5,300 gallons were recovered and treated at the Facility.
- 2.4.1.7. On May 5, 2020, grease deposition in the collection system occurred, and 140 gallons of raw sewage was discharged from a sewer manhole on 104 Ogle Avenue and did not impact surface waters.
- 2.4.1.8. On January 26, 2021, during a period of heavy rainfall, 10,065 gallons of raw sewage was discharged from the same sewer manhole at Painter Street and to the Eel River. An estimated 1,000 gallons were recovered and treated at the Facility.
- 2.4.1.9. On January 27, 2021, during a period of heavy rainfall, 28,750 gallons of raw sewage was discharged from the same sewer manhole at Painter Street and to the Eel River. An estimated 4,800 gallons were recovered and treated at the Facility.
- 2.4.1.10. On December 15, 2021, debris and cloth build-up occurred, 525 gallons of raw sewage was discharged from a sewer manhole in the crosswalk on 2nd

Avenue and Davis Street and did not impact surface waters. An estimated 750 gallons were recovered.

- 2.4.1.11. On December 10, 2022, during a period of excessive rainfall, 585 gallons of raw sewage was discharged from the same sewer manhole at Painter Street and to the Eel River. An estimated 1600 gallons were recovered.
- 2.4.1.12. The Permittee identified the location of these spills as a weak point in the collection system because this point in the collection system receives the flow from two lift stations that pump at maximum capacity during periods of sustained and/or heavy rainfall, and the pipe cannot convey the amount of sewage combined with infiltration and inflow that flows through it during these times.
- 2.4.1.13. On December 26, 2022, during a period of excessive rainfall, 40,675 gallons of raw sewage was discharged from the same sewer manhole at Painter Street and to the Eel River. An estimated 26,300 gallons were recovered.
- 2.4.1.14. On December 30, 2022, during a period of excessive rainfall, 5,500 gallons of raw sewage was discharged from the same sewer manhole at Painter Street and to the Eel River. An estimated 24,200 gallons were recovered.
- 2.4.1.15. On January 4, 2023, during a period of heavy rainfall, 150,100 gallons of raw sewage was discharged from the same sewer manhole at Painter Street and to the Eel River. An estimated 25,300 gallons were recovered and treated at the Facility.
- 2.4.1.16. On January 7, 2023, during a period of heavy rainfall, 211,700 gallons of raw sewage was discharged from the same sewer manhole at Painter Street and to the Eel River. An estimated 39,000 gallons were recovered and treated at the Facility.
- 2.4.1.17. On January 13, 2023, during a period of heavy rainfall, 53,550 gallons of raw sewage was discharged from the same sewer manhole at Painter Street and to the Eel River. An estimated 32,000 gallons were recovered and treated at the Facility.
- 2.4.1.18. On January 14, 2023, during a period of heavy rainfall, 117,500 gallons of raw sewage was discharged from the same sewer manhole at Painter Street and to the Eel River. An estimated 49,700 gallons were recovered and treated at the Facility.
- 2.4.2. The Permittee was consistently unable to meet effluent limitations in the permit term of Order No. R1-2017-0007 for dichlorobromomethane, chlorodibromomethane, Total Trihalomethanes, and Haloacetic Acids. In response, the Regional Water Board Executive Officer issued Time Schedule Order (TSO) No. R1-2017-0045 on October 20, 2017 to address noncompliance with final effluent limitations for these constituents. The TSO

also addressed the Permittee's need to reduce excessive infiltration and inflow (I/I) and therefore prevent future SSO incidents.

- 2.4.3. The Permittee has consistently submitted progress reports and made good faith efforts to address the issues outlined in TSO No. R1-2017-0045. The Permittee will not be able to complete the final compliance requirement within the schedule outlined in TSO No. R1-2017-0045. Therefore, the Permittee shall comply with the compliance schedule outlined in Table 5 of this order.
- 2.4.4. The Permittee sent a letter to the State Water Board's Division of Financial Assistance including a request to amend to the scope, schedule, and budget in response to the 6.4 magnitude earthquake on December 20, 2022. The Permittee noted severe damage to the collection system, which caused an increase in I/I. The damage assessment is not yet complete, but the damage caused by natural disaster will require a more extensive collection system upgrade that will require additional time to complete.
- 2.4.5. Due to the unforeseen challenges of the COVID-19 pandemic such as supply chain issues and lockdown safety measures, City of Rio Dell was unable to complete Tasks I through L of addressed in Table 1 of TSO No. R1-2017-0045 and complete tasks C and D of Table 2 of TSO No. R1-2017-0045. Furthermore, due to the damage inflicted by the natural disasters in December 2022 and January 2023 the Regional Water Quality Control Board finds that additional time is necessary to comply with the effluent limitations for dichlorobromomethane, chlorodibromomethane, Total Trihalomethanes, and Haloacetic Acids and to complete upgrades to comply with Order No. 2022-0103-DWQ, Statewide General WDRs for Sanitary Sewer Systems.

2.5. Planned Changes

- 2.5.1. The Permittee is evaluating chloramine disinfection as an alternative to chlorine disinfection to address disinfection byproduct formation in the treated effluent. Initial bench testing was favorable in terms of achieving compliance with effluent limitations for disinfection byproducts, bacteria, and ammonia. The City is now planning to complete a full-scale chloramine test to evaluate disinfection system effectiveness and ammonia concentrations under simulated discharge conditions. The full-scale test will be conducted during 2023 when treated effluent is applied at the Irrigation Site. The Permittee will install ammonia testing equipment, utilize chloramine at predicted dosages, and evaluate impacts to effluent quality. Upon completion the Permittee will report the results of the full-scale study to the Regional Water Board for approval by the Executive Officer, install permanent equipment, and incorporate any operational changes necessary to achieve compliance with effluent limitations for disinfection byproducts.

3. APPLICABLE PLANS, POLICIES, AND REGULATIONS

The requirements contained in this Order are based on the requirements and authorities described in this section.

3.1. Legal Authorities

This Order is issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the U.S. Environmental Protection Agency (U.S. EPA) and chapter 5.5, division 7 of the California Water Code (commencing with section 13370). It shall serve as a NPDES permit authorizing the Permittee to discharge into waters of the United States at the discharge location described in Table 1, subject to the requirements in this Order. This Order also serves as WDRs pursuant to article 4, chapter 4, division 7 of the Water Code (commencing with section 13260) and water recycling requirements pursuant to article 4, chapter 4, division 7 of the Water Code (commencing with section 13500).

3.2. California Environmental Quality Act (CEQA)

Under Water Code section 13389, this action to adopt an NPDES permit is exempt from the provisions of Chapter 3 of CEQA, (commencing with section 21100) of Division 13 of the Public Resources Code.

The Water Recycling Requirements of this Order are exempt from CEQA pursuant to California Code of Regulations, title 14, section 15301, (existing facilities) as this Order does not authorize an expansion or change of use of recycled water.

3.3. State and Federal Laws, Regulations, Policies, and Plans

3.3.1. Water Quality Control Plan

The Regional Water Board adopted a Water Quality Control Plan for the North Coast Region (hereinafter Basin Plan) that designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve those objectives for all waters addressed through the plan. In addition, the Basin Plan implements State Water Resources Control Board (State Water Board) Resolution No. 88-63, which establishes State policy that all waters, with certain exceptions, should be considered suitable or potentially suitable for municipal or domestic supply (MUN). Beneficial uses applicable to Lower Eel River are summarized in Table F-3, as follows:

Table F-3. Basin Plan Beneficial Uses

Discharge Point	Receiving Water Name	Beneficial Use(s)
001	Lower Eel River within the Ferndale Hydrologic Subarea of the Lower Eel River Hydrologic Area	<u>Existing:</u> Municipal and domestic supply (MUN); Agricultural supply (AGR); Industrial service supply (IND); Groundwater recharge (GWR); Freshwater replenishment (FRSH); Navigation (NAV); Water contact recreation (REC-1); Non-contact water recreation (REC-2); Commercial and Sport Fishing (COMM); Cold freshwater habitat (COLD); Wildlife habitat (WILD); Rare, threatened, or endangered species (RARE); Migration of aquatic organisms (MIGR); and Spawning, reproduction, and/or early development (SPWN); Shellfish Harvesting (SHELL); Native American Culture (CUL). <u>Potential:</u> Industrial process supply (PRO); Marine Habitat (MAR); Hydropower generation (POW), and Aquaculture (AQUA).
003	Groundwater	<u>Existing:</u> Municipal and domestic supply (MUN); Agricultural supply (AGR); Industrial service supply (IND); and Native American culture (CUL). <u>Potential:</u> Industrial process supply (PRO); and Aquaculture (AQUA).

In addition to the beneficial uses set out in the Basin Plan, there are several implementation plans that include actions intended to meet water quality objectives and protect beneficial uses of the North Coast Basin. For the Eel River and its tributaries, no point source waste discharges are allowed during the period of May 15 through September 30, and for all other periods the receiving stream's flow must be at least 100 times greater than the waste flow

unless an exception to the requirement is granted by the Regional Water Board.

Requirements of this Order implement the Basin Plan.

3.3.2. National Toxics Rule (NTR) and California Toxics Rule (CTR)

U.S. EPA adopted the NTR on December 22, 1992, and later amended it on May 4, 1995 and November 9, 1999. About forty criteria in the NTR applied in California. On May 18, 2000, U.S. EPA adopted the CTR. The CTR promulgated new toxics criteria for California and, in addition, incorporated the previously adopted NTR criteria that were applicable in the state. The CTR was amended on February 13, 2001. These rules contain federal water quality criteria for priority pollutants.

3.3.3. State Implementation Policy

On March 2, 2000, the State Water Board adopted the Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California (State Implementation Policy or SIP). The SIP became effective on April 28, 2000, with respect to the priority pollutant criteria promulgated for California by the U.S. EPA through the NTR and to the priority pollutant objectives established by the Regional Water Board in the Basin Plan. The SIP became effective on May 18, 2000, with respect to the priority pollutant criteria promulgated by the U.S. EPA through the CTR. The State Water Board adopted amendments to the SIP on February 24, 2005, that became effective on July 13, 2005. The SIP establishes implementation provisions for priority pollutant criteria and objectives and provisions for chronic toxicity control. Requirements of this Order implement the SIP.

3.3.4. Domestic Water Quality

In compliance with Water Code section 106.3, it is the policy of the State of California that every human being has the right to safe, clean, affordable, and accessible water adequate for human consumption, cooking, and sanitary purposes. This Order promotes that policy by requiring discharges to meet maximum contaminant levels implemented by the Basin Plan that are designed to protect human health and ensure that water is safe for domestic use.

3.3.5. Compliance Schedules and Interim Requirements.

The State Water Board adopted Resolution No. 2008-0025 on April 15, 2008, *titled Policy for Compliance Schedules in National Pollutant Discharge Elimination System Permits*, (Compliance Schedule Policy) which includes compliance schedule policies for pollutants that are not addressed by the SIP. This Policy became effective on August 27, 2008.

The Order includes a compliance schedule for dichlorobromomethane, chlorodibromomethane, total trihalomethanes, and Haloacetic Acids effluent limitations to comply with the Basin Plan (Chapter 3) for the protection of public water supplies at title 22 of the CCR, sections 64431 (Inorganic Chemicals) and 64444 (Organic Chemicals). The compliance schedule is in accordance with the Compliance Schedule Policy as further discussed in section 6.2.11 of this Fact Sheet.

3.3.6. Antidegradation Policy

Federal regulation 40 C.F.R. section 131.12 requires that the state water quality standards include an antidegradation policy consistent with the federal policy. The State Water Board established California's antidegradation policy in State Water Board Resolution 68-16 ("Statement of Policy with Respect to Maintaining High Quality of Waters in California"). Resolution 68-16 is deemed to incorporate the federal antidegradation policy where the federal policy applies under federal law. Resolution 68-16 requires that existing water quality be maintained unless degradation is justified based on specific findings. The Regional Water Board's Basin Plan implements, and incorporates by reference, both the State and federal antidegradation policies. The permitted discharge must be consistent with the antidegradation provision of 40 C.F.R. section 131.12 and State Water Board Resolution 68-16.

3.3.7. Anti-Backsliding Requirements

Sections 402(o) and 303(d)(4) of the CWA and federal regulations at 40 C.F.R. section 122.44(l) restrict backsliding in NPDES permits. These anti-backsliding provisions require that effluent limitations in a reissued permit must be as stringent as those in the previous permit, with some exceptions in which limitations may be relaxed. As discussed in detail in section 4.4.1 of this Fact Sheet, removal or relaxation of effluent limitations is consistent with the anti-backsliding requirements of the CWA and federal regulations.

3.3.8. Endangered Species Act Requirements

This Order does not authorize any act that results in the taking of a threatened or endangered species or any act that is now prohibited, or becomes prohibited in the future, under either the California Endangered Species Act (Fish and Game Code, §§ 2050 to 2097) or the Federal Endangered Species Act (16 U.S.C.A. §§ 151 to 1544). This Order requires compliance with effluent limitations, receiving water limitations, and other requirements to protect the beneficial uses of waters of the state. The Permittee is responsible for meeting all requirements of the applicable Endangered Species Act.

3.3.9. Sewage Sludge and Biosolids

This Order does not authorize any act that results in violation of requirements administered by U.S. EPA to implement 40 C.F.R. Part 503, Standards for the

Use or Disposal of Sewage Sludge. These standards regulate the final use or disposal of sewage sludge that is generated during the treatment of domestic sewage in a municipal wastewater treatment facility. The Permittee is responsible for meeting all applicable requirements of 40 C.F.R. Part 503 that are under U.S. EPA's enforcement authority.

3.4. Impaired Water Bodies on the CWA section 303(d) List

Section 303(d) of the federal CWA requires states to identify waterbodies that do not meet water quality standards and are not supporting their beneficial uses after implementation of technology-based effluent limitations on point sources. Each state must submit an updated list, the 303(d) List of Impaired Waterbodies every two years.

In addition to identifying the waterbodies that are not supporting beneficial uses, the 303(d) list also identifies the pollutant or stressor causing impairment and establishes a schedule for developing a control plan to address the impairment. The CWA requires development of a total maximum daily load (TMDL) or alternate program of implementation for each 303(d) listed pollutant and water body to remedy the impairment. TMDLs establish the maximum quantity of a given pollutant that can be added to a water body from all sources without exceeding the applicable water quality standard for that pollutant and determine wasteload allocations (the portion of a TMDL allocated to existing and future point sources) and load allocations (the portion of a TMDL attributed to existing and future nonpoint sources).

On June 9, 2021, the U.S. EPA provided final approval of the 2018 303(d) List of Impaired Water Bodies prepared by the state. The list identifies the entire Eel River watershed as impaired by sedimentation/siltation and temperature, and the Lower Eel River as impaired by dissolved oxygen and aluminum. Pursuant to CWA section 303(d), the Regional Water Board will develop TMDLs to address the impairments, which will be implemented through various programs, including through provisions of NPDES permits.

Regarding temperature, on December 18, 2007, the North Coast Regional Water Board adopted the Lower Eel River TMDL for temperature and sediment that concludes that most sources of heat in the Lower Eel River are from diffuse, nonpoint sources and result from such factors as removal of stream shade, longer travel time, changes in timing and volume of natural stream flow due to water diversions and impoundments, and increased sediment loads that cause widening of streams. As the critical time period for temperature is in the summer, the TMDL was established for that critical time period, which is also the time period when point source discharges from the Facility are prohibited. The TMDL concludes that, because of the summer discharge prohibition, area facilities such as the Facility do not contribute to temperature loadings to the Lower Eel River Watershed during critical periods, and therefore, the TMDL establishes a "zero" waste load allocation (WLA) to mean that, as long as the Permittee adheres to the

summer discharge prohibition, it will be in compliance with the approved TMDL for temperature.

With regard to sediment, the TMDL establishes a maximum loading of 125 percent of the natural sediment loading for the watershed and further defines that loading rate as 2.5 tons of sediment per square mile of watershed per day on a long-term basis. The TMDL found that nonpoint sources were primarily responsible for excessive sediment loadings to the Lower Eel River, and the TMDL established WLAs for wastewater treatment facilities at levels corresponding to existing permit limitations for suspended and settleable solids. In order to be protective of the Basin Plan water quality objectives for sediment in the Lower Eel River Watershed, this Order retains effluent limitations for TSS and settleable solids from Order No. R1-2017-0007.

Regarding dissolved oxygen, the Permittee monitors the Eel River for dissolved oxygen immediately upstream and downstream of its discharge to the Eel River. Dissolved oxygen monitoring data collected between November 2017 and August 2022 reveal that concentrations range between 9.5 and 11.5 mg/L and that there is little change between the upstream and downstream monitoring data. Regional Water Board therefore concludes that the Permittee's discharge is not expected to contribute to the dissolved oxygen impairment.

Regarding aluminum, the Permittee does not use any aluminum-containing chemicals in its wastewater treatment process, therefore, the Regional Water Board does not anticipate that the Permittee's discharge will contribute to the aluminum impairment. Nevertheless, to confirm this assertion, this Order establishes effluent and receiving water monitoring for aluminum to inform Regional Water Board staff of the reasonable potential for the Permittee to exceed water quality objectives when discharging to the South Fork Eel River.

3.5. Other Plans, Policies and Regulations

- 3.5.1. On December 6, 2022, the State Water Board adopted State Water Board Order No. 2022-0103-DWQ, Statewide General WDRs for Sanitary Sewer Systems. Order No. 2022-0103-DWQ requires that all public agencies that currently own or operate sanitary sewer systems apply for coverage under the General WDRs. The Permittee applied for coverage and is subject to the requirements of Order No. 2022-0103-DWQ and any future revisions thereto for operation of its wastewater collection system. Coverage under the State Water Board Water Quality Order No. 2014-0057-DWQ, NPDES General Permit No. CAS000001, General Permit for Storm Water Discharges Associated with Industrial Activities (Industrial Storm Water General Permit) is not required because the design flow of the Facility is less than 1 mgd.
- 3.5.2. In 1996, the State Water Board and the California Department of Health Services (now State Water Board Division of Drinking Water or DDW) set forth principles, procedures, and agreements to which the agencies committed

themselves relative to the use of recycled water in California, in a document titled Memorandum of Agreement between the Department of Health Services and the State Water Resources Control Board on the Use of Reclaimed Water (MOA). This Order is consistent with the MOA.

- 3.5.3. On February 3, 2009, the State Water Board adopted Resolution No. 2009-0011, Adoption of a Policy for Water Quality Control for Recycled Water (Recycled Water Policy) (Revised December 11, 2018, effective April 8, 2019) for the purpose of increasing the use of recycled water from municipal wastewater sources in a manner that implements state and federal water quality laws. The Recycled Water Policy provides direction to the regional water boards regarding the appropriate criteria to be used in issuing permits for recycled water projects and describes permitting criteria intended to streamline, and provide consistency for, the permitting of the vast majority of recycled water projects. Pertinent provisions and requirements of the Policy have been incorporated into this Order to address conditions specific to the Permittee's plan to implement water recycling.

The Recycled Water Policy recognizes the fact that some groundwater basins in the state contain salts and nutrients that exceed or threaten to exceed water quality objectives in the applicable Basin Plans, and that not all Basin Plans include adequate implementation procedures for achieving or ensuring compliance with the water quality objectives for salt or nutrients. The Recycled Water Policy further recognizes that these conditions can be caused by natural soils/conditions, discharges of waste, irrigation using surface water, groundwater or recycled water, and water supply augmentation using surface or recycled water, and that regulation of recycled water alone will not address these conditions. It is the intent of the Recycled Water Policy that salts and nutrients from all sources be managed on a basin-wide or watershed-wide basis in a manner that ensures attainment of water quality objectives and protection of beneficial uses. The Recycled Water Policy finds that the appropriate way to address salt and nutrient issues is through the development of regional or subregional Salt and Nutrient Management Plans (SNMPs) rather than through imposing requirements solely on individual recycled water projects. This Order is consistent with the requirements of the Recycled Water Policy to implement an SNMP. This Order is consistent with the requirements of the Recycled Water Policy to implement a SNMP. The Recycled Water Policy currently requires monitoring for priority pollutants annually. This Order implements this requirement through the annual CTR priority pollutant monitoring requirement in the MRP that is required of the Permittee pursuant to the SIP.

- 3.5.4. On July 22, 2004, the State Water Board adopted State Water Board Order No. 2004-0012-DWQ, General Waste Discharge Requirements for the Discharge of Biosolids to Land for Use as a Soil Amendment in Agricultural, Silvicultural, Horticultural, and Land Reclamation Activities. The Order requires the Permittee to obtain coverage under Order No. 2004-0012-DWQ prior to any

removal of biosolids from the Facility that will be land disposed on property owned or controlled by the Permittee.

- 3.5.5. When applicable, state law requires dischargers to file a petition with the State Water Board, Division of Water Rights, and receive approval for any change in the point of discharge, place of use, or purpose of use of treated wastewater that decreases the flow in any portion of the watercourse. The State Water Board retains separate jurisdictional authority to enforce any applicable requirements under Water Code section 1211. This is not an NPDES permit requirement.

4. RATIONALE FOR EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

The CWA requires point source dischargers to control the amount of conventional, non-conventional, and toxic pollutants that are discharged into the waters of the United States. The control of pollutants discharged is established through effluent limitations and other requirements in NPDES permits.

There are two principal bases for effluent limitations in the Code of Federal Regulations: 40 C.F.R. section 122.44(a) requires that permits include applicable technology-based limitations and standards; and 40 C.F.R. section 122.44(d) requires that permits include water quality-based effluent limitations (WQBELs) to attain and maintain applicable numeric and narrative water quality criteria to protect the beneficial uses of the receiving water where a reasonable potential to exceed those criteria exists.

4.1. Discharge Prohibitions

- 4.1.1. **Discharge Prohibition 3.1.** The discharge of any waste not disclosed by the Permittee or not within the reasonable contemplation of the Regional Water Board is prohibited.

This prohibition has been retained from Order No. R1-2017-0007 and is based on the Basin Plan and State Water Board Order No. WQO 2002-0012 regarding the petition of WDRs Order No. 01-072 for the East Bay Municipal Utility District and Bay Area Clean Water Agencies. In State Water Board Order No. WQO 2002-0012, the State Water Board found that this prohibition is acceptable in Orders but should be interpreted to apply only to constituents that are either not disclosed by the Permittees, or are not reasonably anticipated to be present in the discharge but have not been disclosed by the Permittees. It specifically does not apply to constituents in the discharge that do not have “reasonable potential” to exceed water quality objectives.

The State Water Board has stated that the only pollutants not covered by this prohibition are those which were “*disclosed to the permitting authority and...can be reasonably contemplated.*” [In re the Petition of East Bay Municipal Utilities District et al., (State Water Board, 2002) Order No. WQO 2002-0012, p. 24]. In

that Order, the State Water Board cited a case which held the Permittee is liable for the discharge of pollutants “*not within the reasonable contemplation of the permitting authority...whether spills or otherwise...*” [*Piney Run Preservation Assn. v. County Commissioners of Carroll County, Maryland* (4th Cir. 2001) 268 F. 3d 255, 268.] Thus, the State Water Board authority provides that, to be permissible, the constituent discharged (1) must have been disclosed by the Permittees and (2) can be reasonably contemplated by the Regional Water Board.

- 4.1.2. **Discharge Prohibition 3.2.** Creation of pollution, contamination, or nuisance, as defined by section 13050 of the Water Code is prohibited.

This prohibition has been retained from Order No. R1-2017-0007 and is based on section 13050 of the Water Code and section 5411 of the California Health and Safety Code.

- 4.1.3. **Discharge Prohibition 3.3.** The discharge of sludge or digester supernatant is prohibited, except as authorized under section 6.3.5.3 of this Order (Sludge Disposal and Handling Requirements).

This prohibition has been retained from Order No. R1-2017-0007 and is based on restrictions on the disposal of sewage sludge found in federal regulations [40 C.F.R. part 503 (Biosolids), part 527, and part 258] and title 27 of the CCR.

- 4.1.4. **Discharge Prohibition 3.4.** The discharge or recycling use of untreated or partially treated waste (receiving a lower level of treatment than described in section 2.1 of the Fact Sheet) from anywhere within the collection, treatment, or disposal systems is prohibited, except as provided for in Attachment D, Standard Provisions 1.7 (Bypass) and 1.8 (Upset).

This prohibition has been retained from Order No. R1-2017-0007 and is based on the Basin Plan to protect the beneficial uses of the receiving water from unpermitted discharges, and the intent of the Water Code sections 13260 through 13264 relating to the discharge of waste to waters of the state without filing for and being issued an Order. This prohibition applies to spills not related to sanitary sewer overflows (SSOs) and other unauthorized discharges of wastewater within the collection, treatment, and disposal facilities. The discharge of untreated or partially treated wastewater from the collection, treatment, or disposal facility represents an unauthorized bypass pursuant to 40 C.F.R. section 122.41(m) or an unauthorized discharge, which poses a threat to human health and/or aquatic life, and therefore is explicitly prohibited by this Order.

- 4.1.5. **Discharge Prohibition 3.5.** The discharge of waste to land that is not owned by the Permittee, governed by City ordinance, or under agreement to use by the Permittee, or for which the Permittee has explicitly permitted such use, is prohibited.

This prohibition is retained from Order No. R1-2017-0007. Land used for the application of wastewater must be owned by the Permittee or be under the control of the Permittee by contract (user agreement) so that the Permittee maintains a means for ultimate disposal of treated wastewater.

- 4.1.6. **Discharge Prohibition 3.6.** The discharge of recycled wastewater to any point not addressed in a DDW-accepted Title 22 Recycled Water Engineering Report is prohibited.

This prohibition is newly established by this Order and is necessary to ensure that the Permittee only discharges recycled water in accordance with WDRs. It is based on sections 301 and 402 of the federal CWA and section 13263 of the Water Code.

- 4.1.7. **Discharge Prohibition 3.7.** The discharge of waste at any point not described in Finding 2.2 of the Fact Sheet or authorized by a permit issued by the State Water Board or another Regional Water Board is prohibited.

This prohibition has been retained from Order No. R1-2017-0007. This prohibition is a general prohibition that allows the Permittee to discharge waste only in accordance with WDRs. It is based on sections 301 and 402 of the federal CWA and section 13263 of the Water Code.

- 4.1.8. **Discharge Prohibition 3.8.** The average dry weather flow of waste through the Facility shall not exceed 0.40 mgd. The average monthly wet weather flow of waste through the Facility shall not exceed 1.25 mgd. The peak daily wet weather flow of waste through the Facility shall not exceed 2.51 mgd. Compliance with this prohibition shall be determined as defined in sections 7.10 and 7.11 of this Order.

The average dry weather flow prohibition is retained from Order No. R1-2017-0007 and is based on the engineering design of the Facility. The peak daily wet weather flow prohibition is also retained from Order No. R1-2017-0007 and based on the peak wet weather treatment capacity of the Facility as stated in the ROWD.

- 4.1.9. **Discharge Prohibition 3.9.** The discharge of waste to the Lower Eel River and its tributaries is prohibited during the period from May 15 through September 30 of each year.

This prohibition is retained from Order No. R1-2017-0007 and is required by the Basin Plan. The Basin Plan prohibits discharges to the Lower Eel River and its tributaries during the period May 15 through September 30 (Chapter 4, Waste Discharge prohibitions for the North Coastal Basin).

- 4.1.10. **Discharge Prohibition 3.10.** During the period from October 1 through May 14, discharges of treated wastewater to the Lower Eel River shall not exceed one percent of the flow of the Lower Eel River, as measured at United States

Geological Survey (USGS) Gage No. 11477000 in the Lower Eel River near Scotia. For the purposes of this Order, compliance with this discharge prohibition shall be determined as follows:

The discharge of secondary treated wastewater shall be adjusted at least once daily to avoid exceeding, to the extent practicable, one percent of the most recent daily flow measurement of the Lower Eel River. Daily flow shall be based on flow meter comparisons reasonably read between the hours of 12:01 am and 12:00 midnight.

In no case shall the total volume of secondary treated wastewater discharged in a calendar month exceed one percent of the total volume of the Lower Eel River's flow that occurs in the same calendar month, as measured per Section 3.11. At the beginning of the discharge season, the monthly flow volume comparison shall be based on the date when the discharge commenced to the end of the calendar month. At the end of the discharge season, the monthly flow volume comparison shall be based on the first day of the calendar month to the date when the discharge ceases for the season.

This prohibition has been retained from Order No. R1-2017-0007 and is required by the Basin Plan (Chapter 4, North Coastal Basin Discharge Prohibition No. 3). The Basin Plan prohibits discharges to the Eel River and its tributaries when the waste discharge flow is greater than one percent of the receiving water's flow. Basin Plan Prohibition No. 3 does not specify how compliance to the one-percent flow requirement will be determined. This prohibition, set forth in Provision 3.11 of this Order, specifies that the discharge may comply with the one percent requirement as a monthly average for the surface water discharge season if USGS Station 11477000 is read at least once daily, and the discharge flow rate shall not be set for greater than one percent of the flow of the river at the time of the daily reading.

- 4.1.12. **Discharge Prohibition 3.11.** The discharge of any radiological, chemical, or biological warfare agent into waters of the state under Water Code section 13375.

This prohibition is retained from Order No. R1-2017-0007 and is based on the discharge prohibitions contained in and Water Code section 13375.

- 4.1.13. **Discharge Prohibition 3.12.** The acceptance of septage to a location other than an approved septage receiving station and in accordance with a septage management program approved by the Regional Water Board Executive Officer is prohibited.

This prohibition is newly established by this Order and is necessary to ensure that septage is not accepted in the absence of a septage management program to ensure that pollutants associated with domestic septage do not pass through or interfere with the operation or performance of the Facility.

4.2. Technology-Based Effluent Limitations

4.2.1. Scope and Authority

- 4.2.1.1. Section 301(b) of the CWA and implementing U.S. EPA permit regulations at 40 C.F.R. section 122.44 require that permits include conditions meeting applicable technology-based requirements at a minimum, and any more stringent effluent limitations necessary to meet applicable water quality standards. The discharge authorized by this Order must meet minimum federal technology-based requirements based on Secondary Treatment Standards at 40 C.F.R. part 133 and Best Professional Judgment (BPJ) in accordance with 40 C.F.R. section 125.3.

Regulations promulgated in 40 C.F.R. section 125.3(a)(1) require technology-based effluent limitations for municipal dischargers to be placed in NPDES permits based on Secondary Treatment Standards or Equivalent to Secondary Treatment Standards.

The Federal Water Pollution Control Act Amendments of 1972 (PL 92-500) established the minimum performance requirements for POTWs [defined in section 304(d)(1)]. Section 301(b)(1)(B) of that Act requires that such treatment works must, as a minimum, meet effluent limitations based on secondary treatment as defined by the U.S. EPA Administrator.

Based on this statutory requirement, U.S. EPA developed secondary treatment regulations, which are specified in 40 C.F.R. part 133. These technology-based regulations apply to all municipal wastewater treatment facilities and identify the minimum level of effluent quality attainable by secondary treatment in terms of BOD₅, TSS, and pH, as follows:

4.2.1.1.1. BOD₅ and TSS

- 4.2.1.1.1.1. The 30-day average shall not exceed 30 mg/L.
- 4.2.1.1.1.2. The 7-day average shall not exceed 45 mg/L.
- 4.2.1.1.1.3. The 30-day average percent removal shall not be less than 85%.

4.2.1.1.2. pH

- 4.2.1.1.2.1. The pH shall be maintained within the limits of 6.0 to 9.0.
- 4.2.1.1.2.2. The more stringent effluent limitation for pH of 6.5 to 8.5 required by this Order is necessary to meet the water quality objective for hydrogen ion concentration (pH) is contained in the Basin Plan, Table 3-1.
- 4.2.1.2. In addition, 40 C.F.R. section 122.45(f) requires the establishment of mass-based effluent limitations for all pollutants limited in Orders, except for 1) pH,

temperature, radiation, or other pollutants which cannot be appropriately expressed by mass, 2) when applicable standards and limitations are expressed in terms of other units of measure, and 3) where the permit limitation is established on a case-by-case basis under 40 C.F.R. section 125.3 and limitations expressed in terms of mass are infeasible because the mass of the pollutant discharged cannot be related to a measure of operation, and permit conditions ensure that dilution will not be used as a substitute for treatment.

Technology-based effluent limitations may be set on a case-by-case basis under section 402(a)(1) of the CWA to the extent that EPA-promulgated effluent limitations are inapplicable based upon the available information and unique factors related to the applicant. A combination of EPA-promulgated effluent limitations and effluent limitations developed under a case-by-case basis scenario may be applied to carry out the provisions of the CWA. “Best Practicable Control Technology” (BPT) requirements may be established by a permitting authority on a case-by-case basis considering the appropriate factors listed at 40 C.F.R. section 125.3(d)(1). Factors to be considered for BPT requirements include:

- 4.2.1.2.1. The total cost of application of the technology in relation to the effluent reduction benefits to be achieved from such application;
- 4.2.1.2.2. The age of equipment and facilities involved;
- 4.2.1.2.3. The process employed;
- 4.2.1.2.4. The engineering aspects of the application of various types of control techniques;
- 4.2.1.2.5. Process changes; and
- 4.2.1.2.6. Non-water quality environmental impacts (including energy requirements).

4.2.2. **Applicable Technology-Based Effluent Limitations**

The effluent limitations in this Order for BOD₅, TSS, and pH not only meet the technology-based requirements for secondary treatment set forth in section 133.102, but they also are required to meet the water quality-based requirements set forth in the Basin Plan.

- 4.2.2.1. **BOD₅ and TSS.** As described above, the secondary treatment standards at 40 C.F.R. part 133 establish the minimum level of effluent quality attainable by secondary treatment in terms of BOD₅, TSS, and pH. Numeric effluent limitations for BOD- and TSS, including the percent removal requirements, are retained from Order No. R1-2017-0007 and reflect the secondary treatment standards at 40 C.F.R. part 133.

4.2.2.2. **pH.** The secondary treatment regulations at 40 C.F.R. part 133 require that pH be maintained between 6.0 and 9.0 standard units. This technology based effluent limitation is applied to discharges from the treatment system at Discharge Point 001. Note that a more stringent effluent limitation range of 6.5 – 8.5 for pH is required to meet the water quality objective for hydrogen ion concentration (pH) in the Lower Eel River contained in Basin Plan, Table 3-1.

4.2.2.3. **Mass-Based Effluent Limitations.** Federal regulations at 40 C.F.R. section 122.45(f) require that, except under certain conditions, all permit limits, standards, or prohibitions be expressed in terms of mass units. Among the conditions exempting the application of mass-based limitations is 40 C.F.R. section 122.45(f)(1)(i), which states “for pH, temperature, and radiation, or other pollutants which cannot appropriately be expressed by mass” and 40 C.F.R. section 122.45(f)(1)(ii), which states “when applicable standards and limitations are expressed in terms of other units of measurement.”

This Order does not include mass-based effluent limitations for the following pollutants pursuant to the exceptions in 40 C.F.R. section 122.45(f)(1)(i) and (ii):

- 4.2.2.3.1. BOD₅ and TSS, because these two parameters are expressed in terms of concentration and percent removal; and
- 4.2.2.3.2. pH and settleable solids, because these parameters cannot appropriately be expressed by mass.

Table F-4. Summary of Technology Based Effluent Limitations – Discharge Point 001 (Monitoring Location EFF-001)

Parameter	Unit	Average Monthly	Average Weekly	Instantaneous Minimum	Instantaneous Maximum
Biochemical Oxygen Demand 5-day @20°C (BOD ₅)	mg/L	30	45	--	--
BOD ₅	% Removal	85	--	--	--
Total Suspended Solids (TSS)	mg/L	30	45	--	--
TSS	% Removal	85	--	--	--
pH	standard units	--	--	6.5 ¹	8.5 ¹

Table Notes

1. This Order includes final instantaneous minimum and maximum effluent limitations for pH of 6.5 and 8.5, respectively, based on the more stringent water quality criteria.

4.3. Water Quality-Based Effluent Limitations (WQBELs)

4.3.1. Scope and Authority

CWA Section 301(b) and 40 C.F.R. section 122.44(d) require that permits include limitations more stringent than applicable federal technology-based requirements where necessary to achieve applicable water quality standards. This Order contains requirements, expressed as a technology equivalence requirement, more stringent than secondary treatment requirements that are necessary to meet applicable water quality standards. The rationale for these requirements, which consist of secondary treatment, is discussed in section 4.3.3 of this Fact Sheet.

Section 122.44(d)(1)(i) of 40 C.F.R. requires that permits include effluent limitations for all pollutants that are or may be discharged at levels that have the reasonable potential to cause or contribute to an exceedance of a water quality standard, including numeric and narrative objectives within a standard. Where reasonable potential has been established for a pollutant, but there is no numeric criterion or objective for the pollutant, water quality-based effluent limitations (WQBELs) must be established using: (1) U.S. EPA criteria guidance under CWA section 304(a), supplemented where necessary by other relevant information; (2) an indicator parameter for the pollutant of concern; or (3) a calculated numeric water quality criterion, such as a proposed state criterion or policy interpreting the state's narrative criterion, supplemented with other relevant information, as provided in section 122.44(d)(1)(vi).

The process for determining reasonable potential and calculating WQBELs when necessary is intended to protect the designated uses of the receiving water as specified in the Basin Plan, and achieve applicable water quality objectives and criteria that are contained in other state plans and policies, or any applicable water quality criteria contained in the CTR and NTR.

4.3.2. Applicable Beneficial Uses and Water Quality Criteria and Objectives

4.3.2.1. **Beneficial Uses.** Beneficial use designations for receiving waters for discharges from the Facility are presented in section 3.3.1 of this Fact Sheet.

4.3.2.2. **Basin Plan Water Quality Objectives.** The Basin Plan contains narrative objectives for color, tastes and odors, floating material, suspended material, settleable material, oil and grease, biostimulatory substances, sediment,

turbidity, pH, dissolved oxygen, bacteria, temperature, toxicity, pesticides, chemical constituents, and radioactivity that apply to inland surface waters, enclosed bays, and estuaries, including the Eel River and its tributaries. For waters designated for use as domestic or municipal supply (MUN), the Basin Plan establishes as applicable water quality criteria the MCLs established by the State Water Board, Division of Drinking Water (DDW) for the protection of public water supplies at title 22 of the CCR section 64431 (Inorganic Chemicals) and section 64444 (Organic Chemicals).

- 4.3.2.3. **SIP, CTR, and NTR.** Water quality criteria and objectives applicable to this receiving water are established by the CTR, established by the U.S. EPA at 40 C.F.R. section 131.38; and the NTR, established by the U.S. EPA at 40 C.F.R. section 131.36. Criteria for most of the 126 priority pollutants are contained within the CTR and the NTR.

The SIP, which is described in section 3.3.3 of this Fact Sheet, includes procedures for determining the need for, and the calculation of, WQBELs and requires Permittees to submit data sufficient to do so.

At title 22, division 4, chapter 15 of the CCR, DDW has established MCLs for certain pollutants for the protection of drinking water. Chapter 3 of the Basin Plan establishes these MCLs as water quality objectives applicable to receiving waters with the beneficial use designation of municipal and domestic supply.

Aquatic life freshwater and saltwater criteria are identified in the CTR and NTR as criterion maximum concentrations (CMC) and criterion continuous concentrations (CCC). The CTR defines the CMC as the highest concentration of a pollutant to which aquatic life can be exposed for a short period of time without deleterious effects and the CCC as the highest concentration of a pollutant to which aquatic life can be exposed for an extended period of time (4 days) without deleterious effects. The CMC is used to calculate an acute or 1-hour average numeric effluent limitation and the CCC is used to calculate a chronic or 4-day average numeric effluent limitation. Aquatic life freshwater criteria were used for the Reasonable Potential Analysis

Human health criteria are further identified as “water and organisms” and “organisms only”. “Water and organism” criteria are designed to address risks to human health from consumption of drinking water, fish and shellfish. The criteria from the “water and organisms” column of CTR were used for the RPA because the Basin Plan identifies that the receiving water, the Eel River, has the beneficial use designation of municipal and domestic supply.

4.3.3. **Determining the Need for WQBELs**

NPDES regulations at 40 C.F.R. section 122.44(d) require effluent limitations to control all pollutants, which are or may be discharged at a level that will cause, have the reasonable potential to cause, or contribute to an excursion above any state water quality standard.

For WQBELs for toxic pollutants, Section 5.2.3 of the EPA Technical Support Document for Water Quality-based Toxic Controls states “in lieu of an Average Weekly Limit (AWL) for POTWs, EPA recommends establishing a Maximum Daily Limit (MDL) (or a maximum test result for chronic toxicity) for toxic pollutants and pollutant parameters in water quality permitting. This is appropriate for at least two reasons. First, the basis for the 7-day average for POTWs derives from the secondary treatment requirements. This basis is not related to the need for assuring achievement of water quality standards. Second, a 7-day average, which could comprise up to seven or more daily samples, could average out peak toxic concentrations and therefore the discharge’s potential for causing acute toxic effects would be missed. A MDL, which is measured by a grab sample, would be toxicologically protective of potential acute toxicity impacts.”

Section 1.4 of the SIP states that maximum daily effluent limitations (MDEL) shall be used for POTWs in place of average weekly effluent limitations (AWEL) for WQBELs. The SIP procedure of calculating an AMEL and an MDEL applies to all CTR pollutants, both those that are for protection of aquatic life and those that are for the protection of human health.

The RPA for discharges to the Lower Eel River at Discharge Point 001 was conducted as follows.

4.3.3.1. **Non-Priority Pollutants**

- 4.3.3.1.1. **pH.** The effluent limitation for pH of 6.5 to 8.5 is retained from Order No. R1-2017-0007. This limitation is based on the water quality objective for all surface waters established in chapter 3, Table 3-1 of the Basin Plan. Federal technology-based requirements prescribed in 40 C.F.R. part 133 are not sufficient to meet these Basin Plan water quality standards.
- 4.3.3.1.2. **Chlorine Residual.** The Basin Plan establishes a narrative water quality objective for toxicity which states “[a]ll waters shall be maintained free of toxic substances in concentrations that are toxic to, or that produce detrimental physiological responses in human, plant, animal, or aquatic life.” The Regional Water Board considers any chlorinated discharge as having the reasonable potential to cause or contribute to exceedances of this water quality objective for toxicity, and therefore this Order includes effluent limitations for chlorine. U.S. EPA has established the following criteria for chlorine-produced oxidants for protection of freshwater aquatic life in

Quality Criteria for Water 1986 (The Gold Book, 1986, EPA 440/5-86-001). The chronic and acute criterion established for residual chlorine are 0.011 mg/L and 0.019 mg/L, respectively. Consistent with Order No. R1-2017-0007, the water quality criteria for total chlorine residual recommended by U.S. EPA have been translated to an AMEL of 0.01 mg/L and an MDEL of 0.02 mg/L in this Order.

- 4.3.3.1.3. **Total Coliform Bacteria.** Coliform bacteria are a pollutant of concern in all wastewaters of domestic origin, and therefore this Order retains the effluent limitations for total coliform bacteria from Order No. R1-2017-0007. Compliance with these effluent limitations will ensure that water quality objectives for bacteria, as established by Chapter 3 of the Basin Plan, will be maintained. These effluent limitations reflect standards for secondary treated water in the Basin Plan (Section 4, Implementation Plans) and as established for secondary-23 recycled water by DDW at title 22, division 4, chapter 3 of the CCR.
- 4.3.3.1.4. **Settleable Solids.** High levels of settleable solids can have an adverse effect on aquatic habitat. Untreated or improperly treated wastewater can contain high amounts of settleable solids. The Lower Eel River and its tributaries are listed as impaired for sediment and settleable solids. Monthly average and maximum daily effluent limitations for settleable solids of 0.1 ml/L and 0.2 ml/L have been retained from Order No. R1-2017-0007. These limitations reflect levels of treatment attainable by secondary treatment facilities. This limitation is necessary to comply with 303(d) list where the Lower Eel River is determined to be impacted by sediment. Furthermore, the Basin Plan prohibits discharge resulting in bottom deposits for all surface waters of the North Coast Region.
- 4.3.3.1.5. **Nitrogen Compounds.** Untreated domestic wastewater contains ammonia nitrogen. Nitrification is a biological process that converts ammonia to nitrite and nitrate. Denitrification is a process that converts nitrate to nitrogen gas, which is then released to the atmosphere. Inadequate or incomplete nitrification may result in the discharge of ammonia to the receiving stream and inadequate or incomplete denitrification may result in the discharge of nitrate to the receiving stream. The Facility is designed to use nitrification to remove ammonia from the waste stream and denitrification to remove nitrate from the waste stream, culminating in an overall reduction in total nitrogen.
- 4.3.3.1.5.1. **Total Nitrogen.** The Basin Plan contains a narrative water quality objective for biostimulatory substances that states, “[w]aters shall not contain biostimulatory substances in concentrations that promote aquatic growths to the extent that such growths cause nuisance or adversely affect beneficial uses.” The Regional Water Board is increasingly concerned about the biostimulatory properties of discharges to surface water in the North Coast Region. Stimulation of biological growth can

deplete dissolved oxygen in receiving water below Basin Plan objectives. The Permittee sampled its discharge at Monitoring Location EFF-001 monthly during periods of discharge between November 2017 and May 2022. Monitoring results for total nitrogen ranged between 0.23 mg/L and 8.9 mg/L based on 30 samples. In order to protect water quality and ensure proper operation of the Facility, an AMEL of 10 mg/L for total nitrogen has been retained from Order No. R1-2017-0007.

- 4.3.3.1.5.2. **Nitrate.** Nitrate is known to cause adverse health effects in humans. For waters designated as domestic or municipal supply, the Basin Plan (Chapter 3) adopts the MCLs, established by DDW for the protection of public water supplies at title 22 of the CCR, sections 64431 (Inorganic Chemicals) and 64444 (Organic Chemicals), as applicable water quality criteria. The MCL for nitrate (10 mg/L as N) is therefore applicable as a water quality criterion. The Permittee sampled its treated wastewater monthly during November 2017 through May 2022. Effluent water monitoring results for nitrate as N ranged between 0.23 mg/L and 7.2 mg/L based on 30 samples. Because nitrate levels in effluent have been measured at concentrations lower than 10 mg/L N, the Regional Water Board concludes that discharges from the Facility do not have a reasonable potential to cause or contribute to exceedances of the applicable water quality criterion for the receiving water for nitrate.

- 4.3.3.1.5.3. **Ammonia.** Ammonia is known to cause toxicity to aquatic organisms in surface waters. The Basin Plan establishes a narrative water quality objective for toxicity, stating that “[a]ll waters shall be maintained free of toxic substances in concentrations that are toxic to, or that produce detrimental physiological responses in human, plant, animal, or aquatic life.” Due to concerns regarding ammonia toxicity, the Regional Water Board relies on U.S. EPA’s recommended water quality criteria for ammonia to interpret the Basin Plan’s narrative objective for toxicity. For freshwater, the recommended criteria are from the April 2013 Aquatic Life Ambient Water Quality Criteria for Ammonia – Freshwater, EPA 822-R-13-001 (2013 Freshwater Criteria). The 2013 Freshwater Criteria is an update to the December 1999 Update of Ambient Water Quality Criteria for Ammonia (1999 Freshwater Criteria).

The 2013 Freshwater Criteria recommends acute and chronic water quality criteria for the protection of aquatic life, including salmonids and sensitive freshwater mussel species in the Family Unionidae that are more sensitive to ammonia than salmonids. Like the 1999 Freshwater Criteria document, the 2013 Freshwater Criteria document recommends acute (1-hour average) criteria based on pH and the presence/absence of salmonids and chronic (30 day average) criteria based on pH and temperature and that no 4-day average concentration should exceed 2.5 times the 30-day chronic criterion. In addition, the 2013 Freshwater

Criteria document recommends these same criteria for sensitive mussel species.

Adequate information is not available to determine if these freshwater mussels are present in the receiving water.

For this Order, the Regional Water Board has changed its approach for evaluating ammonia toxicity. This Order establishes an Ammonia Impact Ratio (AIR) for determining compliance with ammonia effluent limitations. The AIR is calculated as the ratio of the ammonia concentration in the effluent to the applicable 2013 Freshwater Criteria which is based on the receiving water pH and temperature at the time that each effluent sample is collected. See Attachment I of this Order for a sample log to help calculate and record the AIR values and Attachment H for applicable pH- and temperature-dependent criteria.

Receiving water pH of 9.8 and temperature of 20.1°C at Monitoring Location RSW-001 were used to calculate acute and chronic criteria of 0.37 mg/L and 0.15 mg/L, respectively, using the assumptions that salmonids and mussels are both present.

The most stringent water quality objective, as shown in the calculations in the paragraph preceding this, is 0.15 mg/L. The maximum observed effluent ammonia concentration from the Facility was 0.26 mg/L, based on 1 sample collected between November 2017 and August 2022.

Because ammonia in the treated wastewater has been measured at a concentration greater than EPA's 2013 Freshwater Criteria at Monitoring Location EFF-001, the Regional Water Board concludes that discharges from the Facility have a reasonable potential to cause or contribute to exceedances of the Basin Plan's applicable narrative water quality criterion for toxicity. Therefore, this Order includes effluent limitations for ammonia for the protection of aquatic life. This Order establishes an average monthly effluent limitation (AMEL) of 1.0 and a maximum daily effluent limitation of 1.0 as an AIR. Attachment G provides calculations of the ammonia AMEL and MDEL.

- 4.3.3.1.6. **Biostimulatory Substances (Phosphorus and Nitrogen).** The Basin Plan contains a narrative water quality objective for biostimulatory substances that states "[w]aters shall not contain biostimulatory substances in concentrations that promote aquatic growths to the extent that such growths cause nuisance or adversely affect beneficial uses." The Regional Water Board is increasingly concerned about the biostimulatory properties of discharges to surface waters in the North Coast Region. Nutrients, such as phosphorus and nitrogen containing compounds, in treated wastewater stimulate the growth rate of photosynthetic bacteria, algae, and other aquatic plants. The overabundance of nitrogen and phosphorus compounds

in surface water bodies can result in the excessive growth and decay of these organisms, thus accelerating the process of eutrophication. These phenomena cause dissolved oxygen levels to drop below concentrations needed for the survival and health of fish and aquatic life, which in turn negatively affects the aesthetic quality of water bodies and impairs beneficial uses.

At present, for interpretation of the Basin Plan's narrative water quality objective for biostimulatory substances, U.S. EPA has established recommended water quality criteria for nutrients in *Nutrient Criteria Documents for Lakes and Rivers* and *Nutrient Criteria Documents for Rivers and Streams*. U.S. EPA has defined 14 "ecoregions" and further categorized surface waters as lakes and reservoirs or rivers and streams for purposes of defining applicable numeric water quality criteria for nutrients. The State and Regional Water Boards continue to examine other methods of interpreting the Basin Plan's narrative water quality objective for biostimulatory substances. When the Boards determine that U.S. EPA's recommended criteria are appropriate for implementing the Basin Plan objectives, or when a more appropriate and meaningful method is established, the need for limiting nutrients in relation to biostimulatory properties, including phosphorus and nitrogen-containing compounds, in all discharges in the Region will be reassessed. In the meantime, the RPA for nutrients in relation to biostimulatory properties, performed for development of this Order, is inconclusive. The Order retains monitoring requirements for phosphorus and nitrogen containing compounds in discharges from the Facility to allow a determination of reasonable potential analysis at such time as the State and Regional Water Boards select an appropriate method for interpretation of the Basin Plan's narrative objective.

4.3.3.1.7. **Pathogens (*E. coli* bacteria).** On August 7, 2018, the State Water Board adopted Part 3 of the Water Quality Control Plan for Inland Surface Waters, Enclosed Bays, and Estuaries of California – Bacteria Provisions and a Water Quality Standards Variance Policy (Statewide Bacteria Provisions), which establishes water quality objectives for reasonable protection of people that recreate within all surface waters, enclosed bays, and estuaries of the state that have the water contact recreation beneficial use (REC-1). In accordance with the water quality objectives outlined in the Statewide Bacteria Provisions for the protection of freshwaters used for water contact recreation, disinfected effluent shall not result in the exceedance of the following objectives:

4.3.3.1.7.1. The concentration of *E. coli* bacteria shall not exceed 100 colony forming units (cfu) per 100 milliliters (mL) as a six-week rolling geometric mean, calculated weekly.

- 4.3.3.1.7.2. A statistical threshold value (STV) of 320 cfu/100 mL shall not be exceeded by more than 10 percent of the samples collected in a calendar month and calculated in a static manner.

This Order includes effluent limitations for total coliform bacteria based on standards set forth in CCR, title 22, section 60301.225 for disinfected secondary-23 recycled water. Discharges to surface water in compliance with these effluent limitations for recycled water, which are more stringent than the *E. coli* standards set forth in the Statewide Bacteria Standards, will ensure that bacterial standards for water contact recreation are maintained throughout the receiving water.

- 4.3.3.1.8. **Chlorodibromomethane (CDBM).** The CTR establishes a water quality objective for the protection of human health for CDBM of 0.41 µg/L. The Permittee sampled the effluent for CDBM 20 times during the term of Order No. R1-2017-0007 with results ranging from ND (<1 µg/L) to 2.6 µg/L. CDBM was not detected in the receiving water based on one sample. A determination of reasonable potential has been made based on the MEC of the 2.6 µg/L and 11 additional results exceeding the most stringent water quality objective of 0.41 µg/L. Therefore, this order establishes an average monthly effluent limitation of 0.41 µg/L for chlorodibromomethane and a daily maximum limitation of 0.89 µg/L at monitoring locations EFF-001. Fact Sheet section 4.4.4 provides calculations of the CDBM AMEL and MDEL.
- 4.3.3.1.9. **Dichlorobromomethane (DCBM).** The CTR establishes a water quality objective for the protection of human health for DCBM of 0.56 µg/L. The Permittee sampled the effluent for DCBM 20 times during the term of Order No. R1-2017-0007 with results ranging from ND (<1 µg/L) to 16 µg/L. DCBM was not detected in the receiving water based on one sample. A determination of reasonable potential has been made based on the MEC of 15 µg/L and 17 additional results exceeding the most stringent water quality objective of 0.56 µg/L. Therefore, this order establishes an average monthly effluent limitation of 0.56 µg/L for dichlorobromomethane and a daily maximum limitation of 1.1 µg/L at monitoring locations EFF-001. Fact Sheet section 4.4.4 provides calculations of the DCBM AMEL and MDEL.
- 4.3.3.1.10. **4,4-DDT.** The CTR establishes a water quality objective for the protection of human health for 4,4-DDT of 0.00059 µg/L. The Permittee sampled the effluent and receiving water for 4,4-DDT on March 23, 2021. The results for effluent were 0.023 µg/L and ND, respectively. A determination of reasonable potential has been made based on the MEC of 0.023 µg/L exceeding the most stringent water quality objective of 0.00059 µg/L. Therefore, this order establishes an average monthly effluent limitation of 0.00059 µg/L for 4,4-DDT and a daily maximum limitation of 0.00118 µg/L at monitoring locations EFF-001. Fact Sheet section 4.4.4 provides calculations of the 4,4-DDT AMEL and MDEL.

- 4.3.3.1.11. **Antimony.** Antimony is known to cause adverse health effects in humans. For waters designated as domestic or municipal supply, the Basin Plan (Chapter 3) adopts the MCLs, established by DDW for the protection of public water supplies at title 22 of the CCR, sections 64431 (Inorganic Chemicals) and 64444 (Organic Chemicals), as applicable water quality criteria. The MCL for Antimony (6 µg/L) is therefore applicable as a water quality criterion. The effluent and receiving water were sampled for Antimony on March 23, 2021, with a result of 11 µg/L and 10 µg/L, respectively. Because antimony levels in effluent and receiving waters have been measured at concentrations higher than 6 µg/L, the Regional Water Board concludes that the discharge from the Facility does have a reasonable potential to cause or contribute to exceedances of applicable water quality criteria for the receiving water for Antimony. Therefore, this order establishes an average monthly effluent limitation of 6 µg/L for Antimony and a daily maximum limitation of 12 µg/L at monitoring locations EFF-001. Fact Sheet section 4.4.4 provides calculations of the antimony AMEL and MDEL.
- 4.3.3.1.12. **Aluminum.** The 2018 Aluminum Criteria reflect the latest science and allow for development of criteria reflecting the impact of local receiving water chemistry on aluminum toxicity to aquatic life. The updated criteria account for the site-specific bioavailability of aluminum in receiving waters, which is dependent on pH, dissolved organic carbon, and hardness. Due to a lack of sufficient receiving water information (pH, dissolved organic carbon, and hardness) for calculating criteria, the 2018 NAWQC criteria has not been implemented in this permit. Instead, the MRP includes requirements to monitor effluent for aluminum and receiving water for aluminum, pH, dissolved organic carbon, and hardness in order to obtain sufficient data to evaluate for aluminum toxicity based on the 2018 NAWQC.
- 4.3.3.1.13. **Total Trihalomethanes.** Total trihalomethanes include bromoform, chlorodibromomethane, chloroform, and dichlorobromomethane. The CTR does not establish water quality objectives for total trihalomethanes. For waters designated as domestic or municipal supply, the Basin Plan (Chapter 3) adopts the MCLs, established by DDW for the protection of public water supplies at title 22 of the CCR, sections 64431 (Inorganic Chemicals) and 64444 (Organic Chemicals), as applicable water quality criteria. The MCL for total trihalomethanes (80 µg/L) is therefore applicable as a water quality criterion. The Permittee sampled the effluent for total trihalomethanes 22 times during the term of Order No. R1-2017-0007 with results ranging from 24.7 µg/L to 101.39 µg/L. The highest result occurred on March 18, 2020 and calculation of the maximum total trihalomethane effluent concentration of 101.39 µg/L is shown in the following table.

Table F-5. Effluent Trihalomethane Concentrations on March 18, 2020 Sample

Parameter	Effluent Concentration (µg/L)
Bromoform	<0.5
Chlorodibromomethane	1.29
Chloroform	85.6
Dichlorobromomethane	14.5
Sum of Trihalomethanes	101.39

Total Trihalomethanes were not detected in the receiving water based on one sample. A determination of reasonable potential has been made based on the MEC of 101.39 µg/L exceeding the most stringent water quality objective of 80 µg/L. Therefore, this Order establishes an average monthly effluent limitation for total trihalomethanes based on the Primary MCL.

- 4.3.3.1.14. **Haloacetic Acids.** Haloacetic Acids include Monochloroacetic Acid, Dichloroacetic Acid, Trichloroacetic Acid, Monobromoacetic Acid, and Dibromoacetic Acid. For waters designated as domestic or municipal supply, the Basin Plan (Chapter 3) adopts the MCLs, established by DDW for the protection of public water supplies at title 22 of the CCR, sections 64431 (Inorganic Chemicals) and 64444 (Organic Chemicals), as applicable water quality criteria. The MCL for Haloacetic Acids (60 µg/L) is therefore applicable as a water quality criterion. The Permittee sampled the effluent for Haloacetic Acids five times during the term of Order No. R1-2017-0007, with results ranging from 26 µg/L to 170 µg/L. The highest result occurred on November 23, 2021, and calculation of the maximum total Haloacetic Acids effluent concentration of 170 µg/L is shown in the following table.

Table F-6. Effluent Haloacetic Acid Concentrations on November 23, 2021 Sample

Parameter	Effluent Concentration (µg/L)
Monochloroacetic Acid	5.3
Dichloroacetic Acid	60
Trichloroacetic Acid	100
Monobromoacetic Acid	<1
Dibromoacetic Acid	<1
Unidentified Haloacetic Acid	4.7
Sum of Haloacetic Acids	170

Because Haloacetic Acid levels in the effluent have been measured at concentrations greater than 60 µg/L, the Regional Water Board concludes that discharges from the Facility have a reasonable potential to cause or contribute to exceedances of applicable water quality criteria for the receiving water for Haloacetic Acids. Therefore, this Order establishes an

average monthly effluent limitation for Total Haloacetic Acids based on the Primary MCL.

4.3.3.2. **Priority Pollutants**

The SIP establishes procedures to implement water quality criteria from the NTR and CTR and for priority, toxic pollutant objectives established in the Basin Plan. The implementation procedures of the SIP include methods to determine reasonable potential (for pollutants to cause or contribute to excursions above state water quality standards) and to establish numeric effluent limitations, if necessary, for those pollutants showing reasonable potential.

Section 1.3 of the SIP requires the Regional Water Board to use all available, valid, relevant, and representative receiving water and effluent data and information to conduct an RPA. During the term of Order No. R1-2017-0007, priority pollutant sampling on effluent water was conducted on March 23, 2021 at Monitoring Location EFF-001. Additionally, priority pollutant sampling on receiving water was conducted on March 23, 2021 at Monitoring Location RSW-001. These data were used to conduct the RPA.

Hardness: The CTR and the NTR contain water quality criteria for seven metals that vary as a function of hardness; the lower the hardness, the lower the water quality criteria. The SIP requires water quality criteria be properly adjusted for hardness, using the hardness of the receiving water. The hardness-dependent metal criteria include cadmium, copper, chromium (III), lead, nickel, silver, and zinc. The Permittee sampled its receiving water during November 2017 through August 2022. Effluent water monitoring for hardness results ranged between 73 mg/L and 170 mg/L based on 32 samples. The minimum observed receiving water hardness of 73 mg/L was used to calculate the criteria.

To conduct the RPA, Regional Water Board staff identified the maximum effluent concentration (MEC) and maximum background (B) concentration for each priority, toxic pollutant from effluent and receiving water data provided by the Permittee, and compared this information to the most stringent applicable water quality criterion (C) for each pollutant with applicable water quality criteria from the NTR, CTR, and the Basin Plan. Section 1.3 of the SIP establishes three triggers for a finding of reasonable potential.

Trigger 1. If the MEC is greater than C, there is reasonable potential, and an effluent limitation is required.

Trigger 2. If B is greater than C, and the pollutant is detected in effluent (MEC > ND), there is reasonable potential, and an effluent limitation is required.

Trigger 3. After a review of other available and relevant information, a permit writer may decide that a WQBEL is required. Such additional information may include, but is not limited to: the facility type, the discharge type, solids loading analyses, lack of dilution, history of compliance problems, potential toxic impact of the discharge, fish tissue residue data, water quality and beneficial uses of the receiving water, CWA 303(d) listing for the pollutant, and the presence of endangered or threatened species or their critical habitat.

4.3.3.3. Reasonable Potential Determination

Reasonable potential could not be determined for all pollutants, as there are not applicable water quality criteria for all pollutants. The RPA determined that there is either no reasonable potential or there was insufficient information to conclude affirmative reasonable potential for 119 of the 126 priority pollutants.

Table F-7 summarizes the RPA for each pollutant that was reported in detectable concentrations in the effluent or the receiving water. The MECs, most stringent water quality objectives/water quality criteria (WQO/WQCs), and background concentrations (B) used in the RPA are presented, along with the RPA results (Yes or No and which trigger) for each toxic pollutant analyzed. No other pollutants with applicable, numeric water quality criteria from the NTR, CTR, and the Basin Plan were measured above detectable concentrations during the monitoring events conducted by the Permittee. Attachment F-1 to this Order summarizes the RPA for all 126 priority pollutants.

Table F-7. Summary of Reasonable Potential Analysis Results for Priority Pollutants, Ammonia, and Title 22 Pollutants

CTR No.	Pollutant	Unit	C or Most Stringent WQO/WQC	MEC or Minimum DL ¹	B or Minimum DL	RPA Result ²
1	Antimony, Total Recoverable	µg/L	6	11	10	Yes
6	Copper, Total Recoverable	µg/L	64 ³	3.8	1.9	No
23	Chlorodibromomethane	µg/L	0.41	2.6	<0.24	Yes
26	Chloroform	µg/L	80	90.7	<0.33	Yes
27	Dichlorobromomethane	µg/L	0.56	16.3	<0.32	Yes
39	Toluene	µg/L	150	0.65	<0.16	No

CTR No.	Pollutant	Unit	C or Most Stringent WQO/WQC	MEC or Minimum DL ¹	B or Minimum DL	RPA Result ²
13	Zinc, Total Recoverable	µg/L	91	30	<2.7	No
108	4,4-DDT	µg/L	0.00059	0.023	<0.0046	Yes
112	alpha-Endosulfan	µg/L	0.056	0.0042	<0.002	No
N/A	Aluminum	µg/L	50	--	--	Ud
N/A	Ammonia (as N)	mg/L	0.15 ⁴	0.26	--	Yes
N/A	Nitrate (as N)	mg/L	10	7.2	--	No

Table Notes

1. The Maximum Effluent Concentration (MEC) or maximum background concentration (B) is the actual detected concentration unless it is preceded by "<", in which case the value shown is the minimum detection level as the analytical result was reported as not detected (ND).
2. RPA Results:
 - = Yes, if MEC > WQO/WQC, or B > WQO/WQC and MEC is detected.
 - = No, if MEC and B or < WQO/WQC or all effluent data are undetected.
 - = Undetermined (UD).
3. Copper WQO calculated with a water effect ratio (WER) of 8.75 and the most stringent WQO from the CTR using the lowest receiving water hardness of 73 mg/L ($8.75 \times 7.3 \text{ µg/L} = 64 \text{ µg/L}$).
4. Ammonia criteria are determined on a sliding scale based upon temperature and pH. The criterion represented in this table is based upon chronic exposure and a temperature of 20.1°C and a pH of 9.8.

Additional details regarding priority pollutant constituents for which reasonable potential was not found, but warrant further explanation are included in the following paragraphs:

- 4.3.3.3.1. **Copper.** The CTR includes hardness-dependent criteria for the protection of freshwater aquatic life for copper. The criteria for copper are expressed in dissolved concentrations. U.S. EPA recommends conversion factors to translate dissolved concentrations to total concentrations. The U.S. EPA default conversion factors for copper in freshwater are 0.96 for both the acute and the chronic criteria. The default WER used for calculating criteria for copper is 1.0. The Permittee has conducted a WER study to determine the site-specific toxicity of copper in the receiving water at the point of discharge. The Permittee's study concluded that a site-specific WER of 8.75 for total recoverable copper applies to the discharge. Using the worst-case

measured hardness from the receiving water (73 mg/L), the U.S. EPA recommended dissolved-total translator of 0.96, and the site-specific WER of 8.75, the applicable chronic criterion (maximum 4-day average concentration) is 64 µg/L and the applicable acute criterion (maximum 1-hour average concentration) is 92 µg/L. The Permittee sampled the effluent and receiving water for copper once during the term of Order No. R1-2017-0007. Copper was detected in the effluent sample, 3.8 µg/L. Copper was detected in the single receiving water sample at 1.9 µg/L. A determination of no reasonable potential has been made based on the MEC of 3.8 µg/L not exceeding the most stringent water quality objective of 64 µg/L.

4.3.3.3.2. **Alpha-endosulfan.** The CTR establishes a water quality objective for the protection of freshwater aquatic life for alpha-endosulfan. The applicable chronic criterion (maximum 4-day average concentration) is 0.056 µg/L and as the applicable acute criterion (maximum 1-hour average concentration) is 0.22 µg/L. The Permittee sampled the effluent and receiving water for alpha-endosulfan on March 23, 2021. The results for effluent were <0.002 µg/L and, <0.0042 µg/L respectively. A determination of no reasonable potential has been made based on the MEC of 0.0042 µg/L not exceeding the most stringent water quality objective of 0.056 µg/L.

4.3.3.3.3. **Zinc.** The CTR includes hardness-dependent criteria for the protection of freshwater aquatic life for zinc. The criteria for zinc are expressed in dissolved concentrations. U.S. EPA recommends conversion factors to translate dissolved concentrations to total recoverable concentrations. The U.S. EPA default conversion factors for zinc in freshwater are 0.978 for acute criteria and 0.986 for chronic criteria. Using the worst-case measured hardness from the receiving water (73 mg/L), the applicable chronic criterion (maximum 4-day average concentration) is 91 µg/L and the applicable acute criterion (maximum 1-hour average concentration) is 92 µg/L. The Permittee sampled the effluent and receiving water for zinc once during the term of Order No. R1-2017-0007. Zinc was detected in the effluent in the effluent sample, 30 µg/L. Zinc was reported in the single receiving water sample at <2.7 µg/L. A determination of no reasonable potential has been made based on the MEC of 30 µg/L not exceeding the most stringent water quality objective of 91 µg/L.

4.3.4. **WQBEL Calculations**

Final WQBELs have been determined using the methods described in section 1.4 of the SIP.

Step 1: To calculate the effluent limits, an effluent concentration allowance (ECA) is calculated for each pollutant found to have reasonable potential using the following equation, which takes into account dilution and background concentrations:

$$ECA = C + D (C - B),$$

Where:

C = the applicable water quality criterion (adjusted for effluent hardness and expressed as the total recoverable metal, if necessary)

D = dilution credit (here D= 0, as the discharge does not qualify for a dilution credit)

B = background concentration

Here, no credit for dilution is allowed, which results in the ECA being equal to the applicable criterion (ECA = C).

Step 2: For each ECA based on an aquatic life criterion/objective, the long-term average discharge condition (LTA) is determined by multiplying the ECA by a factor (multiplier), which adjusts the ECA to account for effluent variability. The multiplier depends on the coefficient of variation (CV) of the data set and whether it is an acute or chronic criterion/objective. Table 1 of the SIP provides pre-calculated values for the multipliers based on the values of the CV. When the data set contains less than 10 sample results, or when 80 percent or more of the data set is reported as ND, the CV is set equal to 0.6. Derivation of the multipliers is presented in section 1.4 of the SIP.

The reasonable potential analysis described in Fact Sheet section 4.3.3 did not identify the need to calculate effluent limitations for any pollutants with aquatic life criteria, therefore Steps 2 and 3 are included to describe the procedure that would be used in the future if reasonable potential is found for any pollutant(s) with aquatic life criteria.

Step 3: WQBELs, including an AMEL and MDEL, are calculated using the most limiting (lowest) LTA. The LTA is multiplied by a factor that accounts for averaging periods and exceedance frequencies of the effluent limitations, and for the AMEL, the effluent monitoring frequency. The sampling frequency is set equal to 4 ($n = 4$) for the acute criterion and chronic 4-day criterion. The 99th percentile occurrence probability was used to determine the MDEL multiplier and a 95th percentile occurrence probability was used to determine the AMEL multiplier. Since reasonable potential was not found for any pollutants with aquatic life criterion/objectives, no effluent limitations were calculated for this permit.

Step 4: When the most stringent water quality criterion/objective is a human health criterion/objective (as for chlorodibromomethane, dichlorobromomethane, total trihalomethanes, Haloacetic Acids, antimony, and 4,4-DDT), the AMEL is set equal to the ECA. From Table 2 of the SIP, when $CV = 0.6$ and $n = 4$, the MDEL multiplier at the 99th percentile occurrence probability equals 3.11, and the AMEL multiplier at the 95th percentile

occurrence probability equals 1.55 (for 4,4-DDT and antimony). The MDEL for protection of human health is calculated by multiplying the ECA by the ratio of the MDEL multiplier to the AMEL multiplier. Final WQBELs for chlorodibromomethane, dichlorobromomethane, total trihalomethanes, Haloacetic Acids, antimony, and 4,4-DDT are determined as follows.

Table F-8. Determination of Final WQBELs Based on Human Health Criteria

Pollutant	ECA (µg/L)	MDEL/AMEL	MDEL (µg/L)	AMEL (µg/L)
Chlorodibromomethane	0.41	2.16	0.89	0.41
Dichlorobromomethane	0.56	1.71	1.1	0.56
Total Trihalomethanes ¹	80	1.53	122.5	80
Haloacetic Acids ¹	60	1.93	116	60
Antimony	6	2.0	12	6
4,4-DDT	0.00059	2.0	0.0012	0.00059

Table Notes:

1. This Order establishes an AMEL only for Total Trihalomethanes and Haloacetic Acids because the drinking water MCL for these pollutants is based on a long-term average.

4.3.5. Whole Effluent Toxicity (WET)

Effluent limitations for whole effluent toxicity protect the receiving water from the aggregate effect of a mixture of pollutants that may be present in effluent. There are two types of WET tests – acute and chronic. An acute toxicity test is conducted over a short time period and measures mortality. A chronic test is conducted over a longer period of time and may measure mortality, reproduction, and/or growth.

WET requirements in this Order are derived from the CWA, and the Water Quality Control Plan for Inland Surface Waters, Enclosed Bays, and Estuaries of California (Plan). The Plan establishes objectives for water quality and sediment quality that apply to all inland surface waters, enclosed bays, and estuaries and coastal lagoons of the state, including both waters of the United States and surface waters of the state. For compliance with the Plan's water quality objective, this Order requires the Permittee to conduct WET testing for chronic toxicity, as specified in the MRP (Attachment E, section 5).

Test of Significant Toxicity (TST). In 2010, U.S. EPA endorsed the peer-reviewed Test of Significant Toxicity (TST) two-concentration hypothesis testing approach in National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document (EPA 833-R-10-003, 2010) as an improved hypothesis-testing tool to evaluate data from U.S. EPA's toxicity test methods. The TST hypothesis testing approach more reliably identifies toxicity—in

relation to the chronic (0.25 or more) mean response of regulatory management concern—than the NOEC hypothesis-testing approach. The TST hypothesis testing approach more reliably identifies toxicity—in relation to the acute (0.20 or more) mean responses of regulatory management concern—than the No Observed Effect Concentration (NOEC) approach previously used to establish effluent limitations for acute toxicity.

In a letter dated February 12, 2014, the State Water Board submitted an ATP request to USEPA Region 9 for the statewide use of a two-concentration toxicity test design when using the Test of Significant Toxicity (TST) approach. This two-concentration test design is composed of a single effluent concentration and a control concentration. USEPA approved the ATP request on March 17th, 2014. In June 2014, the approval was challenged in court on procedural grounds under the Administrative Procedures Act by the Southern California Alliance of Publicly Owned Treatment Works (SCAP) and the Central Valley Clean Water Association (CVCWA). The U.S. EPA withdrew the approval and notified State Water Board in a memo dated February 11th, 2015.

It is important to note that U.S. EPA's rescission of its approval of the ATP is not based on the substantive TST statistical analysis or the scientific validity of a two-concentration test design. The withdrawal letter also states that currently there is a proposed rulemaking to change the language in the ATP regulations at Code of Federal Regulations, title 40, section 136.

The benefits of requiring the TST in new or amended permits include improving the statistical power of the toxicity test, and simplifying the analysis as compared to traditional hypothesis statistical approaches or point estimates. The calculations are straightforward and provide a clear pass/fail result. With the withdrawal of the two-concentration test design approval, an NPDES permit can still require the TST for statistical analyses. Toxicity test shall be run using a multi-concentration test design in accordance with Code of Federal Regulations, title 40, section 136.3, and the TST shall be utilized with the biological responses from the permitted IWC and the control (effluent concentration of zero). However, even with only two of the five concentration biological responses being used, cost savings in the form of time and effort are still realized for the statistical analysis and data interpretation carried out by the Permittee, lab, and permit manager. This Order requires application of TST for statistical analysis of whole effluent aquatic toxicity data

Test of Significant Toxicity Design. The TST's null hypothesis for chronic toxicity is:

H0: Mean response (In-stream Waste Concentration (IWC) in % effluent) $\leq$ 0.75 mean response (control)

Results are analyzed using the TST approach and an acceptable level of chronic toxicity is demonstrated by rejecting the null hypothesis and reporting “Pass” or “P”.

The chronic IWC (in % effluent) for Discharge Point 001 is 100%. The chronic toxicity trigger for Discharge Point 001 is expressed as a null hypothesis (H0) and regulatory management decision (b value) of 0.75 for the chronic toxicity methods in the MRP. The null hypothesis for this discharge is:

H0: Mean response (100% effluent) $\leq$ 0.75 mean response (control)

Results shall be analyzed using the TST hypothesis testing approach in the MRP. For any chronic aquatic toxicity test method with both lethal and sub-lethal endpoints, the sub-lethal endpoint shall only be required. Compliance with this chronic toxicity limitation is demonstrated by rejecting the null hypothesis and reporting “Pass” or “P”.

When one MDEL or MMEL is not met, but not two in a calendar month, the Permittee must perform an Additional Routine Monitoring Test as specified in the MRP (Attachment E, section 5). If any combination of two or more MDEL or MMEL are not met within a single calendar month or within two successive calendar months, the Permittee will be required to conduct a TRE, as described by the MRP.

Notification requirements for aquatic toxicity testing include a 24-hour notification requirement if test results do not meet an applicable MDEL or MMEL, per the Toxicity Provisions. Verbal notification of aquatic toxicity test results may be left by voice mail if the Regional Water Board staff person is not immediately available by telephone.

4.3.5.1. **Acute Aquatic Toxicity**

Order No. R1-2017-0007 included an effluent limitation for acute toxicity in accordance with the Basin Plan, which requires that the average survival of test organisms in undiluted effluent for any three consecutive 96-hour bioassay tests be at least 90 percent, with no single test having less than 70 percent survival. Furthermore, the permittee was subject to determination of “Pass” or “Fail” and “Percent (%) Effect” from acute toxicity using the Test of Significant Toxicity (TST) approach. The sensitivity species screening conducted during the term of Order No. R1-2017-0007 concluded that the most sensitive species for acute toxicity testing is the water flea (*Ceriodaphnia dubia*). Acute aquatic toxicity test results for the term of Order No. R1-2017-0007 are summarized in Table F-9 below:

Table F-9. Summary of Acute Toxicity Results (Water Flea)

Date	Pass/Fail	Percent Effect (Survival)
4/2/2018	Pass	No Effect
2/25/2019	Pass	5%
2/10/2020	Pass	No Effect
3/2/2020	Pass	No Effect
1/4/2021	Pass	No Effect
1/26/2022	Pass	6.6%

The Toxicity Provisions identify that a discharge has reasonable potential to cause or contribute to an excursion above the acute aquatic toxicity water quality objectives if any of the acute aquatic toxicity tests results in a “fail” at the in-stream waste concentration (IWC), or if any of the acute aquatic toxicity tests have a percent effect at the IWC greater than 10 percent. As shown in Table F-9, acute aquatic toxicity testing did not result in a “fail” and that no resulting percent effect exceeded 10%. As such, it has been determined that a discharge from this Facility does not have reasonable potential to cause or contribute to an exceedance of the water quality objectives for acute toxicity.

4.3.5.2. Chronic Aquatic Toxicity

For Order No. R1-2017-0007, the SIP required the use of short-term chronic toxicity tests to determine compliance with the narrative toxicity objectives for aquatic life in the Basin Plan. Under this monitoring, the Permittee was subject to determination of “Pass” or “Fail” and “Percent (%) Effect” from chronic toxicity using the Test of Significant Toxicity (TST) approach. The sensitivity species screening conducted during the term of Order No. R1-2017-0007 concluded that the most sensitive species for chronic toxicity testing is water flea (*Ceriodaphnia dubia*). Chronic aquatic toxicity test results for the term of Order No. R1-2017-0007 are summarized in Table F-10 below:

Table F-10. Summary of Chronic Toxicity Results (Water Flea)

Date	Pass/Fail	Percent Effect (Survival)	Percent Effect (Reproduction)
4/9/2018	Fail	No Effect	53%
5/2/2018	Fail	No Effect	85.3%

Date	Pass/Fail	Percent Effect (Survival)	Percent Effect (Reproduction)
12/10/2018	Pass	No Effect	-11%
4/15/2019	Fail	10%	47.8%
2/10/2020	Fail	No Effect	95.4%
3/2/2020	Fail	No Effect	95.4%
3/16/2020	Fail	No Effect	25.2%
3/17/2020	Fail	No Effect	24.1%
3/30/2020	Fail	No Effect	30.3%
3/30/2020	Pass	No Effect	16.7%
4/15/2020	Pass	No Effect	12.9%
4/29/2020	Fail	No Effect	42.3%
1/4/2021	Pass	No Effect	-7.6%
2/1/2021	Pass	20%	5.5%
2/15/2021	Pass	No Effect	11.6%
1/26/2022	Fail	No Effect	38.7%

The Toxicity Provisions identify that a discharge has reasonable potential to cause or contribute to an excursion above the chronic aquatic toxicity water quality objectives if any of the chronic aquatic toxicity tests results in a “fail” at the IWC, or if any of the chronic aquatic toxicity tests have a percent effect at the IWC greater than 10 percent. As shown in Table F-10, chronic aquatic toxicity testing 10 samples resulted in a “fail” result and 13 samples resulted in a percent effect for growth that exceeded 10% (a negative percent effect indicates that the effluent sample performed better than the control sample). As such, it has been determined that a discharge from this Facility does have reasonable potential to cause or contribute to an exceedance of the water quality objectives for chronic toxicity and corresponding MDEL and MMEL have been included in this Order, as required by the Toxicity Provisions. Attachment E of this Order requires annual chronic WET monitoring to demonstrate compliance with the Toxicity Provisions and the MDEL and MMEL established in this Order.

4.4. Final Effluent Limitation Considerations

4.4.1. Anti-Backsliding Requirements

Sections 402(o) and 303(d)(4) of the CWA and federal regulations at 40 C.F.R. section 122.44(l) prohibit backsliding in NPDES permits. These anti-

backsliding provisions require effluent limitations in a reissued permit to be as stringent as those in the previous permit, with some exceptions where limitations may be relaxed. All effluent limitations in this Order are at least as stringent as the effluent limitations in the previous Order with the exception of the effluent limitation for total nitrogen and acute toxicity. As previously discussed in section 4.4.3.1.5.1 and 4.4.5.1, respectively. The updated effluent data constitutes new information, which permits the removal of effluent limitations consistent with CWA section 402(o)(2)(B). Therefore, the Order does not retain the effluent limitations for acute toxicity and total nitrogen.

4.4.2. Antidegradation Policies

State Water Board Resolution 68-16, the Statement of Policy with Respect to Maintaining High Quality Waters in California (the Antidegradation Policy) requires that disposal of waste into waters of the state be regulated to achieve the highest water quality consistent with the maximum benefit to the people of the state. The quality of some waters is higher than established by adopted policies and that higher quality water shall be maintained to the maximum extent possible consistent with the Antidegradation Policy. The Antidegradation Policy requires that (1) higher quality water will be maintained until it has been demonstrated to the state that any change will be consistent with the maximum benefit to the people of the state, will not unreasonably affect present and anticipated beneficial use of the water, and will not result in water quality less than that prescribed in the policies; and (2) any activity that produces a waste or may produce waste or increased volume or concentration of waste and discharges to existing high quality water will be required to meet waste discharge requirements that will result in the best practicable treatment or control (BPTC) of the discharge necessary to assure pollution or nuisance will not occur, and the highest water quality consistent with the maximum benefit to the people of the state will be maintained.

Discharges from the Facility are required to maintain protection of the beneficial uses of the receiving water and comply with applicable provisions of the Basin Plan.

This Order is consistent with applicable federal and state antidegradation policies, as it does not authorize the discharge of increased concentrations of pollutants or increased volumes of treated wastewater beyond that which was permitted to discharge in accordance with Order No. R1-2017-0007.

Table F-11. Summary of Effluent Limitations – Discharge Point 001 (Monitoring Location EFF-001)

Parameter	Unit	Average Monthly Effluent Limitation	Average Weekly Effluent Limitation	Maximum Daily Effluent Limitation	Instantaneous Minimum Effluent Limitation	Instantaneous Maximum Effluent Limitation	Basis ¹
Biochemical Oxygen Demand 5-day @ 20°C (BOD ₅)	mg/L	30	45	--	--	--	TT
BOD ₅	% Removal	85	--	--	--	--	CFR
Total Suspended Solids (TSS)	mg/L	30	45	--	--	--	TT
TSS	% Removal	85	--	--	--	--	CFR
pH	Standard Units	--	--	--	6.5	8.5	BP
Ammonia Impact Ratio	Ratio	1	--	1	--	--	NAWQC
Total Coliform Bacteria	MPN/100 mL	--	23 ²	240	--	--	Title 22
Discharge Rate ³	%	--	--	--	--	1	BP
Chronic Toxicity	% Effect	--	--	--	--	50%	BP
Settleable Solids	ml/L	0.1	--	0.2	--	--	BP
Chlorine, Total Residual	mg/L	0.01	--	0.02	--	--	BP
Chlorodibromomethane	µg/L	0.41	--	0.80	--	--	CTR
Dichlorobromomethane	µg/L	0.56	--	1.1	--	--	CTR
Total Trihalomethanes	µg/L	80	--	--	--	--	BP
Haloacetic Acids	µg/L	60	--	--	--	--	BP
Nitrogen, Total (as N)	mg/L	10	--	--	--	--	Title 22

Parameter	Unit	Average Monthly Effluent Limitation	Average Weekly Effluent Limitation	Maximum Daily Effluent Limitation	Instantaneous Minimum Effluent Limitation	Instantaneous Maximum Effluent Limitation	Basis ¹
4,4-DDT	ug/L	0.00059	--	0.00118	--	--	CTR
Antimony, Total Recoverable	ug/L	6	--	12	--	--	BP

Table Notes

1. Definitions of acronyms in Table F-11:

BP - Basin Plan

CFR – 40 C.F.R. part 133

CTR – California Toxics Rule

NAWQC – National Ambient Water Quality Criteria

TP - Toxicity Provisions

TT - Based on the treatment capability of the Facility.

Title 22 - Based on DDW Reclamation Criteria, CCR, Division 4, Chapter 3 (title 22).

2. The number of total coliform bacteria shall not exceed a Most Probable Number (MPN) of 23 per 100 milliliters (mL), in more than one sample in any 30-day period.

3. During the period from October 1 through May 14, discharge of treated wastewater shall not exceed 1 percent (1:100) of the upstream receiving water flow. For purposes of this Order, the flow in the Lower Eel River shall be measured at USGS Gauge No. 1147000 .

4.4.3. **Stringency of Requirements for Individual Pollutants**

This Order contains both technology-based and water quality-based effluent limitations for individual pollutants. The technology-based effluent limitations consist of restrictions on BOD₅ and TSS. Restrictions on these pollutants are discussed in section 4.2 of this Fact Sheet. This Order's technology-based pollutant restrictions implement the minimum, applicable federal technology-based requirements. In addition, this Order contains effluent limitations for ammonia, pH, and total coliform bacteria that are more stringent than the minimum, federal technology-based requirements but are necessary to meet water quality standards. These requirements are discussed in section 4.3.3 of the Fact Sheet.

Water quality-based effluent limitations have been derived to implement water quality objectives that protect beneficial uses. Both the beneficial uses and the water quality objectives have been approved pursuant to federal law and are the applicable federal water quality standards. To the extent that toxic pollutant water quality-based effluent limitations were derived from the CTR, the CTR is the applicable standard pursuant to 40 C.F.R. section 11.38. The procedures for calculating the individual water quality-based effluent limitations for priority pollutants are based on the CTR implemented by the SIP, which was approved by U.S. EPA on May 18, 2000. Most beneficial uses and water quality objectives contained in the Basin Plan were approved under state law and submitted to and approved by U.S. EPA prior to May 30, 2000. Any water quality objectives and beneficial uses submitted to U.S. EPA prior to May 30, 2000, but not approved by U.S. EPA before that date, are nonetheless "applicable water quality standards for purposes of the CWA" pursuant to 40 C.F.R. section 11.21(c)(1). The remaining water quality objectives and beneficial uses implemented by this Order were approved by U.S. EPA and are applicable water quality standards pursuant to section 11.21(c)(2). Collectively, this Order's restrictions on individual pollutants are no more stringent than required to implement the requirements of the CWA.

The Regional Water Board has considered the factors in Water Code section 13263, including the provisions of Water Code section 13241, in establishing these requirements.

4.5. **Interim Effluent Limitations**

- 4.5.1. **Concentration Based Effluent Limitations for dichlorobromomethane, chlorodibromomethane, total trihalomethanes, and haloacetic acids.** Interim effluent limitations for dichlorobromomethane, chlorodibromomethane, total trihalomethanes, and haloacetic acids for this Order are derived based on Facility performance using available effluent monitoring data at Discharge Point 001, the point of discharge to the Eel River. This performance-based effluent limitation was calculated using statistical methodology described in the U.S. EPA Technical Support Document for Water Quality-based Toxics Control

(TSD) and a statistical tool, RP Calc, developed by State Water Resources Control Board staff to assist State and Regional Water Board staff in the development of interim effluent limitations. The 95th percentile concentration for dichlorobromomethane, chlorodibromomethane, total trihalomethanes, and haloacetic acids was calculated at the 95 percent confidence level to determine the interim effluent limitation. The calculated interim effluent limitations in Table 3 were rounded to the nearest decimal for data collected during the term of the previous permit, WDRs Order No. R1-2017-0007.

- 4.5.2. Despite diligent efforts to comply with the time schedule, the City of Rio Dell was unable to complete Tasks I through L addressed in Table 1 of Order No. R1-2017-0045 by their respective compliance dates. Progress was delayed due to unforeseen challenges including supply chain breakdowns and lockdown safety measures during the statewide COVID-19 emergency response. Furthermore, damage inflicted by the December 20, 2022 earthquake centered along coastal Humboldt County and prolonged flooding impacting the area in December 2022 and January 2023 delayed compliance projects. As a result, the Regional Water Board finds that it is reasonable to provide additional time to comply with the effluent limitations for dichlorobromomethane, chlorodibromomethane, Total Trihalomethanes, and Haloacetic Acids.
- 4.5.3. The compliance schedules established in this Order are intended to be as short as possible. The compliance schedule for dichlorobromomethane, chlorodibromomethane, total trihalomethanes, and haloacetic acids accounts for the interrelationship between ammonia, bacteria, and disinfection byproducts, the time necessary to complete construction, testing, and fine tuning of the Permittee's new phased Chloramine Disinfection Project. The Regional Water Board recommends that the Permittee continue to evaluate resources to identify a means to shorten the time frame for achieving compliance with effluent limitations for these pollutants.

4.6. Land Discharge Specifications

This Order does not authorize discharges of waste to land.

4.7. Recycling Specifications

This Order authorizes the Permittee to discharge treated municipal wastewater to the Irrigation Site (Discharge Point 003) that complies with the Water Recycling Specifications and Requirements contained in section 4.3 of the Order. Water recycling specifications are based on the technical capabilities of the wastewater treatment system and standards required by the Basin Plan and Title 22.

4.7.1. Scope and Authority

Section 13263 of the Water Code requires the Regional Water Board to prescribe requirements for proposed discharges, existing discharges, or material change in an existing discharge based upon the conditions of the

disposal area or receiving waters upon or into which the discharge is made or proposed. The prescribed requirements shall implement any relevant water quality control plans that have been adopted and shall take into consideration the beneficial uses to be protected, the water quality objectives reasonably required for that purpose, other waste discharges, the need to prevent nuisance, and the provisions of Water Code section 13241. In prescribing requirements, the Regional Water Board is not obligated to authorize the full waste assimilation capacities of the receiving water.

Here, the Regional Water Board considered all of these factors when developing the WDRs for the recycled water discharge. Limitations for BOD₅, TSS, and pH were scientifically derived to implement water quality objectives that protect beneficial uses. Both beneficial uses and the water quality objectives have been approved pursuant to state law. In addition, discharge prohibitions were included to prohibit the discharge of untreated or partially treated waste, in order to protect public health and prevent nuisance.

Water Code section 13241 requires the Regional Water Board to establish water quality objectives in water quality control plans as in its judgment will ensure the reasonable protection of beneficial uses and prevention of nuisance, recognizing that it may be possible for the quality of water to be changed to some degree without unreasonably affecting beneficial uses. The Basin Plan establishes water quality objectives specific to the North Coast Region for the protection of past, present, and probable future beneficial uses of water. Factors required for consideration during development of applicable water quality objectives, such as the characteristics of the hydrographic unit under consideration, economic considerations, and other factors required in accordance with section 13241 were considered during the Basin Planning and adoption process.

The Permittee did not submit any evidence regarding whether the waste discharge requirements for recycled water discharges would interfere with the development of needed housing within the region or the costs of compliance, particularly anything to show that the costs of compliance with the Order would be unmanageable.

Groundwater monitoring conducted semi-annually between November 2017 and August 2022 does not reveal any evidence of impacts to groundwater from the Permittee's recycled water discharge. The Permittee has not proposed any changes in discharge practices for the term of this Order.

4.7.2. Applicable Beneficial Uses and Water Quality Criteria and Objectives

- 4.7.2.1. **Beneficial Uses.** Beneficial use designations for groundwater established in the Basin Plan include MUN, AGR, IND, and PRO.

- 4.7.2.2. **Basin Plan Water Quality Objectives.** The Basin Plan contains narrative objectives for tastes and odors, bacteria, radioactivity, and chemical constituents (including those chemicals that adversely affect agricultural water supply) that apply to groundwater.

4.7.3. **Determining the Need for Requirements for Water Recycling**

- 4.7.3.1. The Water Recycling Specifications are established in this Order to conform to requirements contained in the California Code of Regulations, title 22, division 4, chapter 3 for the production of disinfected secondary recycled water. The Permittee is required to comply with applicable state and local requirements regarding the production and use of recycled water, including requirements of Water Code sections 13500 – 13577 (Water Reclamation) and DDW regulations at title 22, sections 60301 – 60357 of the California Code of Regulations (Water Recycling Criteria). The requirement is newly established in this Order.
- 4.7.3.2. **BOD₅ and TSS.** Consistent with Order No. R1-2017-0007, this Order establishes discharge limitations for BOD₅ and TSS based on technology-based effluent limitations that consist of a monthly average of 30 mg/L and a weekly average of 45 mg/L. These levels are technically achievable based on the capability of the secondary wastewater treatment system. These limits are included in the Order to ensure that discharges to the water recycling system receive proper treatment.
- 4.7.3.3. **Coliform Bacteria.** The Order includes effluent limitations for total coliform bacteria of 23 MPN/100 mL as a monthly median and 240 MPN/100 mL as a daily maximum. These limitations are based on regulations for secondary-23 recycled water contained in title 22, division 4, chapter 3 of the CCR to ensure that the quality of the water discharged to land is protective of human health. Although discharge to the irrigation site only requires undisinfected secondary recycled water based to meet the requirements for fodder for non-dairy producing livestock, coliform limitations established in this Order are appropriate for this use to ensure protection of groundwater beneficial use (MUN). These limitations can be reasonably achieved through proper operation of the Permittee's wastewater treatment facilities.
- 4.7.3.4. **pH.** Consistent with Order No. R1-2017-0007, this Order includes instantaneous minimum and maximum effluent limitations for pH of 6.5 and 8.5, Note that a more stringent effluent limitation range of 6.5 – 8.5 for pH is required to meet the water quality objective for hydrogen ion concentration (pH) in the South Fork Eel River contained in Basin Plan, Table 3-1.
- 4.7.3.5. **Nitrate and Total Nitrogen.** Nitrate is known to cause adverse health effects in humans. For waters designated as domestic or municipal supply, the Basin Plan (Chapter 3) adopts MCLs, established by DDW for the protection of public water supplies at title 22, section 64431 (Inorganic

Chemicals) and 64444 (Organic Chemicals) of the CCR, as applicable water quality criteria. The MCL for nitrate (10 mg/L) is therefore applicable as a water criterion.

The Permittee sampled its discharge at Monitoring Location EFF-003 monthly during periods of land application between June 2018 and June 2022. Monitoring results for nitrate ranged between 0.48 mg/L and 9.0 mg/L based on 10 samples and for total nitrogen ranged between 1 mg/L and 10 mg/L. These monitoring results reveal that discharges to the land application site met total nitrogen and nitrate land specifications in Order No. R1-2017-0007 for the entire period of record.

Nitrate and total nitrogen results were also analyzed for periods of discharge to surface waters in section 4.4.3.1.5 of this Fact Sheet. In addition, the antidegradation analysis described in section 3.3.6 of this Fact Sheet relied on the recycled water application discharge not exceeding a total nitrogen concentration of 10 mg/L. To retain consistency and in order to protect water quality and ensure proper operation of the Facility, an AMEL of 10 mg/L for total nitrogen from Order No. R1-2017- 0007 has been retained in this order for discharges to the Irrigation Site.

- 4.7.3.6. In addition, the MRP in Attachment E of this Order requires ongoing groundwater monitoring for nitrogen, salts and coliform bacteria to ensure that concentrations of pollutants will not adversely impact beneficial uses of groundwater.
- 4.7.3.7. This Order is consistent with the maximum benefit to people of the State because it allows continued operation of an existing wastewater treatment system, and it requires monitoring of groundwater to assess potential impacts from the recycled water use.

4.7.4. **Other Requirements**

The Order contains additional specifications that apply to the Facility regardless of the disposal method (surface water discharge or water recycling), including:

- 4.7.5. **Residual Chlorine.** Consistent with Order No. R1-2017-0007, this Order requires the Permittee to maintain a chlorine residual concentration that ensures the discharge meets the total coliform effluent limitations at the end of the disinfection process so that adequate pathogen reduction is continuously achieved at Discharge Points 001 and 003.

Continuous monitoring analyzers for chlorine residual or for dechlorination agent residual in the effluent are appropriate methods for compliance determination. A positive residual dechlorination agent in the effluent indicates that chlorine is not present in the discharge, which demonstrates compliance with the effluent limitations. This type of monitoring can also be used to prove that some chlorine residual exceedances are false positives. Continuous

monitoring data showing either a positive dechlorination agent residual or a chlorine residual at or below the prescribed limit are sufficient to show compliance with the total residual chlorine effluent limitation prescribed in section 4.1.1.1, Table 2, provided that the instruments are maintained and calibrated in accordance with the manufacturer's recommendations.

5. RATIONALE FOR RECEIVING WATER LIMITATIONS

5.1. Surface Water

CWA section 303(a-c) requires states to adopt water quality standards, including criteria where they are necessary to protect beneficial uses. The Regional Water Board adopted water quality criteria as water quality objectives in the Basin Plan. The Basin Plan states that "[t]he numerical and narrative water quality objectives define the least stringent standards that the Regional [Water] Board will apply to regional waters in order to protect the beneficial uses." The Basin Plan includes numeric and narrative water quality objectives for various beneficial uses and water bodies. This Order contains Receiving Surface Water Limitations based on the Basin Plan numerical and narrative water quality objectives for biostimulatory substances, bacteria, chemical constituents, color, dissolved oxygen, floating material, oil and grease, pH, pesticides, radioactivity, sediment, settleable material, suspended material, tastes and odors, temperature, toxicity, and turbidity.

5.2. Groundwater

- 5.2.1. The beneficial uses of the underlying groundwater are municipal and domestic supply, industrial service supply, industrial process supply, agricultural supply, and freshwater replenishment to surface waters.
- 5.2.2. Groundwater limitations are required to protect the beneficial uses of the underlying groundwater.
- 5.2.3. Discharges from the Facility shall not cause exceedance of applicable water quality objectives or create adverse impacts to beneficial uses of groundwater.
- 5.2.4. The Basin Plan requires that waters designated for use as MUN shall not contain concentrations of chemical constituents in excess of the limits specified in CCR, title 22, division 4, chapter 15, article 4.1, section 64431, and article 5.5, section 64444.

6. RATIONALE FOR PROVISIONS

6.1. Standard Provisions

6.1.1. Federal Standard Provisions

Standard Provisions, which apply to all NPDES permits in accordance with 40 C.F.R. section 122.41, and additional conditions applicable to specified categories of permits in accordance with 40 C.F.R. section 122.42, are provided in Attachment D to the Order. The Permittee must comply with all standard provisions and with those additional conditions that are applicable under 40 C.F.R. section 122.42. The rationale for the special conditions contained in the Order is provided in section 6.2, below.

Sections 122.41(a)(1) and (b) through (n) of 40 C.F.R. establish conditions that apply to all state issued NPDES permits. These conditions must be incorporated into the permits either expressly or by reference. If incorporated by reference, a specific citation to the regulations must be included in the Order. Section 123.25(a)(12) allows the state to omit or modify conditions to impose more stringent requirements. In accordance with 40 C.F.R. section 123.25, this Order omits federal conditions that address enforcement authority specified in 40 C.F.R. sections 122.41(j)(5) and (k)(2) because the enforcement authority under the Water Code is more stringent. In lieu of these conditions, this Order incorporates by reference Water Code section 13387(e).

6.1.2. Regional Water Board Standard Provisions

In addition to the Federal Standard Provisions (Attachment D), the Permittee shall comply with the Regional Water Board Standard Provisions provided in Standard Provisions 6.1.2 of the Order.

- 6.1.2.1. Order Provision 6.1.2.1 identifies the state's enforcement authority under the Water Code, which is more stringent than the enforcement authority specified in the federal regulations (e.g., 40 C.F.R. sections 122.41(j)(5) and (k)(2)).
- 6.1.2.2. Order Provision 6.1.2.2 requires the Permittee to notify Regional Water Board staff, orally and in writing, in the event that the Permittee does not comply or will be unable to comply with any Order requirement. This provision requires the Permittee to make direct contact with a Regional Water Board staff person.

6.2. Special Provisions

6.2.1. Reopener Provisions

- 6.2.1.1. **Standard Revisions (Special Provision 6.3.1.1).** Conditions that necessitate a major modification of a permit are described in 40 C.F.R. section 122.62, which include the following:
 - 6.2.1.1.1. When standards or regulations on which the permit was based have been changed by promulgation of amended standards or regulations or by judicial decision. Therefore, if revisions of applicable water quality standards are promulgated or approved pursuant to section 303 of the CWA or

amendments thereto, the Regional Water Board will revise and modify this Order in accordance with such revised standards.

- 6.2.1.1.2. When new information that was not available at the time of permit issuance would have justified different permit conditions at the time of issuance.
- 6.2.1.2. **Reasonable Potential (Special Provision 6.3.1.2).** This provision allows the Regional Water Board to modify, or revoke and reissue, this Order if present or future investigations demonstrate that the Permittee governed by this Permit is causing or contributing to excursions above any applicable priority pollutant criterion or objective, or adversely impacting water quality and/or the beneficial uses of receiving waters.
- 6.2.1.3. **Species Sensitivity Screening (Special Provision 6.3.1.3).** This provision allows the Regional Water Board to modify this Order if the species sensitivity screening identifies a most sensitive species that is different than the most sensitive species already identified in the Order.
- 6.2.1.4. **Whole Effluent Toxicity (Special Provision 6.3.1.4).** This Order requires the Permittee to investigate the causes of and identify corrective actions to reduce or eliminate effluent toxicity through a TRE. This Order may be reopened to include a numeric chronic toxicity limitation, new acute toxicity limitation, and/or a limitation for a specific toxicant identified in the TRE.
- 6.2.1.5. **Acute Aquatic Toxicity (Special Provision 6.3.1.5).** This provision allows the Regional Water Board to reopen this Order to include a MDEL and MMEL for acute aquatic toxicity, based on the reevaluation of the reasonable potential for the Permittee to cause or contribute to an exceedance of the acute aquatic toxicity water quality objective.
- 6.2.1.6. **303(d)-Listed Pollutants (Special Provision 6.3.1.6).** This provision allows the Regional Water Board to reopen this Order to modify existing effluent limitations or add effluent limitations for pollutants that are the subject of any future TMDL action.
- 6.2.1.7. **Water Effects Ratios (WERs) and Metal Translators (Special Provision 6.3.1.7).** This provision allows the Regional Water Board to reopen this Order if future studies undertaken by the Permittee provide new information and justification for applying a WER or metal translator to a water quality objective for one or more priority pollutants.
- 6.2.1.8. **Nutrients (Special Provision 6.3.1.8).** This Order contains effluent limitations for ammonia and effluent monitoring for nutrients (ammonia, nitrate, nitrite, organic nitrogen, and phosphorus). This provision allows the Regional Water Board to reopen this Order if future monitoring data indicates the need for new or revised effluent limitations for any of these parameters.

- 6.2.1.9. **Salt and Nutrient Management Plans (Special Provision 6.3.1.9).** This provision allows the Regional Water Board to reopen this Order if it adopts a regional or subregional SNMP that is applicable to the Permittee.
- 6.2.1.10. **Title 22 Recycled Water Engineering Report (Special Provision 6.3.1.10).** This provision allows the Regional Water Board to reopen this Order to adequately implement title 22, if necessary, based on the Permittee's title 22 engineering report.
- 6.2.1.11. **Mixing Zone Study.** (Special Provision 6.3.1.11) This provision allows the Regional Water Board to reopen this Order if the Permittee demonstrates to the satisfaction of the Regional Water Board Executive Officer that it has evaluated all reasonable alternatives for compliance with human health-based effluent limitations, if imposed, and conducts a mixing zone study that provides a basis for determining that permit conditions (i.e. effluent limitations and receiving water monitoring locations) should be considered for modification.
- 6.2.2. **Best Management Practices and Pollution Prevention**
 - 6.2.2.1. **Pollutant Minimization Program (Special Provision 6.3.3.1).** This provision is included in this Order pursuant to section 2.4.5 of the SIP. The Regional Water Board includes standard provisions in all NPDES permits requiring development of a Pollutant Minimization Program when there is evidence that a toxic pollutant is present in the effluent at a concentration greater than an applicable effluent limitation.
- 6.2.3. **Construction, Operation, and Maintenance Specifications**
 - 6.2.3.1. **Operation and Maintenance (Special Provisions 6.3.4.1 and 6.3.4.2).** 40 C.F.R. section 122.41(e) requires proper operation and maintenance of permitted wastewater systems and related facilities to achieve compliance with permit conditions. An up-to-date operation and maintenance manual, as required by Provision 6.3.4.2 of this Order, is an integral part of a well-operated and maintained facility.
- 6.2.4. **Special Provisions for Publicly Owned Treatment Works (POTWs)**
 - 6.2.4.1. **Wastewater Collection Systems (Special Provision 6.3.5.1)**
 - 6.2.4.1.1. **Statewide General WDRs for Sanitary Sewer Systems.** On December 6, 2022, the State Water Board adopted General Waste Discharge Requirements for Sanitary Sewer Systems, Water Quality Order No. 2022-0103-DWQ (General Order). The General Order requires public agencies that own or operate sanitary sewer systems with greater than one mile of pipes or sewer lines to enroll for coverage under the General Order. The General Order requires agencies to develop sanitary sewer management plans (SSMPs) and report all SSOs, among other requirements and

prohibitions. The Permittee has enrolled under the General Order as required.

- 6.2.4.2. **Source Control Provisions (Special Provision 6.3.5.2).** Pursuant to Special Provision 6.3.5.2.1, the Permittee shall implement the necessary legal authorities to monitor and enforce source control standards, restrict discharges of toxic materials to the collection system, and inspect facilities connected to the system.

40 C.F.R. section 403.8(a) requires POTWs with a total design flow greater than 5 mgd and receiving pollutants which pass through or interfere with the operation of the POTW to establish a POTW Pretreatment Program. The Regional Water Board may also require that a POTW with a design flow of 5 mgd or less develop a POTW Pretreatment Program if the nature or volume of the industrial influent, treatment process upsets, violations of POTW effluent limitations, contamination of municipal sludge, or other circumstances warrant in order to prevent interference or pass through. The Permittee did not report any known industrial wastes subject to regulation under the NPDES Pretreatment Program being discharged to the Facility in section 4 of EPA Application Form 2A and the permitted flow of the Facility is less than 5 mgd; therefore, the Order does not require the Permittee to develop a pretreatment program that conforms to federal regulations. However, in order to prevent interference with the POTW or pass through of pollutants to the receiving water, the Order requires the Permittee to implement a source control program.

Water Code section 13263.3(d)(1) allows the Regional Water Board to require a discharger to complete and implement a pollution prevention plan if pollution prevention is necessary to achieve a water quality objective, to include, pursuant to Water Code section 13263.3(d)(3), an analysis of the methods that could be used to prevent the discharge of the pollutants into the POTW. These methods can include application of local limits to industrial or commercial dischargers, pollution prevention techniques, public education and outreach, or other innovative and alternative approaches to reduce discharges of pollutants to the POTW. The analysis also shall identify sources, or potential sources, not within the ability or authority of the POTW to control, such as pollutants in the potable water supply, airborne pollutants, pharmaceuticals, or pesticides, and estimate the magnitude of those sources, to the extent feasible. This Order includes requirements for the Permittee to implement a source identification and reduction program.

A key component of an effective source control program is the identification and location of possible industrial users within the POTW's wastewater collection system. This information is typically obtained by the POTW through industrial waste surveys. The following types of resources can be consulted in compiling a master list of industrial users:

- 6.2.4.2.1. Water and sewer billing records
- 6.2.4.2.2. Applications for sewer service
- 6.2.4.2.3. Local telephone directories
- 6.2.4.2.4. Chamber of Commerce and local business directories
- 6.2.4.2.5. Business license records
- 6.2.4.2.6. POTW and wastewater collection personnel and field observations
- 6.2.4.2.7. Business associations
- 6.2.4.2.8. The internet
- 6.2.4.2.9. Industrial and non-residential sewer use permit records

In addition, the Regional Water Board recognizes that some form of source control is prudent to ensure the efficient operation of the Facility, the safety of Facility staff, and to ensure that pollutants do not pass through the treatment Facility to impair the beneficial uses of the receiving water. The proposed Order includes prohibitions for the discharge of pollutants that may interfere, pass through, or be incompatible with treatment operations, interfere with the use of disposal of sludge, or pose a health hazard to personnel.

- 6.2.5. **Sludge Disposal and Handling Requirements (Special Provision 6.3.5.3).** The disposal or reuse of wastewater treatment screenings, sludges, or other solids removed from the liquid waste stream is regulated by 40 C.F.R. parts 257, 258, 501, and 503, and the State Water Board promulgated provisions of title 27 of the CCR.
- 6.2.6. **Biosolids Management (Special Provision 6.3.5.4).** This provision requires the Permittee to comply with the State's regulations relating to the discharge of biosolids to the land. The discharge of biosolids through land application is not regulated under this Order. The Permittee is required to obtain coverage under the State Water Board Order No. 2004-0012-DWQ, General Waste Discharge Requirements for the Discharge of Biosolids to Land as a Soil Amendment in Agricultural, Silvicultural, Horticultural, and Land Reclamation Activities (General Order). Coverage under the General Order, as opposed to coverage under this NPDES permit or individual WDRs, implements a consistent statewide approach to regulating this waste discharge.
- 6.2.7. **Operator Certification (Special Provision 6.3.5.5).** This provision requires the Facility to be operated by supervisors and operators who are certified as required by title 23, section 3680 of the CCR.

6.2.8. **Adequate Capacity (Special Provision 6.3.5.6).** The goal of this provision is to ensure appropriate and timely planning by the Permittee to ensure adequate capacity for the protection of public health and water quality.

6.2.9. **Other Special Provisions**

6.2.9.1. **Storm Water (Special Provision 6.3.6.1).** This provision requires the Permittee, if applicable, to obtain coverage under the State Water Board's Water Quality Order No. 2014-0057-DWQ, NPDES General Permit No. CAS000001, General Permit for Storm Water Discharges Associated with Industrial Activities (or subsequent renewed versions of the NPDES General Permit CAS000001). Currently, the Facility is exempted from these requirements based on the size of the Facility (less than 1 mgd). In addition, all storm water that falls within the Facility is captured, treated, and disposed of within the Facility's NPDES permitted wastewater process.

The Order requires the Permittee to implement and maintain BMPs to control the run-on of storm water to the Facility and to describe the effectiveness of these storm water BMPs, as well as activities to maintain and upgrade these BMPs during the previous year, in its Annual Facility Report to the Regional Water Board.

6.2.10. **Compliance Schedules**

6.2.10.1. Compliance schedule for the final effluent limitations for dichlorobromomethane, chlorodibromomethane, total trihalomethanes, and haloacetic acids (Special Provision 6.3.6.2). Consistent with Order No. R1-2023-0030, this Order includes a compliance schedule for the Permittee to achieve compliance with final effluent limitations for dichlorobromomethane, chlorodibromomethane, total trihalomethanes, and haloacetic acids. The compliance schedule is needed because the Order includes final effluent limitations for dichlorobromomethane, chlorodibromomethane, total trihalomethanes, and haloacetic acids that will require the Permittee to implement actions, such as designing and constructing facilities or implementing new or significantly expanded programs and securing financing to comply with the permit limitations that are included in this Order to implement water quality objectives.

6.2.10.2. The compliance schedule is in accordance with the State Water Board Compliance Schedule Policy based on the Permittee providing written documentation demonstrating that it needs additional time to complete tasks needed to comply with the more stringent final dichlorobromomethane, chlorodibromomethane, total trihalomethanes, and haloacetic acids limitations. The Permittee needs time to design facilities and/or develop a program and secure financing for final upgrades. Regional Water Board staff determined that the Permittee should be able to complete the proposed tasks in a period of six years if the Permittee finds chloramination to be

effective during the full-scale testing period. If the full-scale testing report concludes that final compliance cannot be achieved using chloramination, then an additional four years have been granted to determine and implement an alternate strategy to comply with final effluent limitations. The compliance schedule authorized in Order No. R1- 2023-0031 and continued in this Order now provides a total of ten years for the Permittee to achieve compliance with the final effluent limitation for dichlorobromomethane, chlorodibromomethane, total trihalomethanes, and haloacetic acids. The compliance schedule in this Order authorizes interim effluent limitations set forth in Table 3, because the Permittee has continued to make diligent efforts to quantify pollutant levels in the discharge and implement effective process control. The compliance schedule will result in the highest discharge quality that can be achieved until final compliance is attained. The compliance schedules established in this Order provides the shortest time possible to achieve the final effluent limitations for dichlorobromomethane, chlorodibromomethane, total trihalomethanes, and haloacetic acids given the challenges in implementing facility improvements, and/or alternative strategies that will result in compliance with final disinfection byproduct limitations. No interim compliance dates exceed one year.

7. RATIONALE FOR MONITORING AND REPORTING REQUIREMENTS

CWA section 308 and 40 C.F.R. sections 122.41(h), (j)-(l), 122.44(i), and 122.48 require that all NPDES permits specify monitoring and reporting requirements. Water Code sections 13267 and 13383 also authorize the Regional Water Board to establish monitoring, inspection, entry, reporting, and recordkeeping requirements. The Monitoring and Reporting Program (MRP), Attachment E of this Order establishes monitoring, reporting, and recordkeeping requirements that implement federal and state requirements. The following provides the rationale for the monitoring and reporting requirements contained in the MRP for this facility.

7.1. Influent Monitoring

- 7.1.1. Influent monitoring requirements at Monitoring Location INF-001 for BOD₅ and TSS are retained from Order No. R1-2017-0007 and are necessary to determine compliance with the Order's 85 percent removal requirement for these parameters.
- 7.1.2. Influent monitoring requirements for flow at Monitoring Location INF-001 are retained from Order No. R1-2017-0007 and are necessary to determine compliance with Discharge Prohibition 3.9.

7.2. Effluent Monitoring

- 7.2.1. Effluent monitoring requirements are necessary to determine compliance with prohibitions and/or effluent limitations established by the Order. Monitoring at Monitoring Location EFF-001 is necessary to demonstrate compliance with

effluent limitations and demonstrate whether or not the discharge poses reasonable potential for a pollutant to exceed any numeric or narrative water quality objectives.

- 7.2.1.1. Effluent monitoring frequencies and sample types for flow, BOD₅, TSS, pH, temperature, total coliform bacteria, hardness, ammonia, residual chlorine, dissolved oxygen, nitrate, chlorodibromomethane, dichlorobromomethane, total trihalomethanes, Haloacetic Acids, and phosphorus at Monitoring Location EFF-001 have been retained from Order No. R1-2017-0007.
- 7.2.1.2. This Order includes a prohibition of discharges that exceed one percent of the flow of the Lower Eel River. Therefore, this Order requires the Permittee to calculate and report the discharge flow rate as a percentage of receiving water flow.
- 7.2.1.3. Effluent monitoring for *E. coli* bacteria has been established at Monitoring Location EFF-001 in this Order to inform Regional Water Board staff of the reasonable potential for the Permittee to exceed water quality objectives when discharging to Lower Eel River.
- 7.2.1.3.1. There are 5 distinct six-week periods between October 1 through May 14 and the Order establishes a monitoring regime that allows the Discharger to select a minimum of three six-week periods annually over the authorized discharge period to monitor its effluent and the receiving water to determine compliance with the bacteria objectives. This monitoring regime allows the Discharger to monitor the receiving water for approximately 60 percent of the discharge season instead of monitoring every week over the 32-week discharge season, reducing the cost of permit compliance for a disadvantaged community, and allowing the Discharger to plan and implement its monitoring program more efficiently.
- 7.2.1.4. Monitoring data collected over the term of Order No. R1-2017-0007 demonstrated that the discharge exhibits reasonable potential to cause or contribute to an exceedance of water quality criteria for antimony, and 4,4-DDT. Therefore, this Order establishes monitoring requirements for antimony, and 4,4-DDT at Monitoring Location EFF-001 to determine compliance with the applicable effluent limitations.
- 7.2.1.5. This Order is consistent with the requirements of the Recycled Water Policy to implement a SNMP. The Recycled Water Policy currently requires monitoring for priority pollutants annually. This Order requires the Permittee to monitor priority pollutants once per permit term.
- 7.2.1.6. This Order requires calculation of BOD₅ and TSS percent removal, in order to assess compliance with the BOD and TSS percent removal effluent limitations in Effluent Limitation sections 4.1.1.2 of the Order.

7.3. Toxicity Testing Requirements

Effluent monitoring data collected during the term of Order No. R1-2017-0007 indicates that the discharge does not exhibit reasonable potential to cause or contribute to an exceedance of water quality objectives for acute aquatic toxicity. Therefore, this Order discontinues quarterly effluent monitoring requirements for acute aquatic toxicity. Effluent data indicates that the discharge exhibits reasonable potential to cause or contribute to an exceedance of water quality objectives for chronic aquatic toxicity. This Order includes annual effluent monitoring requirements for chronic aquatic toxicity, as required by the Toxicity Provisions.

In addition to routine chronic toxicity monitoring, this Order requires the Permittee to maintain and update its TRE Work Plan, in accordance with appropriate U.S. EPA guidance to ensure that the Permittee has a plan to immediately move forward with the initial tiers of a TRE in the event effluent toxicity is encountered in the future. The TRE is initiated by evidence of a pattern of toxicity demonstrated through routine and MMEL testing for chronic aquatic toxicity.

The Toxicity Provisions allow the Regional Water Board to use a species sensitivity screening generated within ten years prior to the renewal of this Permit when the data are representative of the effluent, the Regional Water Board accepts use of the data, the data are analyzed using the TST, and the data are from chronic aquatic testing of, at minimum, one vertebrate, one invertebrate, and one plant/algae from Table 1 of Section IV.B.1.b. of the Toxicity Provisions. The Regional Water Board has determined that the species sensitivity screening conducted between March 9 and 13 of 2018, meets the above requirements, and the species used for chronic toxicity monitoring shall be *Ceriodaphnia dubia*.

7.4. Recycled Water Monitoring Requirements

This Order requires that the Permittee comply with applicable state and local requirements regarding the production of recycled water.

Recycled water monitoring requirements at Monitoring Location EFF-003 for flow, BOD₅, TSS, total coliform bacteria, and pH have been retained from Order No. R1-2017-0007.

7.5. Receiving Water Monitoring

7.5.1. Surface Water Monitoring

- 7.5.1.1. Receiving water monitoring is required to demonstrate compliance with the Receiving Water Limitations.
- 7.5.1.2. Monitoring requirements at Monitoring Location RSW-001 for flow, BOD₅, pH, hardness, temperature, turbidity, nitrate, phosphorus, and CTR priority pollutants have been retained from Order No. R1-2017-0007.

- 7.5.1.3. Monitoring Requirements at Monitoring Location RSW-002 for pH, temperature, dissolved oxygen, electrical conductivity, total dissolved solids, and turbidity have been retained from Order No. R1-2017-0007.
- 7.5.1.4. Receiving water monitoring for *E. coli* bacteria has been established in this Order to determine background levels of *E. coli* in the receiving water and to determine compliance with bacteria water quality objectives.
 - 7.5.1.4.1. There are 5 distinct six-week periods between October 1 through May 14 and the Order establishes a monitoring regime that allows the Discharger to select a minimum of three six-week periods annually over the authorized discharge period to monitor its effluent and the receiving water to determine compliance with the bacteria objectives. This monitoring regime allows the Discharger to monitor the receiving water for approximately 60 percent of the discharge season instead of monitoring every week over the 32-week discharge season, reducing the cost of permit compliance for a disadvantaged community, and allowing the Discharger to plan and implement its monitoring program more efficiently. Receiving water monitoring requirements include three monitoring events concurrent with effluent monitoring to provide a representative characterization of effluent and receiving water conditions at the time of discharge.

7.5.2. **Groundwater Monitoring**

- 7.5.2.1. Monitoring requirements at Monitoring Locations MW-001 through MW-004 for depth to groundwater, nitrate, total coliform, chloroform, and total dissolved solids have been retained from Order No. R1-2017-0007.
- 7.5.2.2. The Regional Water Board finds that this Order's effluent monitoring requirement for the title 22 pollutants that have been identified in the effluent, will provide sufficient information to characterize the impacts of the discharge to groundwater. Therefore, monitoring requirements at Monitoring Locations MW-001 through MW-004 have been added for 4,4-DDT, and antimony because these pollutants were detected at concentrations in excess of applicable MCLs in effluent and/or groundwater in title 22 monitoring conducted between November 2017 and August 2022.

7.6. **Other Monitoring Requirements**

- 7.6.1. **Monitoring Location INT-001.** Internal monitoring at the end of the chlorine contact chamber is required to measure chlorine residual in lieu of daily coliform monitoring to assure adequate disinfection on a daily basis by demonstrating that the appropriate chlorine residual concentration is maintained in the effluent at INT-001 at all times.
- 7.6.2. **Visual Monitoring (Monitoring Locations EFF-001 and RSW-001).** Visual monitoring requirements are retained from the previous Order and are

necessary to ensure compliance with Receiving Water Limitations sections 5.1.6, 5.1.7, 5.1.0, 5.1.10, 5.1.11, and 5.1.16 of this Order.

- 7.6.3. **Sludge Monitoring (MRP section 9.4).** Sludge monitoring requirements at Monitoring Location BIO-001 serve as a basis for the Permittee to develop the Sludge Handling and Disposal Activity Report that is required as part of the Annual Report pursuant to section 10.4.3.7 of the MRP.
- 7.6.4. **Discharge Monitoring Report-Quality Assurance (DMR-QA) Study Program (MRP section 1.6).** Under the authority of section 308 of the CWA (33 U.S.C. § 118), U.S. EPA requires major and selected minor dischargers under the NPDES Program to participate in the annual DMR-QA Study Program. The DMR-QA Study evaluates the analytical ability of laboratories that routinely perform or support self-monitoring analyses required by NPDES permits. There are two options to satisfy the requirements of the DMR-QA Study Program: (1) The Discharger can obtain and analyze a DMR-QA sample as part of the DMR-QA Study; or (2) Per the waiver issued by U.S. EPA to the State Water Board, the Discharger can submit the results of the most recent Water Pollution Performance Evaluation Study from its own laboratories or its contract laboratories. A Water Pollution Performance Evaluation Study is similar to the DMR-QA Study. Thus, it also evaluates a laboratory's ability to analyze wastewater samples to produce quality data that ensure the integrity of the NPDES Program. The Discharger shall ensure that the results of the DMR-QA Study or the results of the most recent Water Pollution Performance Evaluation Study are submitted annually to the State Water Board. The State Water Board's Quality Assurance Program Officer will send the DMR-QA Study results or the results of the most recent Water Pollution Performance Evaluation Study to U.S. EPA's DMR-QA Coordinator and Quality Assurance Manager.
- 7.6.5. **Accelerated Monitoring Requirements.** Table E-4 includes accelerated monitoring requirements for parameters that are required to be monitored weekly, monthly, quarterly, and annually. Accelerated monitoring is required for all pollutants with effluent limitations.
- 7.6.6. **Flow Monitoring.** Sections 1.4 and 10.4.3.5 of the MRP require proper installation, calibration, operation, and maintenance of flow metering devices.
- 7.6.7. **Spill Notification.** The MRP that is part of this Order establishes requirements for reporting spills and unauthorized discharges, with the exception of SSOs which must be reported in accordance with the requirements of State Water Board Order No. 2022-0103-DWQ.
- 7.6.8. **Notification and Reporting for Recycled Water Spills.** Section 10.5.4 of the MRP includes reporting requirements for spills of secondary treated water in excess of 50,000 gallons. This requirement implements Water Code section 13529.2.

- 7.6.9. **Disaster Preparedness Assessment Report and Action Plan** (MRP section 10.4.2). Natural disasters, extreme weather events, sea level rise, and shifting precipitation patterns, some of which are projected to intensify due to climate change, have significant implications for wastewater treatment and operations. Some natural disasters are expected to become more frequent and extreme according to the current science on climate change. In order to ensure that Facility operations are not disrupted, compliance with conditions of this Order are achieved, and receiving waters are not adversely impacted by permitted and unpermitted discharges, this Order requires the Permittee to submit a Disaster Preparedness Assessment Report and Action Plan and implement the necessary control measures in accordance with an approved schedule of implementation.

8. PUBLIC PARTICIPATION

The California Regional Water Quality Control Board, North Coast Region (North Coast Regional Water Board) has considered the issuance of waste discharge requirements (WDRs) that will serve as a National Pollutant Discharge Elimination System (NPDES) permit for the City of Rio Dell Wastewater Treatment Facility. As a step in the WDR adoption process, the Regional Water Board staff has developed tentative WDRs. The Regional Water Board encourages public participation in the WDR adoption process.

8.1. Notification of Interested Parties

The Regional Water Board notified the Permittee and interested agencies and persons of its intent to prescribe waste discharge requirements for the discharge and provided an opportunity to submit written comments and recommendations. Notification was provided through the following posting on the Notification was provided through the following posting on the [Regional Water Board's site](https://www.waterboards.ca.gov/northcoast/public_notices/public_hearings/npdes_permits_and_wdrs/) at: (https://www.waterboards.ca.gov/northcoast/public_notices/public_hearings/npdes_permits_and_wdrs/).

The public had access to the agenda and any changes in dates and locations through the [Regional Water Board's site](https://www.waterboards.ca.gov/northcoast/public_notices/public_hearings/npdes_permits_and_wdrs/) at: (https://www.waterboards.ca.gov/northcoast/public_notices/public_hearings/npdes_permits_and_wdrs/).

8.2. Written Comments

Interested persons were invited to submit written comments concerning these tentative WDRs as provided through the notification process. Comments were due to the Regional Water Board Executive Office electronically via e-mail to NorthCoast@waterboards.ca.gov or on disk (CD or DCD) in Portable Document Format (PDF) file in lieu of paper-sourced documents. The guidelines for electronic submittal of documents can be found on the [Regional Water Board website](https://www.waterboards.ca.gov/northcoast/) at (<https://www.waterboards.ca.gov/northcoast/>).

To be fully responded to by staff and considered by the Regional Water Board, the written comments were due at the Regional Water Board office by 5:00 p.m. on **June 18, 2023**.

8.3. Public Hearing

The Regional Water Board held a public hearing on the tentative WDRs during its regular Board meeting on the following date and time and at the following location:

Date: **October 5, 2023**
Time: 9:00 a.m. or as announced in the Regional Water Board's agenda
Location: Regional Water Board Hearing Room
5550 Skylane Boulevard, Suite A
Santa Rosa, CA 95403

Interested persons were invited to attend. At the public hearing, the Regional Water Board heard testimony, pertinent to the discharge, WDRs, and permit. For accuracy of the record, important testimony was requested in writing.

Please be aware that dates and venues may change. On the [Regional Water Board website](https://www.waterboards.ca.gov/northcoast/) (<https://www.waterboards.ca.gov/northcoast/>) you can access the current agenda for changes in dates and locations.

8.4. Reconsideration of Waste Discharge Requirements

Any person aggrieved by this action of the Regional Water Board may petition the State Water Board to review the action in accordance with Water Code section 13320 and California Code of Regulations, title 23, sections 2050 and following. The State Water Board must receive the petition by 5:00 p.m., within 30 calendar days of the date of adoption of this Order at the following address, except that if the thirtieth day following the date of this Order falls on a Saturday, Sunday, or state holiday, the petition must be received by the State Water Board by 5:00 p.m. on the next business day:

State Water Resources Control Board
Office of Chief Counsel
P.O. Box 100, 1001 I Street
Sacramento, CA 95812-0100

Or by email at waterqualitypetitions@waterboards.ca.gov

For instructions on how to file a petition for review, see the at [Water Quality Petitions Website](http://www.waterboards.ca.gov/public_notices/petitions/waterquality/wqpetition_instr.shtml) (http://www.waterboards.ca.gov/public_notices/petitions/waterquality/wqpetition_instr.shtml)

8.5. Information and Copying

The Report of Waste Discharge, other supporting documents, and comments received are on file and may be inspected at the address above at any time between 8:30 a.m. and 4:45 p.m., Monday through Friday. Copying of documents may be arranged through the Regional Water Board by calling (707) 576-2220.

8.6. Register of Interested Persons

Any person interested in being placed on the mailing list for information regarding the WDRs and NPDES permit should contact the Regional Water Board, reference this Facility, and provide a name, address, and phone number.

8.7. Additional Information

Requests for additional information or questions regarding this order should be directed to Sabrina Cegielski at Sabrina.Cegielski@waterboards.ca.gov or (707) - 543-7126.

Table F-12. Wastewater Treatment Facility RPA Summary

Constituent Name	Units		MEC ¹		B	C	CMC	CCC	Water & Org. ²	Org. Only ³	MCL	RP ^{4,5}
Antimony, Total Recoverable	ug/L		11		10	6	--	--	14	4300	6	Yes
Arsenic, Total Recoverable	ug/L	<	6.9	<	6.9	10	340	150	--	--	10	No
Beryllium, Total Recoverable	ug/L	<	0.42	<	0.42	4	--	--	--	--	4	No
Cadmium, Total Recoverable	ug/L	<	0.55	<	0.55	1.9	3.2	1.9	--	--	5	No
Copper, Total Recoverable	ug/L		3.8		1.9	7.1	10	7.1	1300	--	1000	No
Lead, Total Recoverable	ug/L	<	2.9	<	2.9	15	55	2.1	--	--	15	No
Mercury, Total Recoverable	ug/L	<	0.16	<	0.16	2	--	--	--	--	2	No
Nickel, Total Recoverable	ug/L		3.1	<	3.1	40	360	40	610	4600	100	No
Selenium, Total Recoverable	ug/L	<	9.5	<	9.5	5	20	5	170	4200	50	Uo
Silver, Total Recoverable	ug/L	<	2.8	<	--	100	--	--	--	--	100	No
Silver, Total Recoverable	ug/L	<	--	<	2.8	2.4	2.4	--	--	--	100	Uo
Thallium, Total Recoverable	ug/L	<	3.4	<	3.4	1.7	--	--	1.7	6.3	2	Uo
Zinc, Total Recoverable	ug/L		30	<	2.7	92	92	92	7400	26000	5000	No
Cyanide, Total (as CN)	mg/L	<	0.0007	<	0.0007	5.2	22	5.2	700	220000	150	No

Constituent Name	Units		MEC ¹		B	C	CMC	CCC	Water & Org. ²	Org. Only ³	MCL	RP ^{4,5}
Asbestos	MFL	<	0.18	<	0.18	7	--	--	7	--	7	No
2,3,7,8-TCDD (Dioxin)	pg/L	<	0.368	<	0.368	0.013	--	--	0.013	0.013	0.00003	Uo
Acrolein	ug/L	<	0.71	<	0.71	3	3	3	320	780	--	No
Acrylonitrile	ug/L	<	0.19	<	0.19	0.059	--	--	0.059	0.66	--	Uo
Benzene	ug/L	<	0.28	<	0.28	1	--	--	1.2	71	1	No
Bromoform	ug/L	<	0.32	<	0.32	4.3	--	--	4.3	360	80	No
Carbon tetrachloride	ug/L	<	0.44	<	0.44	0.25	--	--	0.25	4.4	0.5	Uo
Chlorobenzene	ug/L	<	0.2	<	0.2	70	--	--	680	21000	70	No
Chlorodibromo-methane	ug/L		2.6	<	0.24	0.41	--	--	0.41	34	80	Yes
Chloroethane	ug/L	<	0.13	<	0.13	No Criteria	--	--	--	--	--	No
2-Chloroethylvinyl Ether	ug/L	<	0.33	<	0.33	No Criteria	--	--	--	--	--	No
Chloroform	ug/L		90.7	<	0.33	80	--	--	--	470	80	Yes
Dichlorobromo-methane	ug/L		16.3	<	0.32	0.56	--	--	0.56	46	80	Yes
1,1-Dichloroethane	ug/L	<	0.29	<	0.29	5	--	--	--	--	5	No
1,2-Dichloroethane	ug/L	<	0.32	<	0.32	0.38	--	--	0.38	99	0.5	No
1,1-Dichloroethylene	ug/L	<	0.33	<	0.33	0.057	--	--	0.057	3.2	6	Uo
1,2-Dichloropropane	ug/L	<	0.25	<	0.25	0.52	--	--	0.52	39	5	No
1,3-Dichloropropylene	ug/L	<	0.17	<	0.17	0.5	--	--	10	1700	0.5	No
Ethylbenzene	ug/L	<	0.2	<	0.2	300	--	--	100	29000	300	No

Constituent Name	Units		MEC ¹		B	C	CMC	CCC	Water & Org. ²	Org. Only ³	MCL	RP ^{4,5}
Methyl Bromide	ug/L	<	0.24	<	0.24	48	--	--	48	4000	--	No
Methyl Chloride	ug/L	<	0.15	<	0.15	No Criteria	--	--	--	--	--	No
Methylene Chloride	ug/L	<	0.14	<	0.14	4.7	--	--	4.7	1600	5	No
1,1,2,2-Tetrachloroethane	ug/L	<	0.16	<	0.16	0.17	--	--	0.17	11	1	No
Tetrachloroethylene	ug/L	<	0.23	<	0.23	0.8	--	--	0.8	8.85	5	No
Toluene	ug/L		0.65	<	0.16	150	--	--	6800	200000	150	No
trans-1,2-Dichloroethylene	ug/L	<	0.26	<	0.26	10	--	--	700	140000	10	No
1,1,1-Trichloroethane	ug/L	<	0.31	<	0.31	200	--	--	--	--	200	No
1,1,2-Trichloroethane	ug/L	<	0.21	<	0.21	0.6	--	--	0.6	42	5	No
Trichloroethylene	ug/L	<	0.25	<	0.25	2.7	--	--	2.7	81	5	No
Vinyl Chloride	ug/L	<	0.07	<	0.07	0.5	--	--	2	525	0.5	Uo
2-Chlorophenol	ug/L	<	1.8	<	1.9	120	--	--	120	400	--	No
2,4-Dichlorophenol	ug/L	<	2.2	<	2.2	93	--	--	93	790	--	No
2,4-Dimethylphenol	ug/L	<	1.6	<	1.6	540	--	--	540	2300	--	No
2-Methyl-4,6-Dinitrophenol	ug/L	<	6	<	6	13.4	--	--	13.4	765	--	No
2,4-Dinitrophenol	ug/L	<	14	<	14	70	--	--	70	14000	--	No

Constituent Name	Units		MEC ¹		B	C	CMC	CCC	Water & Org. ²	Org. Only ³	MCL	RP ^{4,5}
2-Nitrophenol	ug/L	<	1.8	<	1.9	No Criteria	--	--	--	--	--	No
4-Nitrophenol	ug/L	<	8	<	8	No Criteria	--	--	--	--	--	No
4-Chloro-3-methylphenol	ug/L	<	1.9	<	1.9	No Criteria	--	--	--	--	--	No
Pentachlorophenol	ug/L	<	23	<	23	0.28	5.28	4.05	0.28	8.2	1	Uo
Phenol, Single Compound	ug/L	<	0.9	<	0.93	21000	--	--	21000	5E+06	--	No
2,4,6-Trichlorophenol	ug/L	<	2.1	<	2.1	2.1	--	--	2.1	6.5	--	Uo
Acenaphthene	ug/L	<	1.8	<	1.8	1200	--	--	1200	2700	--	No
Acenaphthylene	ug/L	<	1.6	<	1.7	No Criteria	--	--	--	--	--	No
Anthracene	ug/L	<	2.0	<	2.0	9600	--	--	9600	110000	--	No
Benzidine	ug/L	<	22	<	22	0.00012	--	--	0.00012	0.0005	--	Uo
Benzo(a)anthracene	ug/L	<	1.5	<	1.5	0.0044	--	--	0.0044	0.049	--	Uo
Benzo(a)pyrene	ug/L	<	2.3	<	2.3	0.0044	--	--	0.0044	0.049	0.2	Uo
Benzo(b)fluoranthene	ug/L	<	2.6	<	2.6	0.0044	--	--	0.0044	0.049	--	Uo
Benzo(ghi)perylene	ug/L	<	1.7	<	1.7	No Criteria	--	--	--	--	--	No
Benzo(k)fluoranthene	ug/L	<	2	<	2	0.0044	--	--	0.0044	0.049	--	Uo
Bis (2-Chloroethoxy) Methane	ug/L	<	1.7	<	1.7	No Criteria	--	--	--	--	--	No
Bis (2-Chloroethyl) Ether	ug/L	<	2.3	<	2.3	0.01	--	--	0.01	1.4	--	Uo

Constituent Name	Units		MEC ¹		B	C	CMC	CCC	Water & Org. ²	Org. Only ³	MCL	RP ^{4,5}
Bis (2-Chloroisopropyl) Ether	ug/L	<	4	<	4	1400	--	--	1400	170000	--	No
Bis (2-Ethylhexyl) Phthalate	ug/L	<	2	<	2	1.8	--	--	1.8	5.9	4	Uo
4-Bromophenyl Phenyl Ether	ug/L	<	1.7	<	1.7	No Criteria	--	--	--	--	--	No
Butylbenzyl Phthalate	ug/L	<	1.6	<	1.6	3000	--	--	3000	5200	--	No
2-Chloronaphthalene	ug/L	<	1.8	<	1.8	1700	--	--	1700	4300	--	No
4-Chlorophenyl Phenyl Ether	ug/L	<	1.7	<	1.7	No Criteria	--	--	--	--	--	No
Chrysene	ug/L	<	1.8	<	1.8	0.0044	--	--	0.0044	0.049	--	Uo
Dibenzo(a,h)anthracene	ug/L	<	1.9	<	1.9	0.0044	--	--	0.0044	0.049	--	Uo
1,2-Dichlorobenzene	ug/L	<	2.1	<	2.1	600	--	--	2700	17000	600	No
1,3-Dichlorobenzene	ug/L	<	2	<	2	400	--	--	400	2600	--	No
1,4-Dichlorobenzene	ug/L	<	2.3	<	2.3	5	--	--	400	2600	5	No
3,3-Dichlorobenzidine	ug/L	<	1.8	<	1.8	0.04	--	--	0.04	0.077	--	Uo
Diethyl Phthalate	ug/L	<	2.0	<	2.0	23000	--	--	23000	120000	--	No
Dimethyl Phthalate	ug/L	<	1.9	<	1.9	13000	--	--	13000	3E+06	--	No
Di-n-butyl Phthalate	ug/L	<	2.1	<	2.2	2700	--	--	2700	12000	--	No
2,4-Dinitrotoluene	ug/L	<	1.9	<	1.9	0.11	--	--	0.11	9.1	--	Uo
2,6-Dinitrotoluene	ug/L	<	2	<	2.1	No Criteria	--	--	--	--	--	No
Di-n-octyl Phthalate	ug/L	<	7.7	<	7.9	No Criteria	--	--	--	--	--	No
1,2-Diphenylhydrazine	ug/L	<	0.9	<	0.9	0.04	--	--	0.04	0.54	--	Uo

Constituent Name	Units		MEC ¹		B	C	CMC	CCC	Water & Org. ²	Org. Only ³	MCL	RP ^{4,5}
Fluoranthene	ug/L	<	2.1	<	2.1	300	--	--	300	370	--	No
Fluorene	ug/L	<	1.9	<	2	1300	--	--	1300	14000	--	No
Hexachlorobenzene	ug/L	<	2.3	<	2.3	0.00075	--	--	0.00075	0.0008	1	Uo
Hexachlorobutadiene	ug/L	<	2.5	<	2.6	0.44	--	--	0.44	50	--	Uo
Hexachlorocyclopentadiene	ug/L	<	16	<	17	50	--	--	240	17000	50	No
Hexachloroethane	ug/L	<	2.2	<	2.2	1.9	--	--	1.9	8.9	--	Uo
Indeno (1,2,3-cd) Pyrene	ug/L	<	1.7	<	1.8	0.0044	--	--	0.0044	0.049	--	Uo
Isophorone	ug/L	<	1.8	<	1.8	8.4	--	--	8.4	600	--	No
Naphthalene	ug/L	<	1.9	<	2	No Criteria	--	--	--	--	--	No
Nitrobenzene	ug/L	<	2.8	<	2.8	17	--	--	17	1900	--	No
N-Nitrosodimethylamine	ug/L	<	1.7	<	1.7	0.00069	--	--	0.00069	8.1	--	Uo
N-Nitrosodi-n-Propylamine	ug/L	<	2.3	<	2.4	0.005	--	--	0.005	1.4	--	Uo
N-Nitrosodiphenylamine	ug/L	<	2.3	<	2.3	5	--	--	5	16	--	No
Phenanthrene	ug/L	<	1.9	<	2	No Criteria	--	--	--	--	--	No
Pyrene	ug/L	<	1.7	<	0.8	960	--	--	960	11000	--	No
1,2,4-Trichlorobenzene	ug/L	<	1.9	<	1.9	5	--	--	35	70	5	No
Aldrin	ug/L	<	0.002	<	0.002	0.00013	3	--	0.00013	0.0001	--	Uo
alpha-BHC	ug/L	<	0.002	<	0.002	0.0039	--	--	0.0039	0.013	--	No
beta-BHC	ug/L	<	0.005	<	0.005	0.014	--	--	0.014	0.046	--	No

Constituent Name	Units		MEC ¹		B	C	CMC	CCC	Water & Org. ²	Org. Only ³	MCL	RP ^{4,5}
gamma-BHC	ug/L	<	0.003	<	0.003	0.019	0.95	0.08	0.019	0.063	0.2	No
delta-BHC	ug/L	<	0.003	<	0.003	No Criteria	--	--	--	--	--	No
Chlordane	ug/L	<	0.019	<	0.019	0.00057	2.4	0.0043	0.00057	0.0006	0.1	Uo
4,4-DDT	ug/L		0.023	<	0.005	0.00059	1.1	0.001	0.00059	0.0006	--	Yes
4,4-DDD	ug/L	<	0.002	<	0.002	0.00083	1.1	0.001	0.00083	0.0008	--	Uo
alpha-Endosulfan	ug/L		0.004	<	0.002	0.056	0.22	0.056	110	240	--	No
beta-Endosulfan	ug/L	<	0.002	<	0.00014	0.056	0.22	0.056	110	240	--	No
Endosulfan Sulfate	ug/L	<	0.002	<	0.002	110	--	--	110	240	--	No
Endrin	ug/L	<	0.002	<	0.002	0.036	0.086	0.036	0.76	0.81	2	No
Endrin Aldehyde	ug/L	<	0.015	<	0.015	0.76	--	--	0.76	0.81	--	No
Heptachlor	ug/L	<	0.002	<	0.002	0.00021	0.52	0.0038	0.00021	0.0002	0.01	Uo
Heptachlor Epoxide	ug/L	<	0.001	<	0.001	0.0001	0.52	0.0038	0.0001	0.0001	0.01	Uo
Toxaphene	ug/L	<	0.036	<	0.036	0.0002	0.73	0.0002	0.00073	0.0008	3	Uo
Chromium (VI)	ug/L	<	1.9	<	1.9	11	16	11	--	--	--	No
PCB(1)	ug/L	<	0.11	<	0.11	0.00017	--	0.014	0.00017	0.0002	0.5	Uo
Aluminum	ug/L		140		--	No Criteria	750	87	--	--	50	Ud
Chromium (III)	ug/L	<	10	<	10	160	1300	160	--	--	--	No
Ammonia (as N)	mg/L		0.26		--	0.151790	0.37292	0.151791	--	--	--	Yes
Nitrate (as N)	mg/L		7.2		--	10	--	--	10	--	10	Uo
Nitrite (as N)	mg/L	<	0.14		--	1	--	--	--	--	1	Uo

Table Notes

1. MEC = Maximum Effluent Concentration
2. Water & Org = CTR Water Quality Criteria for Human Health for Consumption of Water & Organisms
3. Org. Only = CTR Water Quality Criteria for Human Health for Organisms Only
4. RP = Reasonable Potential
5. Ud = Undetermined, Insufficient Data
6. Uo = Undetermined, No Water Quality Criteria.

ATTACHMENT G - AMEL AND MDEL AMMONIA STANDARDS BASED ON 2013 FRESHWATER ACUTE CRITERIA

Table G-1. pH and Temperature Dependent AMEL Ammonia Criteria

pH	Temp (°C)																	
	0	14.0	15.0	16.0	17.0	18.0	19.0	20.0	21.0	22.0	23.0	24.0	25.0	26.0	27.0	28.0	29.0	30.0
6.5	12	12	12	12	11	10	9.4	8.6	7.9	7.3	6.7	6.2	5.7	5.3	4.8	4.4	4.1	3.8
6.6	12	12	12	12	11	10	9.0	8.3	7.6	7.0	6.5	5.9	5.5	5.0	4.6	4.3	3.9	3.6
6.7	11	11	11	11	10	9.3	8.6	7.9	7.3	6.7	6.1	5.7	5.2	4.8	4.4	4.1	3.7	3.4
6.8	11	11	11	10	9.5	8.8	8.1	7.4	6.8	6.3	5.8	5.3	4.9	4.5	4.2	3.8	3.5	3.2
6.9	10	10	10	9.6	8.9	8.2	7.5	6.9	6.4	5.9	5.4	5.0	4.6	4.2	3.9	3.6	3.3	3.0
7.0	9.2	9.2	9.2	8.9	8.2	7.5	6.9	6.4	5.9	5.4	5.0	4.6	4.2	3.9	3.6	3.3	3.0	2.8
7.1	8.4	8.4	8.4	8.1	7.5	6.9	6.3	5.8	5.3	4.9	4.5	4.2	3.8	3.5	3.3	3.0	2.8	2.5
7.2	7.5	7.5	7.5	7.3	6.7	6.2	5.7	5.2	4.8	4.4	4.1	3.7	3.5	3.2	2.9	2.7	2.5	2.3
7.3	6.7	6.7	6.7	6.5	5.9	5.5	5.0	4.6	4.3	3.9	3.6	3.3	3.1	2.8	2.6	2.4	2.2	2.0
7.4	5.8	5.8	5.8	5.7	5.2	4.8	4.4	4.1	3.7	3.4	3.2	2.9	2.7	2.5	2.3	2.1	1.9	1.8
7.5	5.1	5.1	5.1	4.9	4.5	4.2	3.8	3.5	3.2	3.0	2.7	2.5	2.3	2.1	2.0	1.8	1.7	1.5
7.6	4.3	4.3	4.3	4.2	3.9	3.6	3.3	3.0	2.8	2.6	2.3	2.2	2.0	1.8	1.7	1.6	1.4	1.3
7.7	3.7	3.7	3.7	3.6	3.3	3.0	2.8	2.6	2.4	2.2	2.0	1.8	1.7	1.6	1.4	1.3	1.2	1.1
7.8	3.1	3.1	3.1	3.0	2.8	2.5	2.3	2.1	2.0	1.8	1.7	1.5	1.4	1.3	1.2	1.1	1.0	0.9
7.9	2.6	2.6	2.6	2.5	2.3	2.1	1.9	1.8	1.6	1.5	1.4	1.3	1.2	1.1	1.0	0.9	0.8	0.8
8.0	2.1	2.1	2.1	2.1	1.9	1.8	1.6	1.5	1.4	1.3	1.2	1.1	1.0	0.9	0.8	0.8	0.7	0.6
8.1	1.8	1.8	1.8	1.7	1.6	1.5	1.3	1.2	1.1	1.0	1.0	0.9	0.8	0.7	0.7	0.6	0.6	0.5
8.2	1.5	1.5	1.5	1.4	1.3	1.2	1.1	1.0	0.9	0.9	0.8	0.7	0.7	0.6	0.6	0.5	0.5	0.4
8.3	1.2	1.2	1.2	1.2	1.1	1.0	0.9	0.8	0.8	0.7	0.7	0.6	0.6	0.5	0.5	0.4	0.4	0.4
8.4	1.0	1.0	1.0	1.0	0.9	0.8	0.7	0.7	0.6	0.6	0.5	0.5	0.5	0.4	0.4	0.4	0.3	0.3
8.5	0.8	0.8	0.8	0.8	0.7	0.7	0.6	0.6	0.5	0.5	0.4	0.4	0.4	0.3	0.3	0.3	0.3	0.2
8.6	0.7	0.7	0.7	0.7	0.6	0.6	0.5	0.5	0.4	0.4	0.4	0.3	0.3	0.3	0.3	0.2	0.2	0.2
8.7	0.6	0.6	0.6	0.5	0.5	0.5	0.4	0.4	0.4	0.3	0.3	0.3	0.3	0.2	0.2	0.2	0.2	0.2
8.8	0.5	0.5	0.5	0.5	0.4	0.4	0.4	0.3	0.3	0.3	0.3	0.2	0.2	0.2	0.2	0.2	0.2	0.1
8.9	0.4	0.4	0.4	0.4	0.4	0.3	0.3	0.3	0.3	0.2	0.2	0.2	0.2	0.2	0.2	0.1	0.1	0.1
9.0	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.2	0.2	0.2	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.1

Table G-2. pH and Temperature Dependent MDEL Ammonia Criteria

pH	Temp (°C)																	
	0	14.0	15.0	16.0	17.0	18.0	19.0	20.0	21.0	22.0	23.0	24.0	25.0	26.0	27.0	28.0	29.0	30.0
6.5	32	32	32	31	29	27	25	23	21	19	18	16	15	14	13	12	11	10
6.6	31	31	31	30	28	26	24	22	20	18	17	16	14	13	12	11	10	9
6.7	30	30	30	29	26	24	22	21	19	17	16	15	14	13	12	11	10	9
6.8	28	28	28	27	25	23	21	19	18	16	15	14	13	12	11	10	9	8
6.9	26	26	26	25	23	21	20	18	17	15	14	13	12	11	10	9	9	8
7.0	24	24	24	23	21	20	18	17	15	14	13	12	11	10	9.3	8.6	7.9	7.3
7.1	22	22	22	21	19	18	16	15	14	13	12	11	10	9.2	8.5	7.8	7.2	6.6
7.2	20	20	20	19	18	16	15	14	13	12	11	9.8	9.0	8.3	7.6	7.0	6.5	6.0
7.3	17	17	17	17	16	14	13	12	11	10	9.4	8.7	8.0	7.4	6.8	6.2	5.7	5.3
7.4	15	15	15	15	14	13	12	11	9.8	9.0	8.3	7.6	7.0	6.5	5.9	5.5	5.0	4.6
7.5	13	13	13	13	12	11	10	9.2	8.5	7.8	7.2	6.6	6.1	5.6	5.1	4.7	4.4	4.0
7.6	11	11	11	11	10	9.3	8.6	7.9	7.2	6.7	6.1	5.6	5.2	4.8	4.4	4.1	3.7	3.4
7.7	9.6	9.6	9.6	9.3	8.6	7.9	7.3	6.7	6.1	5.7	5.2	4.8	4.4	4.1	3.7	3.4	3.2	2.9
7.8	8.1	8.1	8.1	7.8	7.2	6.6	6.1	5.6	5.2	4.8	4.4	4.0	3.7	3.4	3.1	2.9	2.7	2.4
7.9	6.7	6.7	6.7	6.5	6.0	5.5	5.1	4.7	4.3	4.0	3.7	3.4	3.1	2.8	2.6	2.4	2.2	2.0
8.0	5.6	5.6	5.6	5.4	5.0	4.6	4.2	3.9	3.6	3.3	3.0	2.8	2.6	2.4	2.2	2.0	1.8	1.7
8.1	4.6	4.6	4.6	4.5	4.1	3.8	3.5	3.2	3.0	2.7	2.5	2.3	2.1	2.0	1.8	1.7	1.5	1.4
8.2	3.8	3.8	3.8	3.7	3.4	3.1	2.9	2.6	2.4	2.2	2.1	1.9	1.7	1.6	1.5	1.4	1.3	1.2
8.3	3.1	3.1	3.1	3.0	2.8	2.6	2.4	2.2	2.0	1.8	1.7	1.6	1.4	1.3	1.2	1.1	1.0	1.0
8.4	2.6	2.6	2.6	2.5	2.3	2.1	1.9	1.8	1.7	1.5	1.4	1.3	1.2	1.1	1.0	0.9	0.9	0.8
8.5	2.1	2.1	2.1	2.1	1.9	1.7	1.6	1.5	1.4	1.3	1.2	1.1	1.0	0.9	0.8	0.8	0.7	0.6
8.6	1.8	1.8	1.8	1.7	1.6	1.4	1.3	1.2	1.1	1.0	1.0	0.9	0.8	0.7	0.7	0.6	0.6	0.5
8.7	1.5	1.5	1.5	1.4	1.3	1.2	1.1	1.0	0.9	0.9	0.8	0.7	0.7	0.6	0.6	0.5	0.5	0.4
8.8	1.2	1.2	1.2	1.2	1.1	1.0	0.9	0.9	0.8	0.7	0.7	0.6	0.6	0.5	0.5	0.4	0.4	0.4
8.9	1.0	1.0	1.0	1.0	0.9	0.8	0.8	0.7	0.7	0.6	0.6	0.5	0.5	0.4	0.4	0.4	0.3	0.3
9.0	0.9	0.9	0.9	0.9	0.8	0.7	0.7	0.6	0.6	0.5	0.5	0.4	0.4	0.4	0.3	0.3	0.3	0.3

ATTACHMENT H - EXAMPLE AMMONIA IMPACT RATIO (AIR) CALCULATOR

[illegible]